12-Sep-2025

# SEC-T 0x10sion
# Music Synthesizer Badge

## Assembly Instructions

(with re-programming how-to)




V 0.2

open source hardware

SEC-T 0X10SION

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CORNFIELD ELECTRONICS

# Learn To Solder



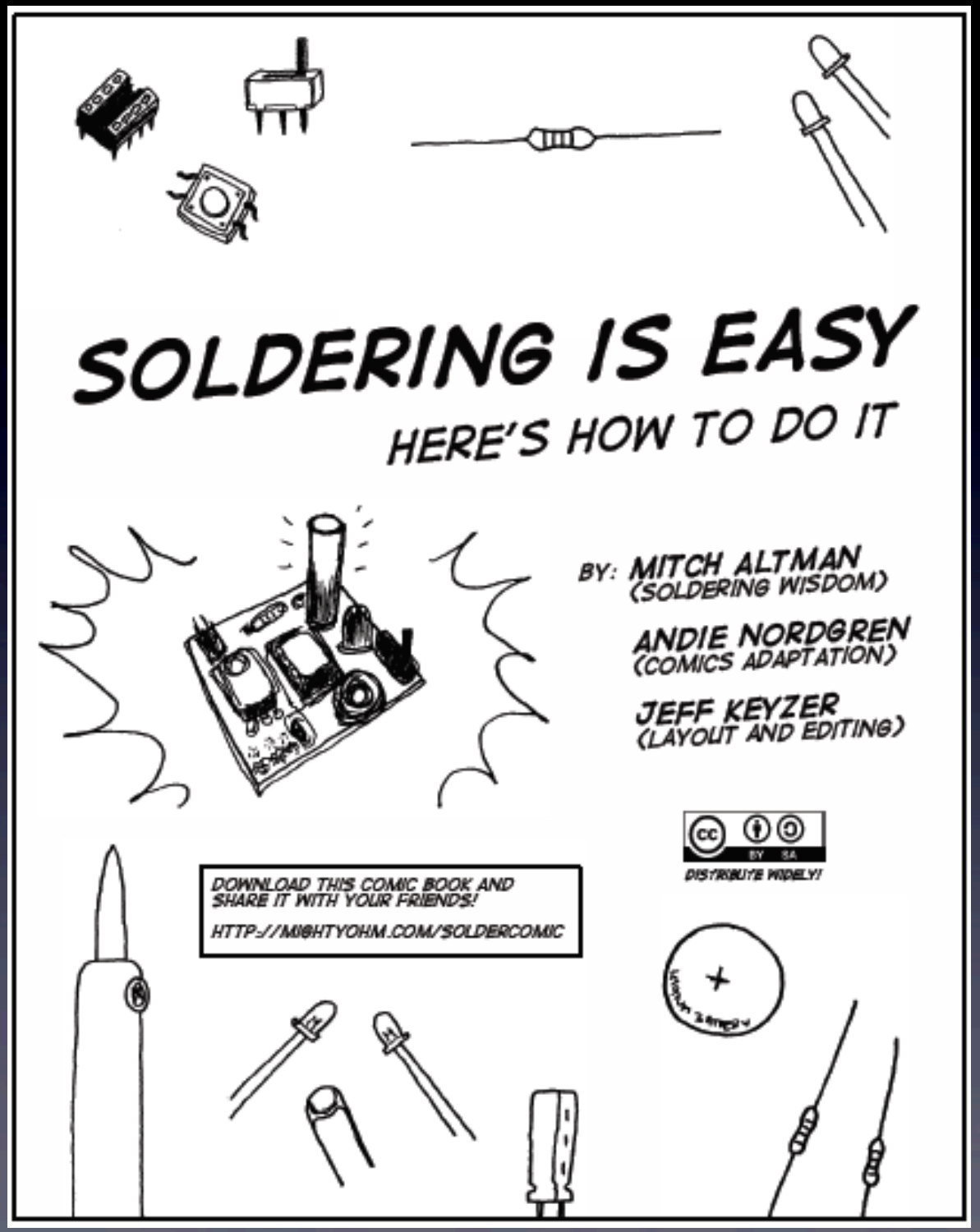

The following photos will show you how to solder.

But feel free to download the “Soldering Is Easy” comic book for free!

(In many different languages.)

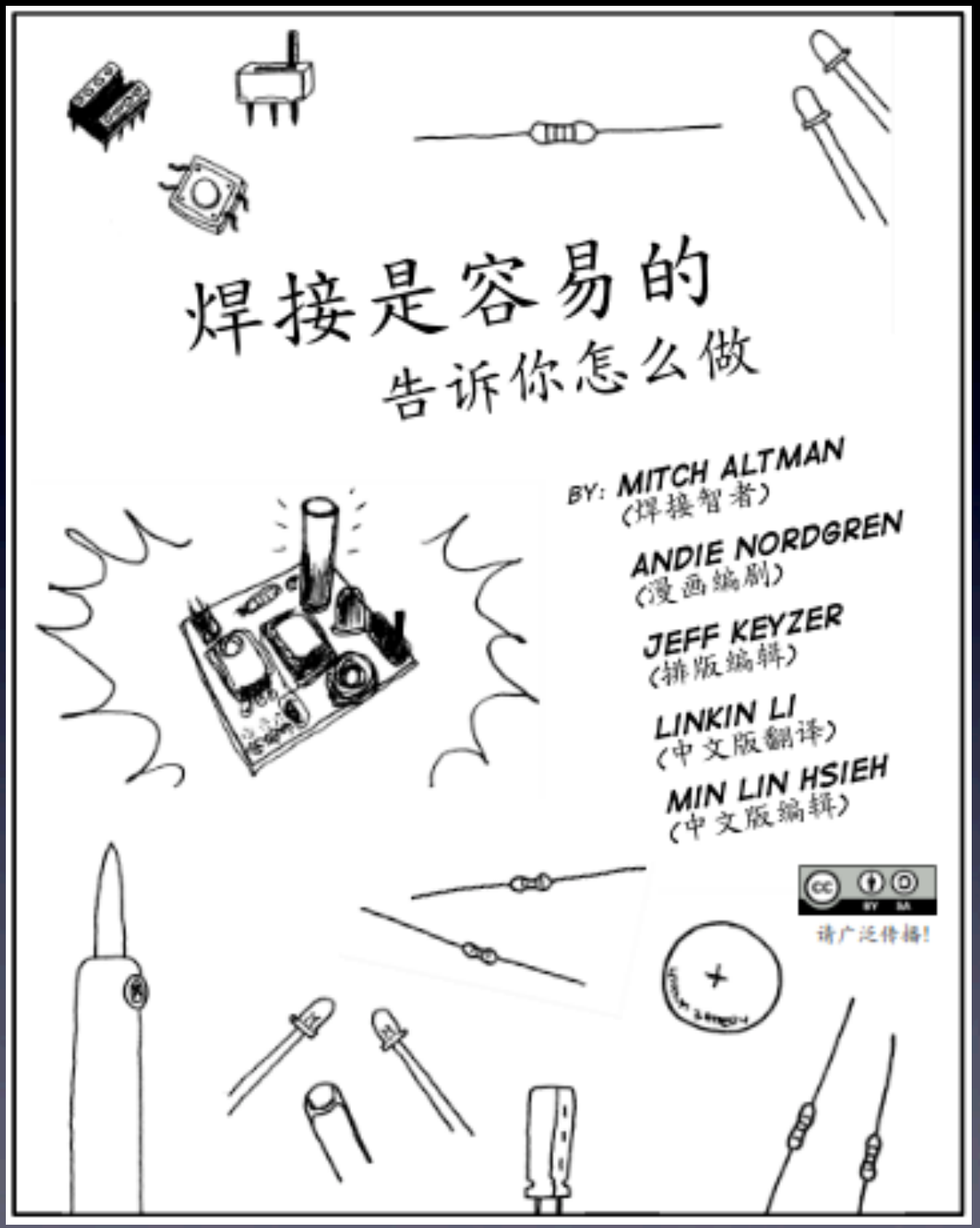
download for free at:
http://mightyohm.com/soldercomic

Learn To Solder



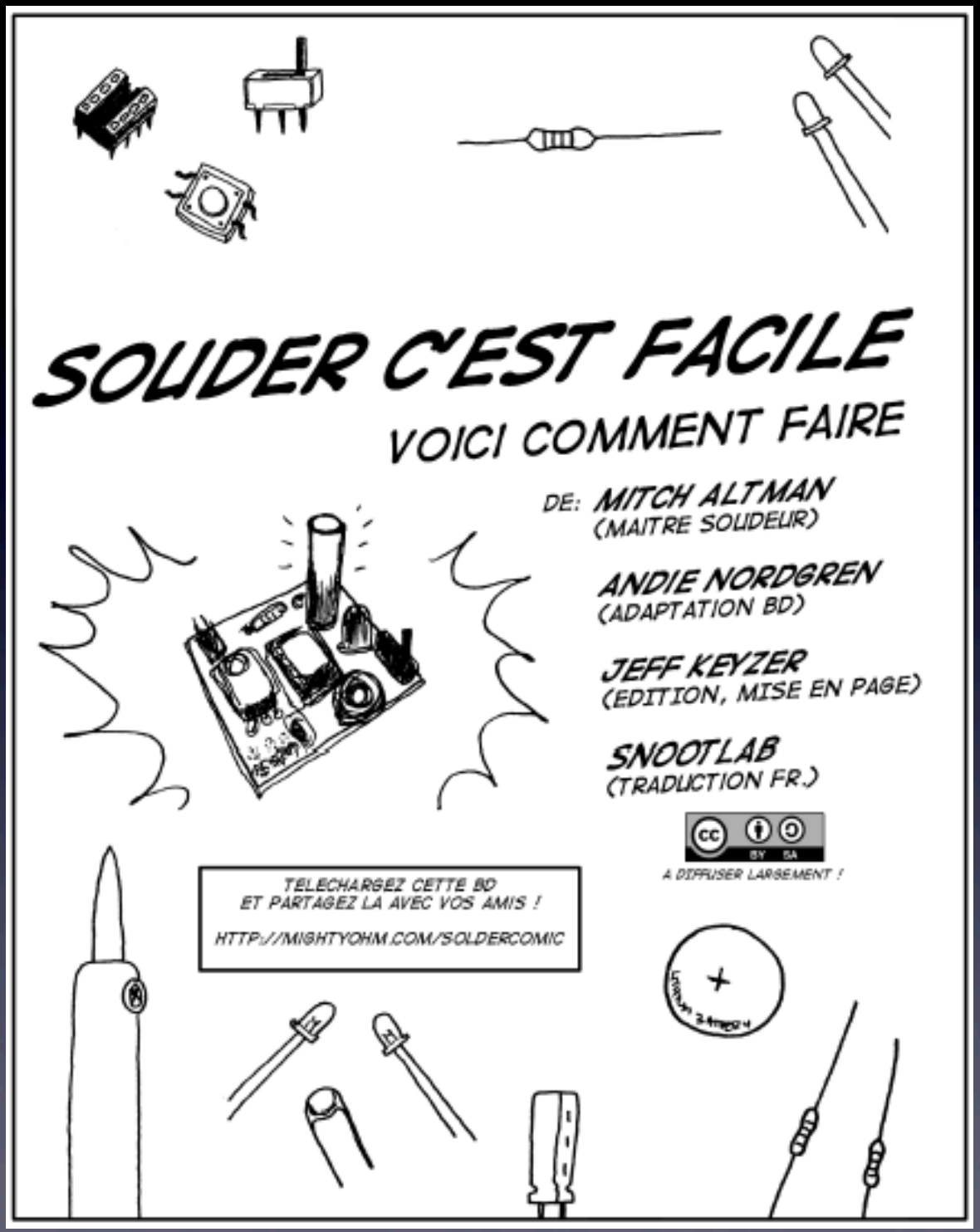

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Learn To Solder

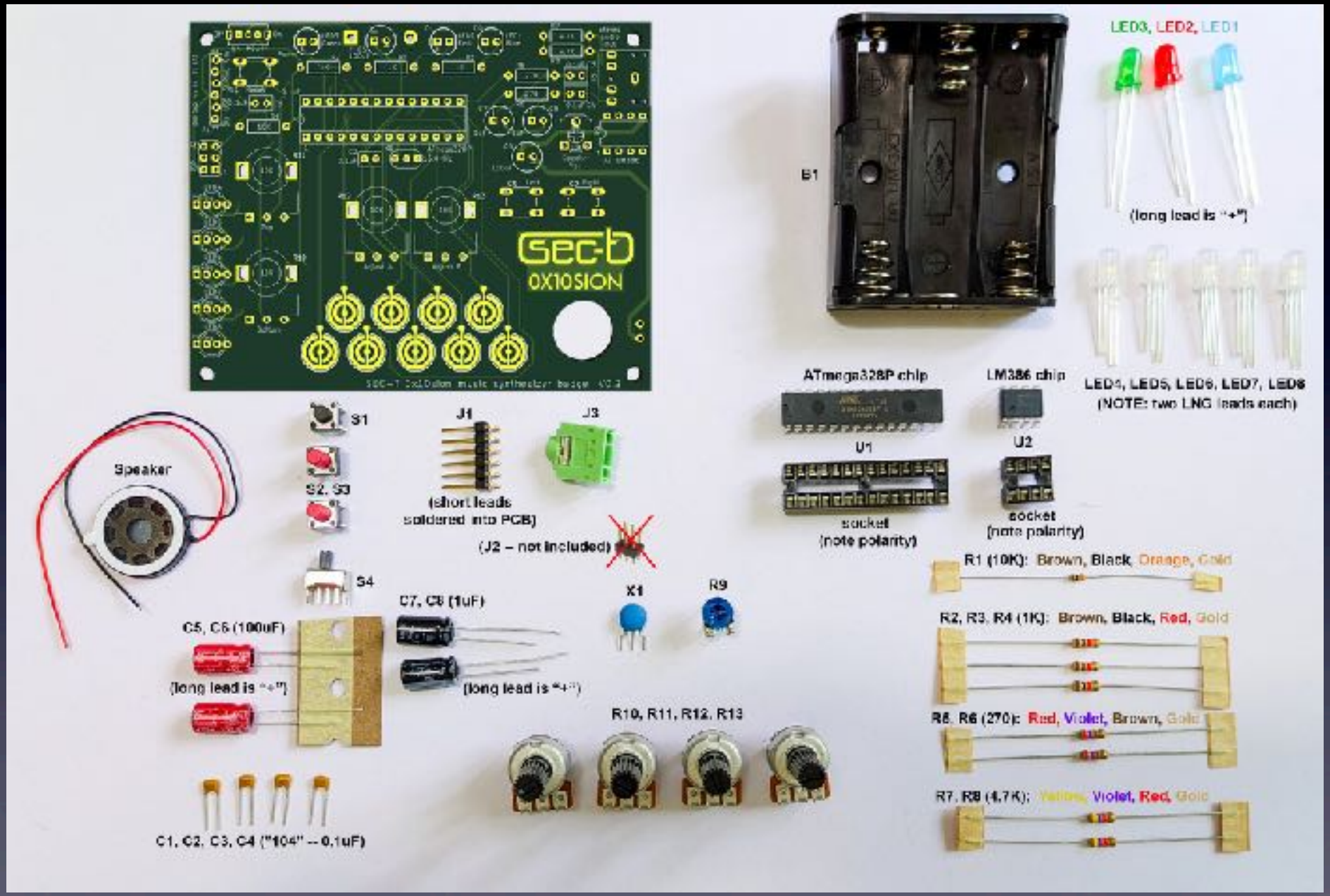



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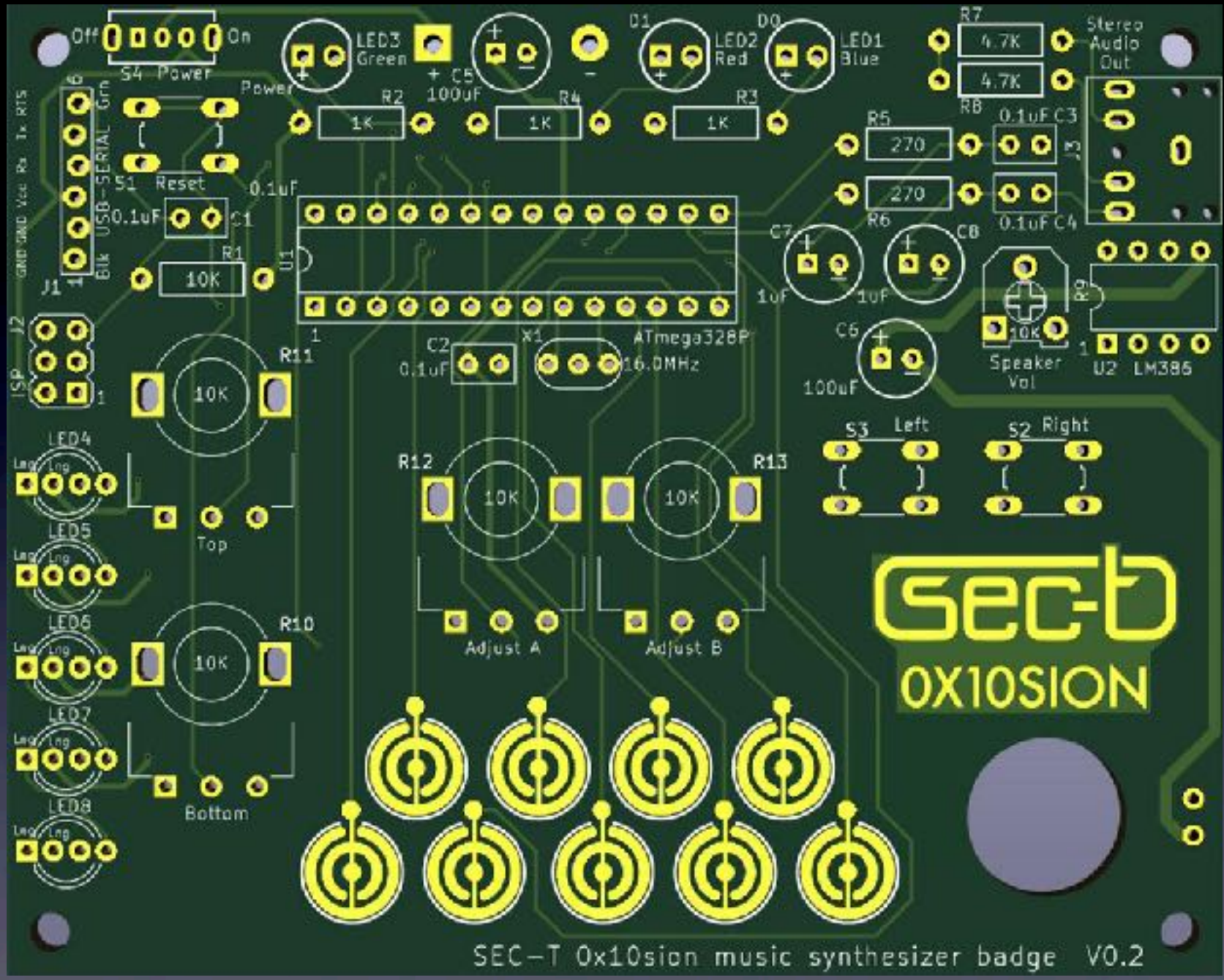
Learn To Solder



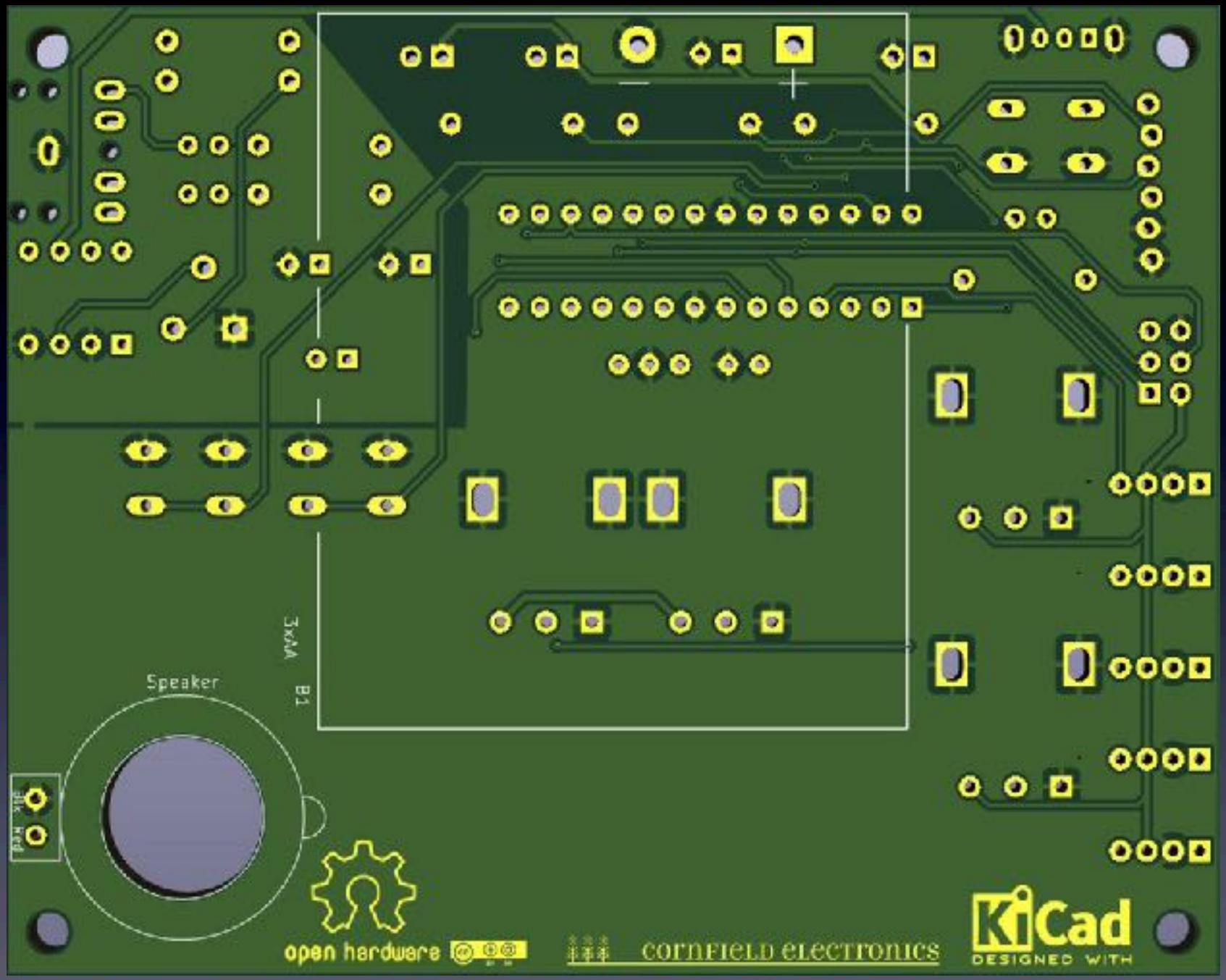

download for free at:
http://mightyohm.com/soldercomic
(In many different languages.)




All of the parts




The board we'll solder the parts to (front)




The board we'll solder the parts to (back)





**Note:**

Since we will use Lead-Free solder it is helpful to also have flux paste in a syringe and Isopropyl Alcohol




## The tools you'll need:

- soldering Iron (35W or less)
- solder (*more details coming*)
- soldering iron stand
- cellulose kitchen sponge *(not plastic!)*
- *small* wire cutter

# Our first part

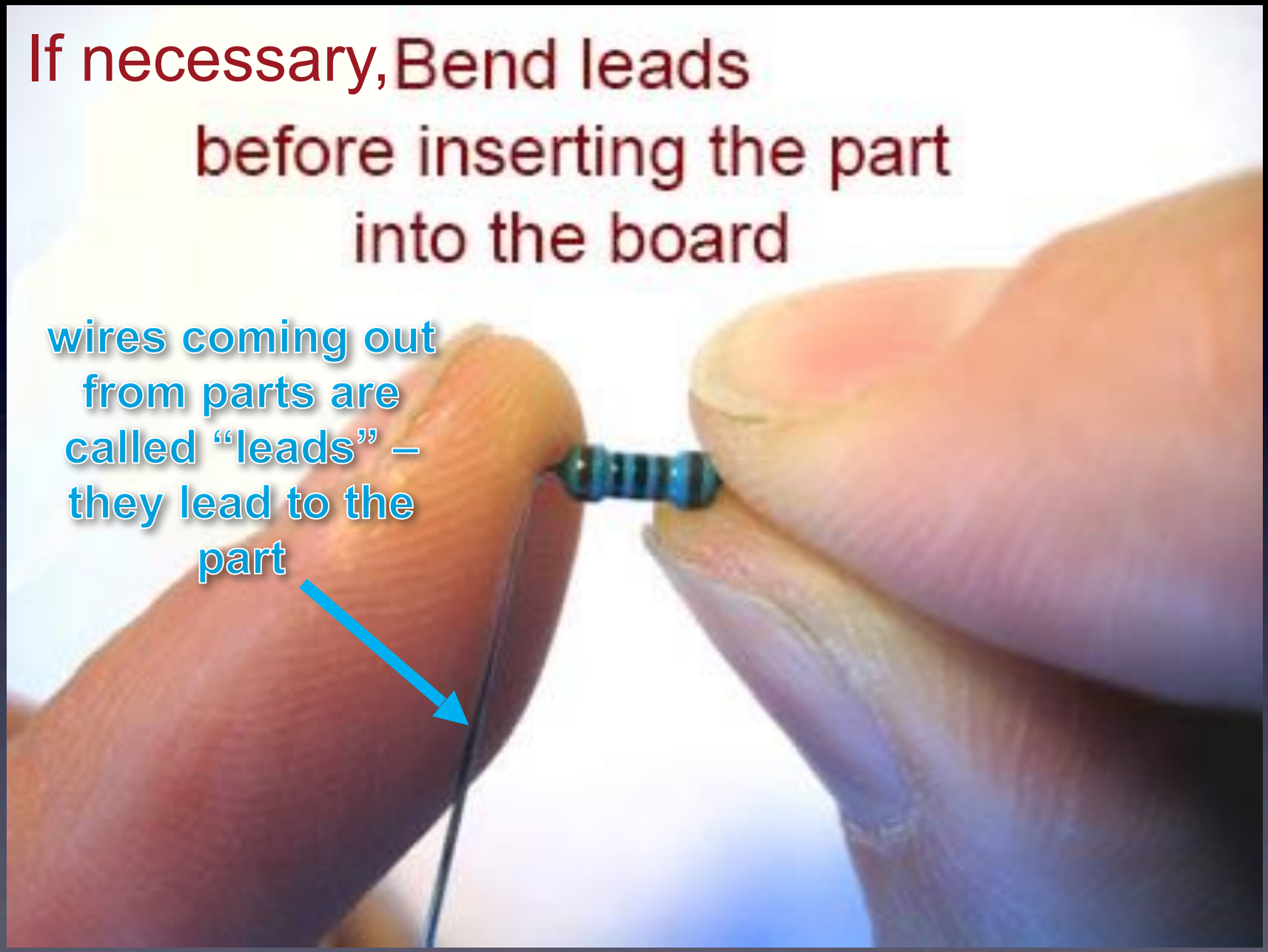


**R1: Brown, Black, Orange**

(the small one)

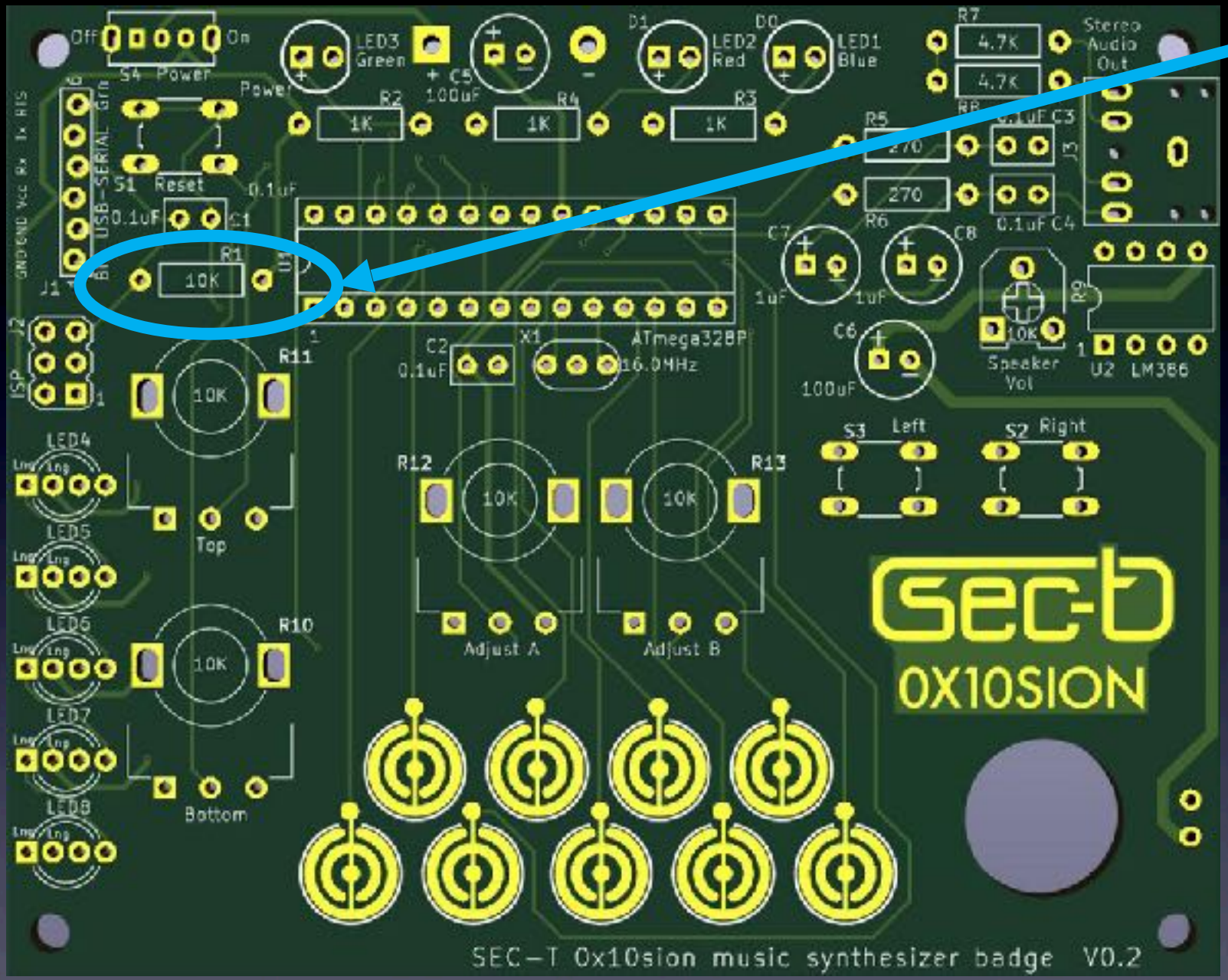
(*not* Brown, Black, ~~Red~~)

# Some parts, such as resistors, need their leads bent first

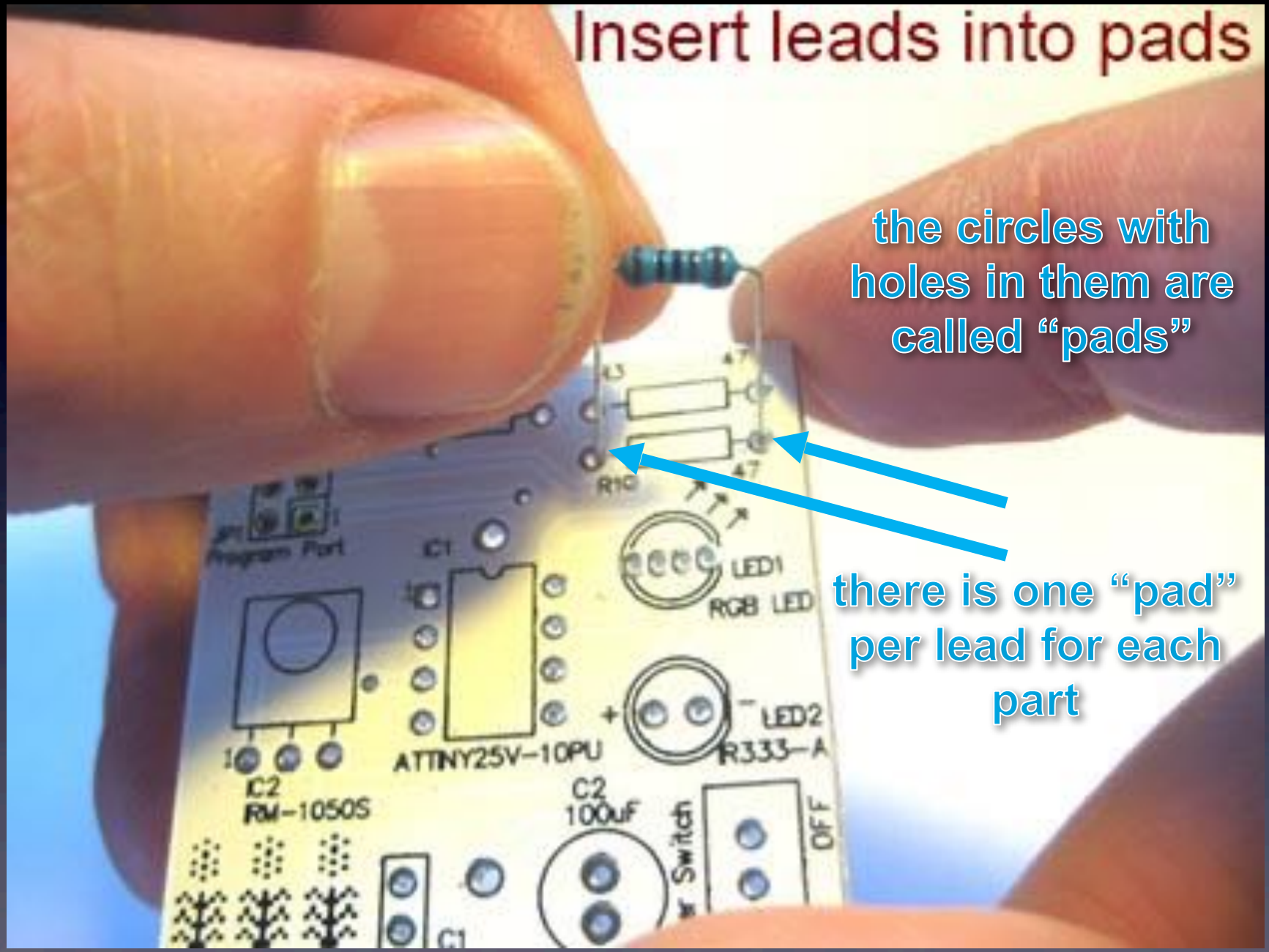


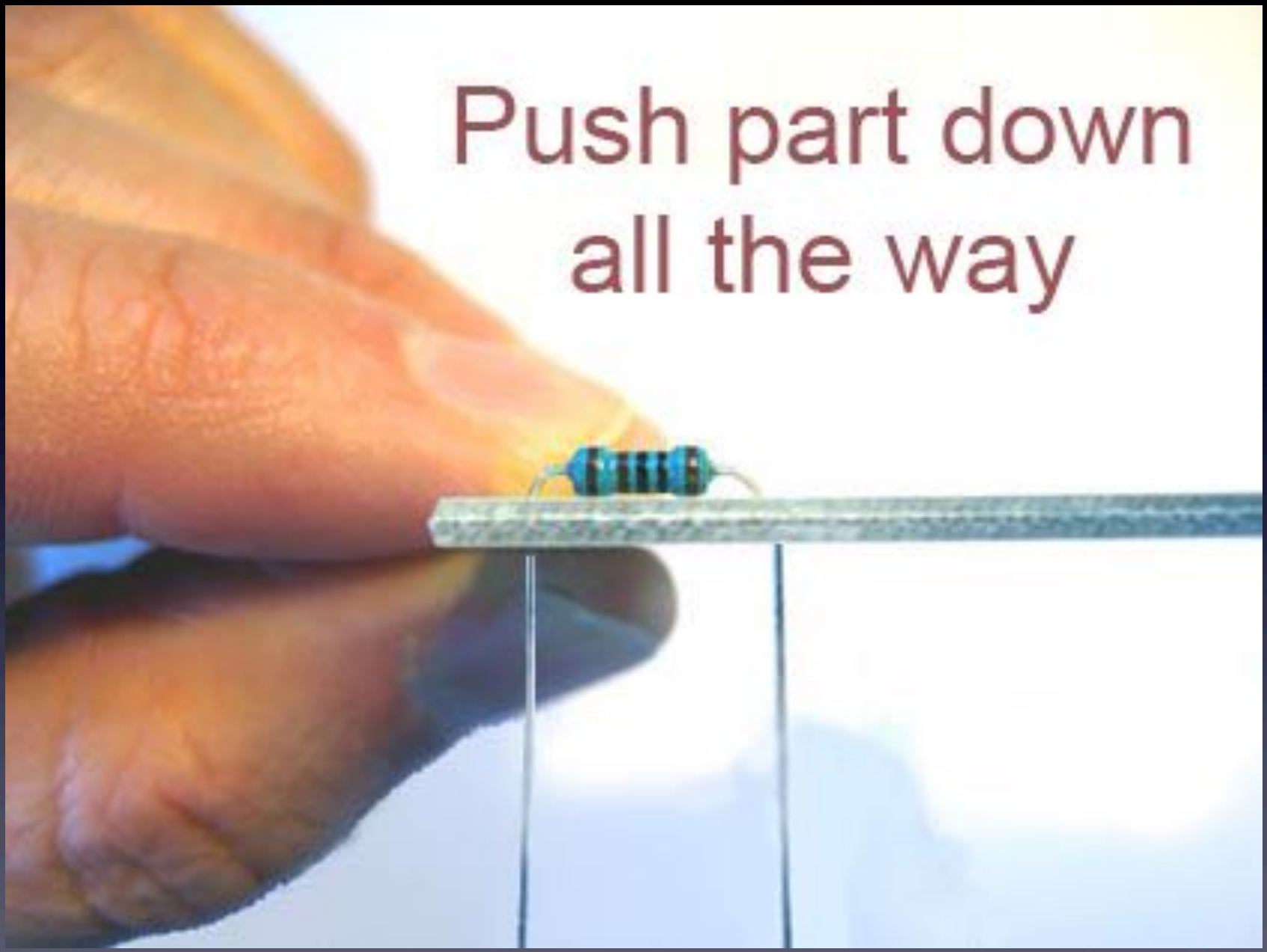


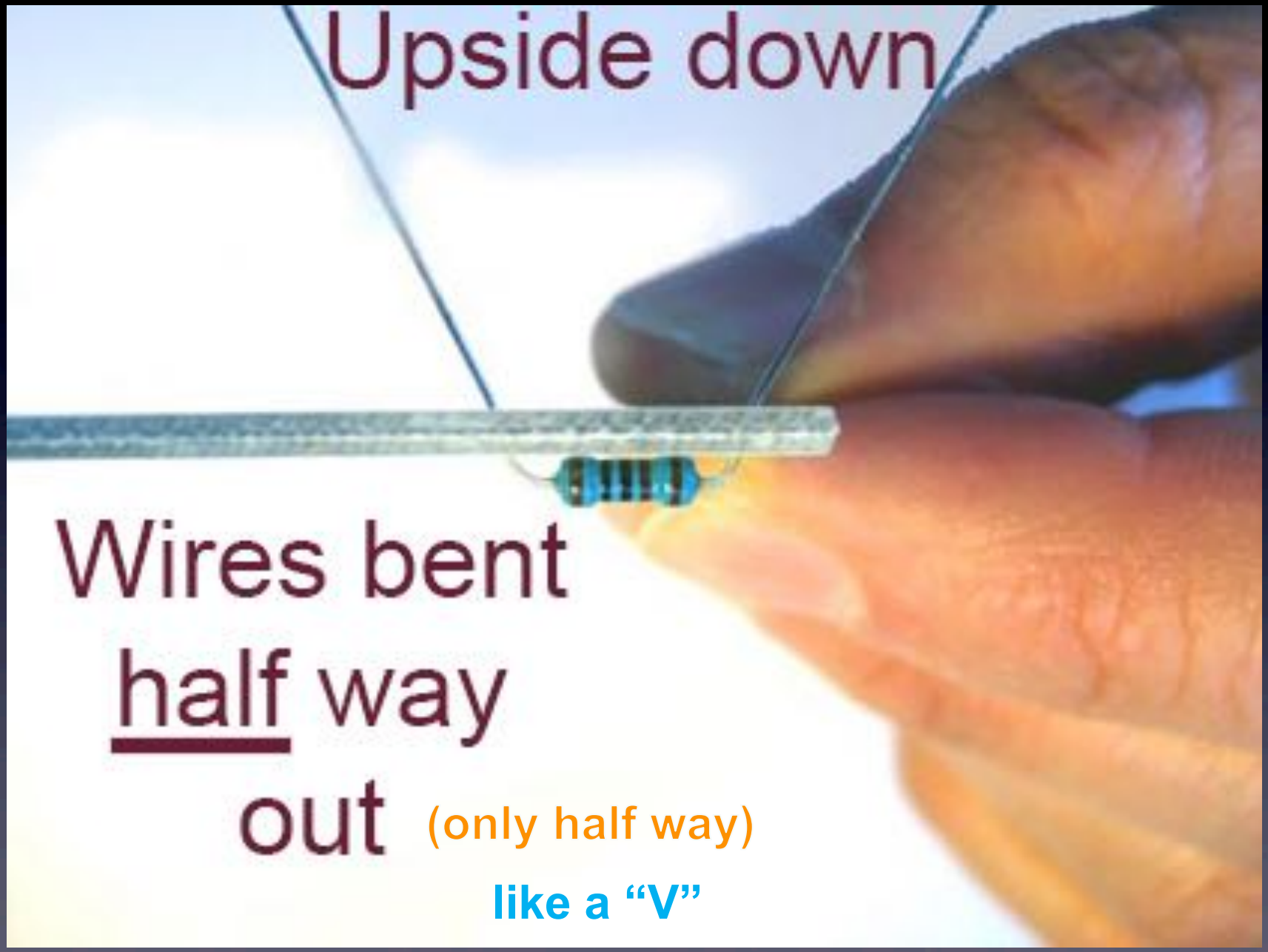
**R1 – this is how it will look *before* inserting it into the board**



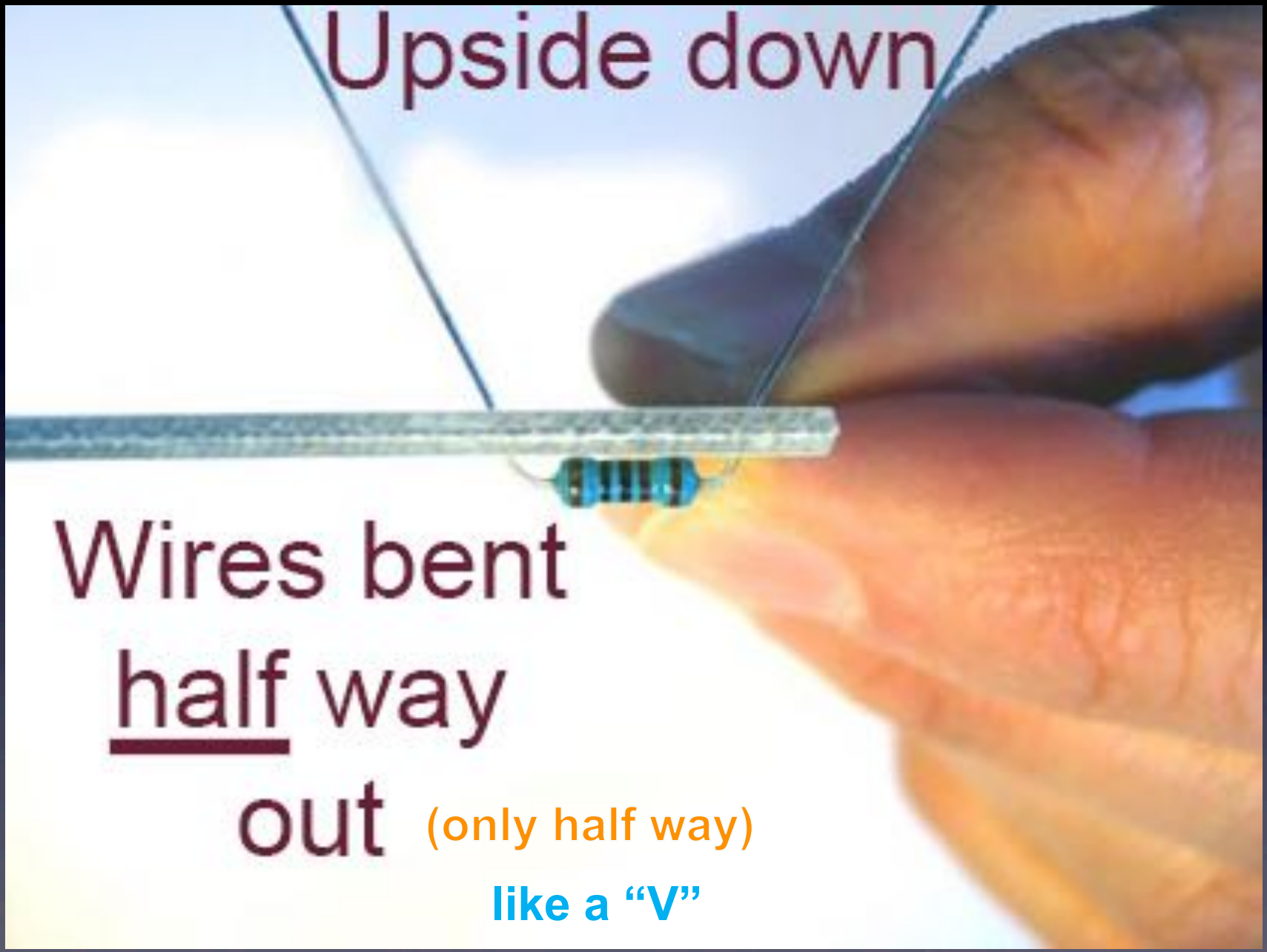

# R1 – this is where it goes


Insert leads into pads
the circles with holes in them are called “pads”
there is one “pad” per lead for each part
R3
47
R10
47
LED1
RGB LED
IC1
ATTINY25V-10PU
LED2
R333-A
IC2
RM-1050S
C2
100uF
C1
Switch
OFF

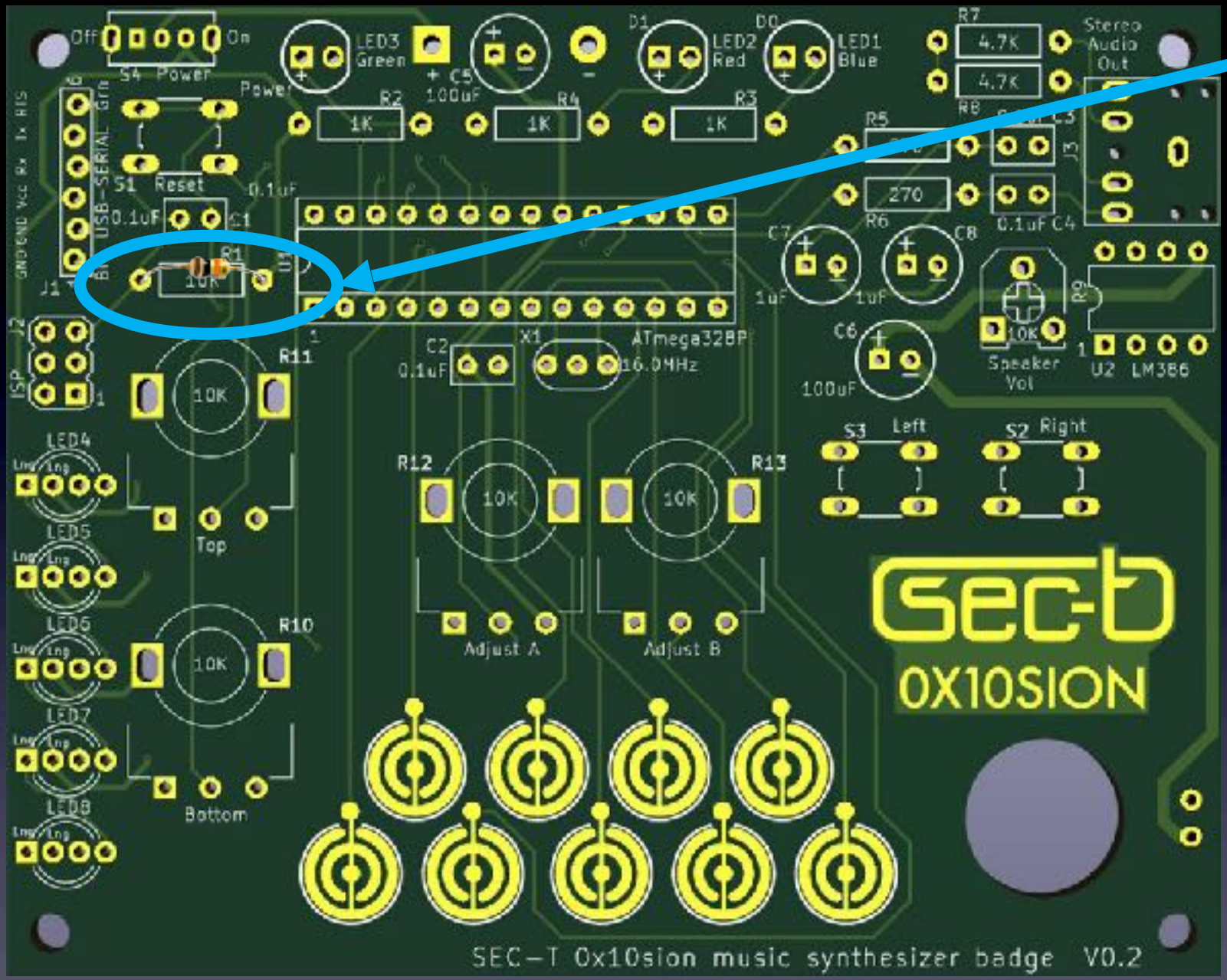

Push part down
all the way




so that the part won’t fall out while soldering it




# Ready to Solder !




# R1 – inserted into the board



# How to hold a soldering iron

(Like a pencil – held from underneath)

Important

The best kind of solder for DIY electronics:

*(Sn – Tin / Pb – Lead)*

63/37 rosin core,

0.031” (0.8mm) diameter (or smaller)

*(60/40 is also good)*

**Note**:

**Most**
***Lead-Free*** **solder**
**has poisonous fumes!**

# This is what we will use:

A good kind of solder for DIY electronics:

***This is the only good*** Lead-Free ***solder I have found!***

*(after years of searching)*




**Kester**
**K100LD Rosin**
*(<u>not</u> "No Clean")*
0.031" diameter (0.8mm)

# This is what we will use:

## A good kind of solder for DIY electronics:

***This is the only good Lead-Free solder I have found!***

*(after years of searching)*



Kester **K100LD Rosin** Solder

0.031" diameter (0.8mm)







**Note:**

**Since we will use *Lead-Free* solder**
**it is *helpful***
**to also have**
***flux paste* in a syringe**
***and Isopropyl Alcohol***

# Another good kind of solder for DIY electronics:

***This is another good Lead-Free solder I have found!***




**Duratool**
**D01685 Rosin**

0.7mm diameter

***(as good as the Kester K100LD Rosin)***

# Another good kind of solder for DIY electronics:

***This is another good Lead-Free solder I have found!***




**MG Chemicals**
**4900 Rosin (112g, 227g, 454g)**

0.8mm diameter

***(as good as the Kester K100LD Rosin)***

# Another good kind of solder for DIY electronics:

***This is another good Lead-Free solder I have found!***




**iFixit**
**IF145-077-2 (12g)**
1.0mm diameter

***(as good as the Kester K100LD Rosin)***

# 3 Safety Tips…

# Safety Tip #1:

## Hot!!

(When you touch the tip,
you *will* let go quickly every time!)

# Safety Tip #2:

Soldering chemicals are toxic

But they easily wash off your hands with soap and water

# Safety Tip #3:

*(coming soon…)*

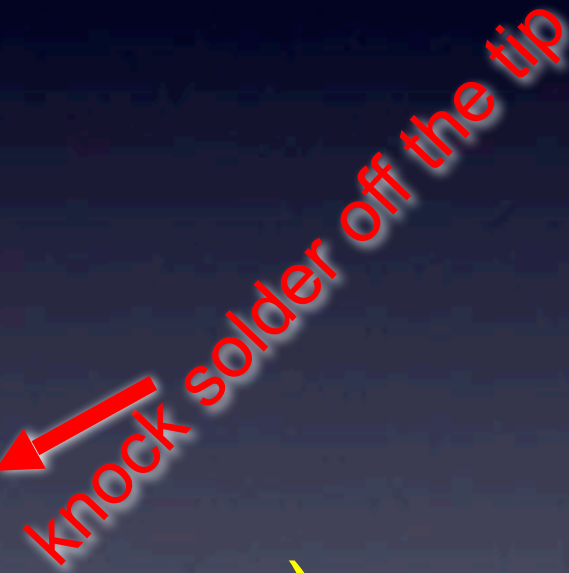
# 2 secrets
# to good soldering…

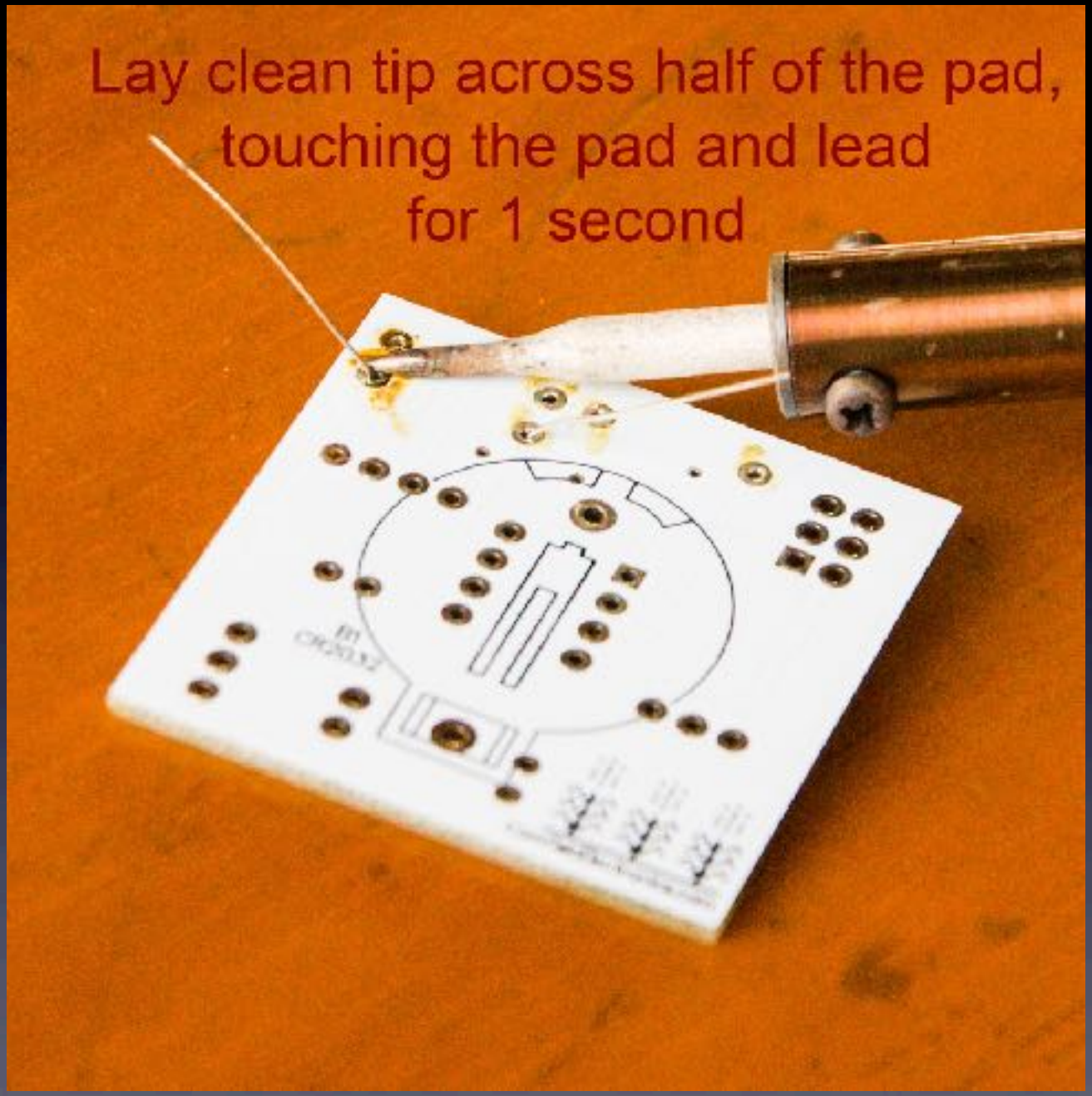
## Secret #1:

Clean the tip!
(before every solder connection)

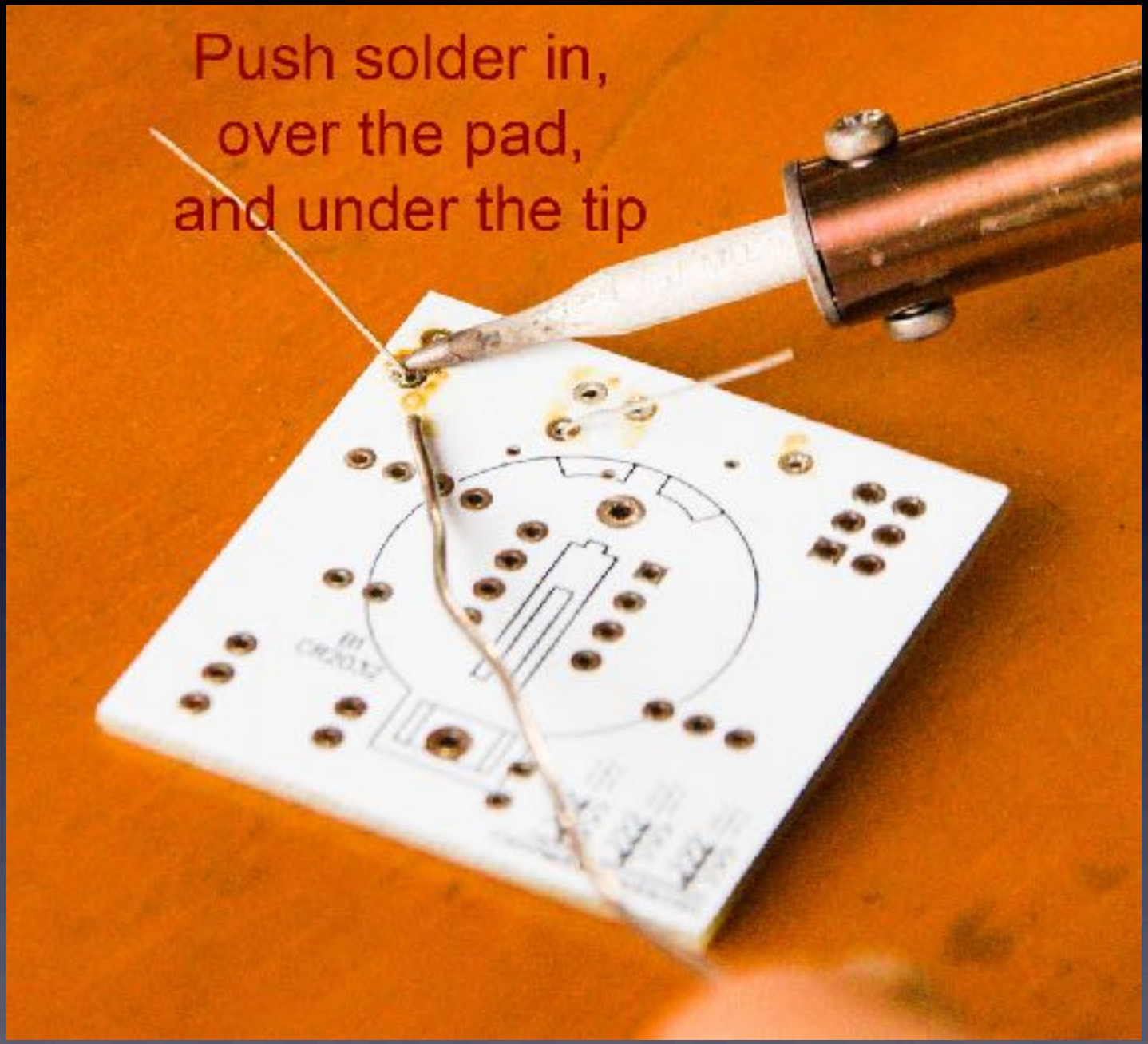
Bang (lightly) 3 times,
Swipe, Rotate, Swipe (on the sponge):

*Keep the tip shiny silver!*



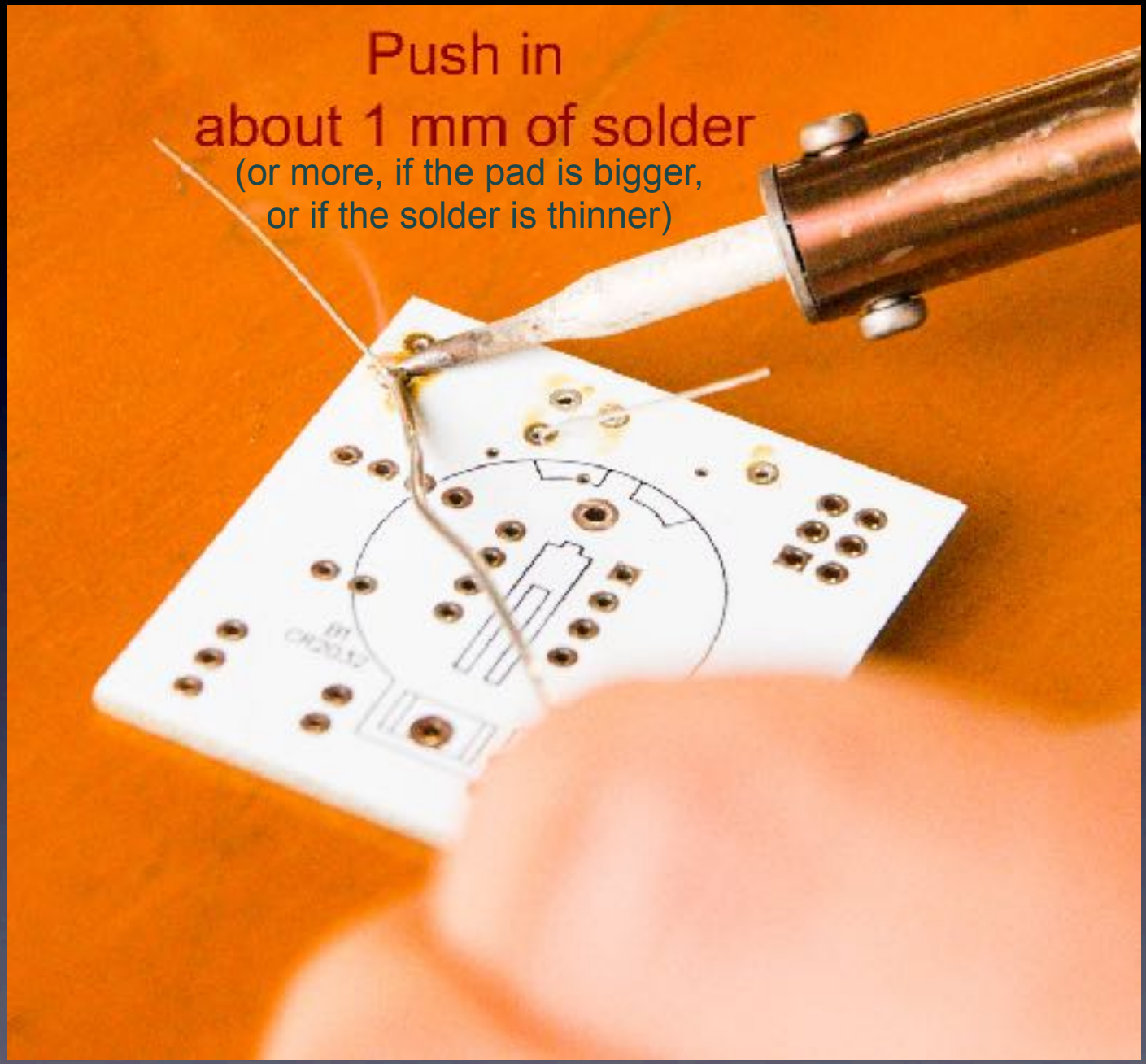

Lay clean tip across half of the pad,
touching the pad and lead
for 1 second

Do this quickly (slowly doesn't work well) – solder in & out in about 1 second




**IMPORTANT:** Make sure solder melts on the underside of the soldering iron tip (not the side or top of the soldering iron tip)!

Do this quickly (slowly doesn't work well) – solder in & out in about 1 second




Make sure solder melts on the <u>underside</u> of the soldering iron tip
(not the side or top of the soldering iron tip)!

HEY !!!

KEEP HOLDING TIP DOWN FOR 1 MORE SECOND !!

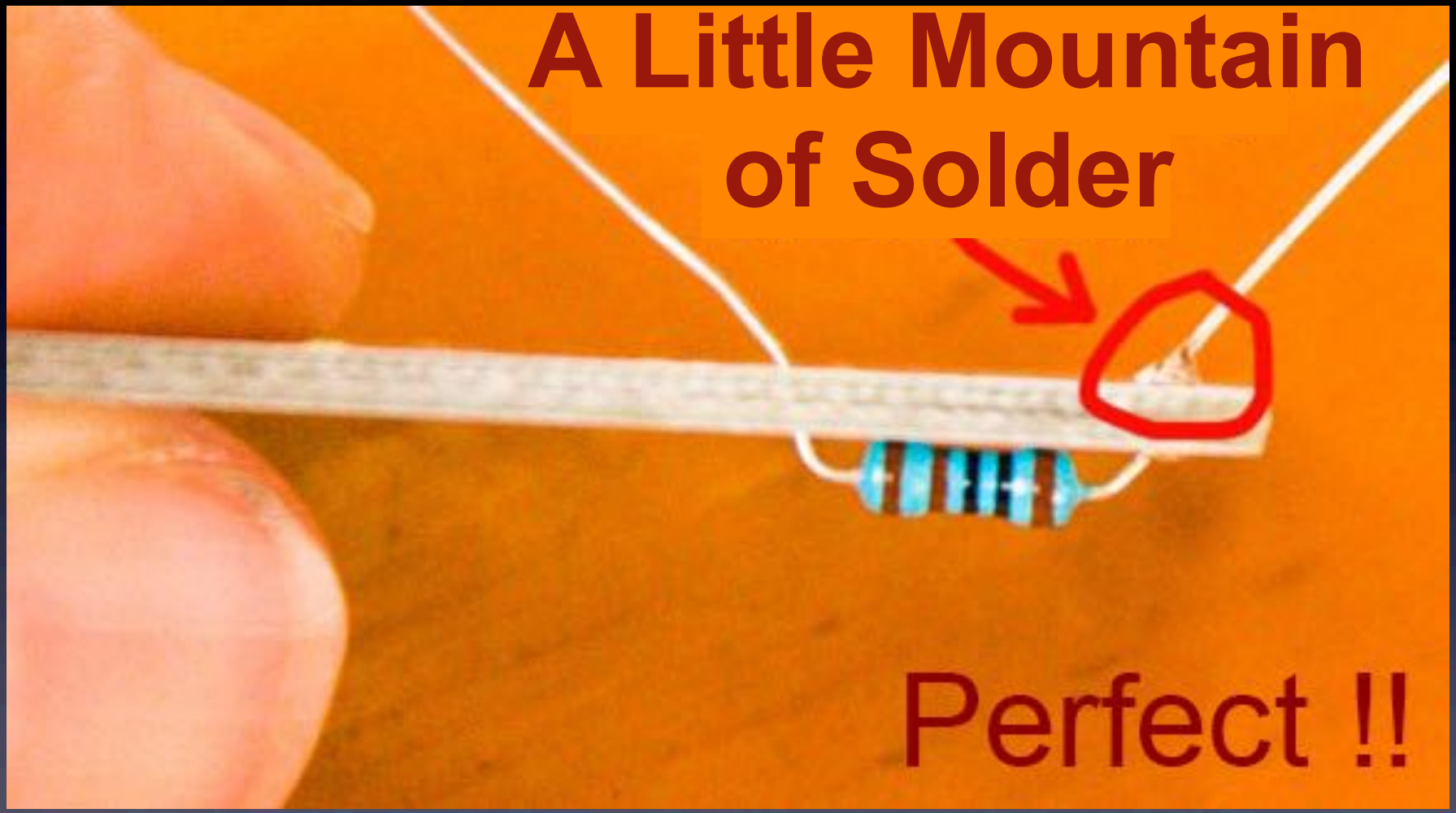


**WAIT !!**
Don’t lift the tip !!

## Secret #2:

Keep hot tip down
1 second
For solder to flow !!


Now
Lift soldering iron




If you can see any of the pad, or the hole, you need more solder – so, just do all the steps again to make it perfect.

# The Rhythm !

is just as important as the preceding steps!

# The Rhythm !

and speed  (about 1 second per step)



# The Rhythm !

and speed  (about 1 second per step)

# Clean the tip



# The Rhythm !

and speed  (about 1 second per step)



## Tip Down

# The Rhythm !

and speed  (about 1 second per step)



Solder In

# The Rhythm !

and speed  (about 1 second per step)



Solder Out

# The Rhythm !

and speed  (about 1 second per step)



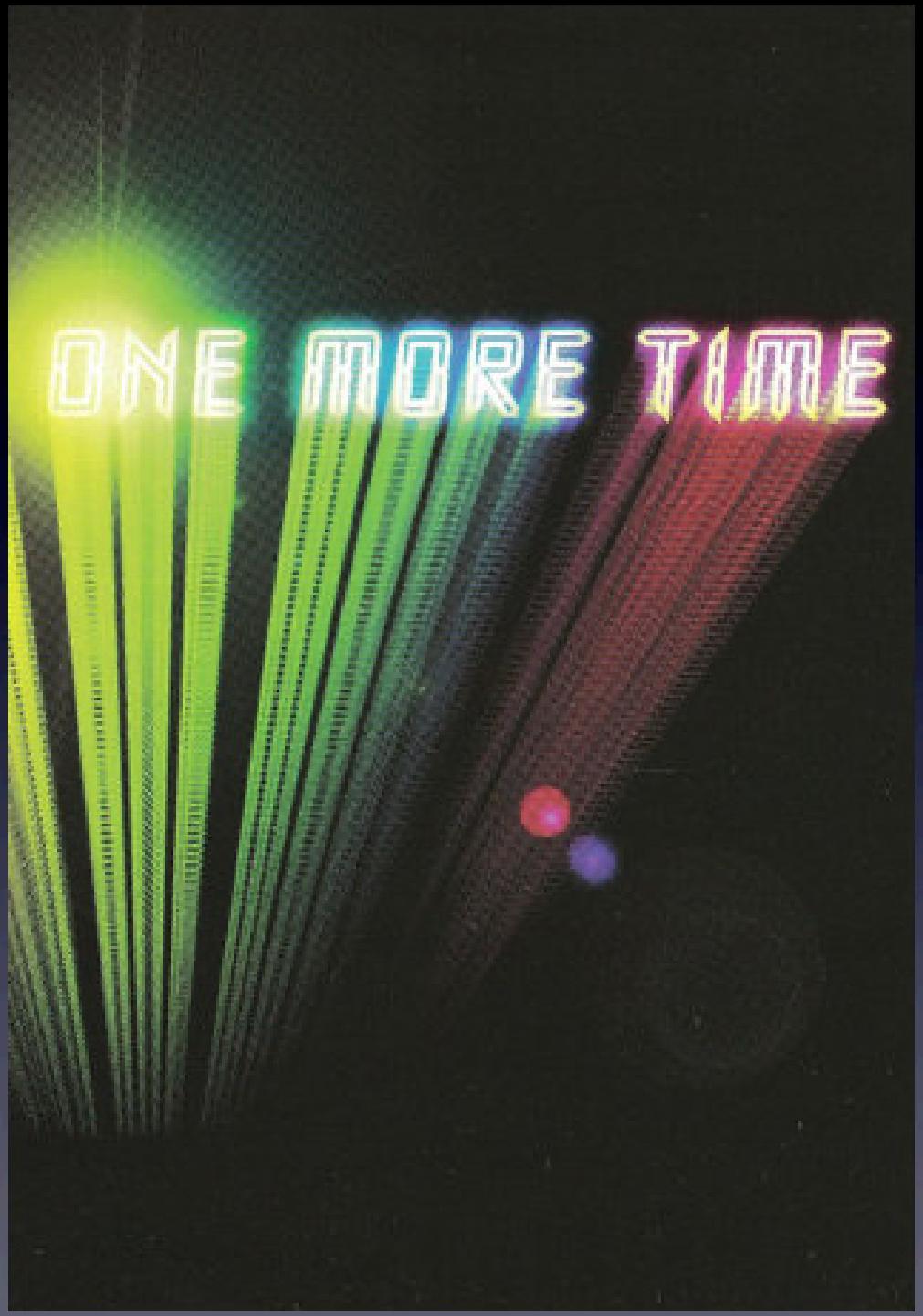
***WAIT !***

# The Rhythm !

and speed  (about 1 second per step)



Lift Tip


ONE MORE TIME

# The Rhythm !

and speed  (about 1 second per step)



# The Rhythm !

and speed  (about 1 second per step)

# Clean the tip



# The Rhythm !

and speed  (about 1 second per step)



## Tip Down

# The Rhythm !

and speed  (about 1 second per step)



Solder In

# The Rhythm !

and speed  (about 1 second per step)



Solder Out

# The Rhythm !

and speed  (about 1 second per step)



***WAIT !***

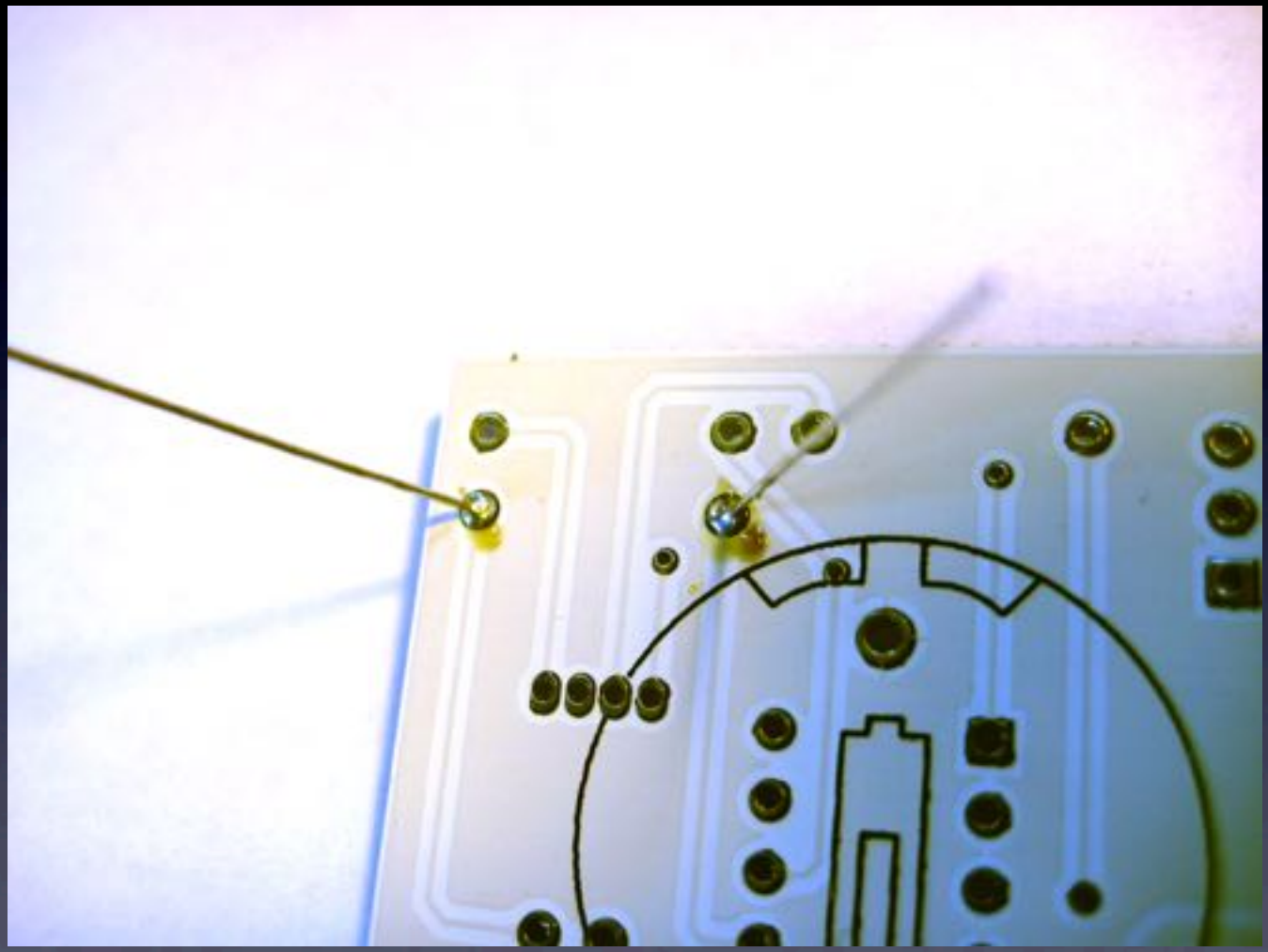
# The Rhythm !

and speed  (about 1 second per step)



Lift Tip

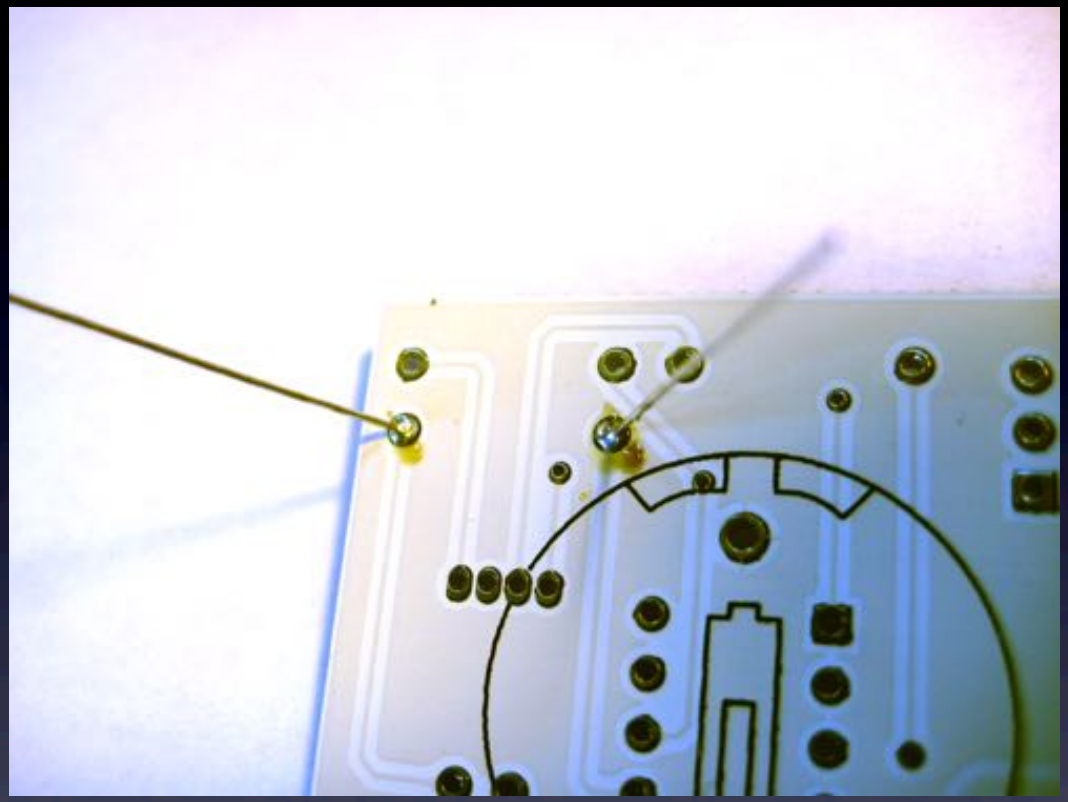
# Solder all of the leads of the part to the board



For this part, there are two leads

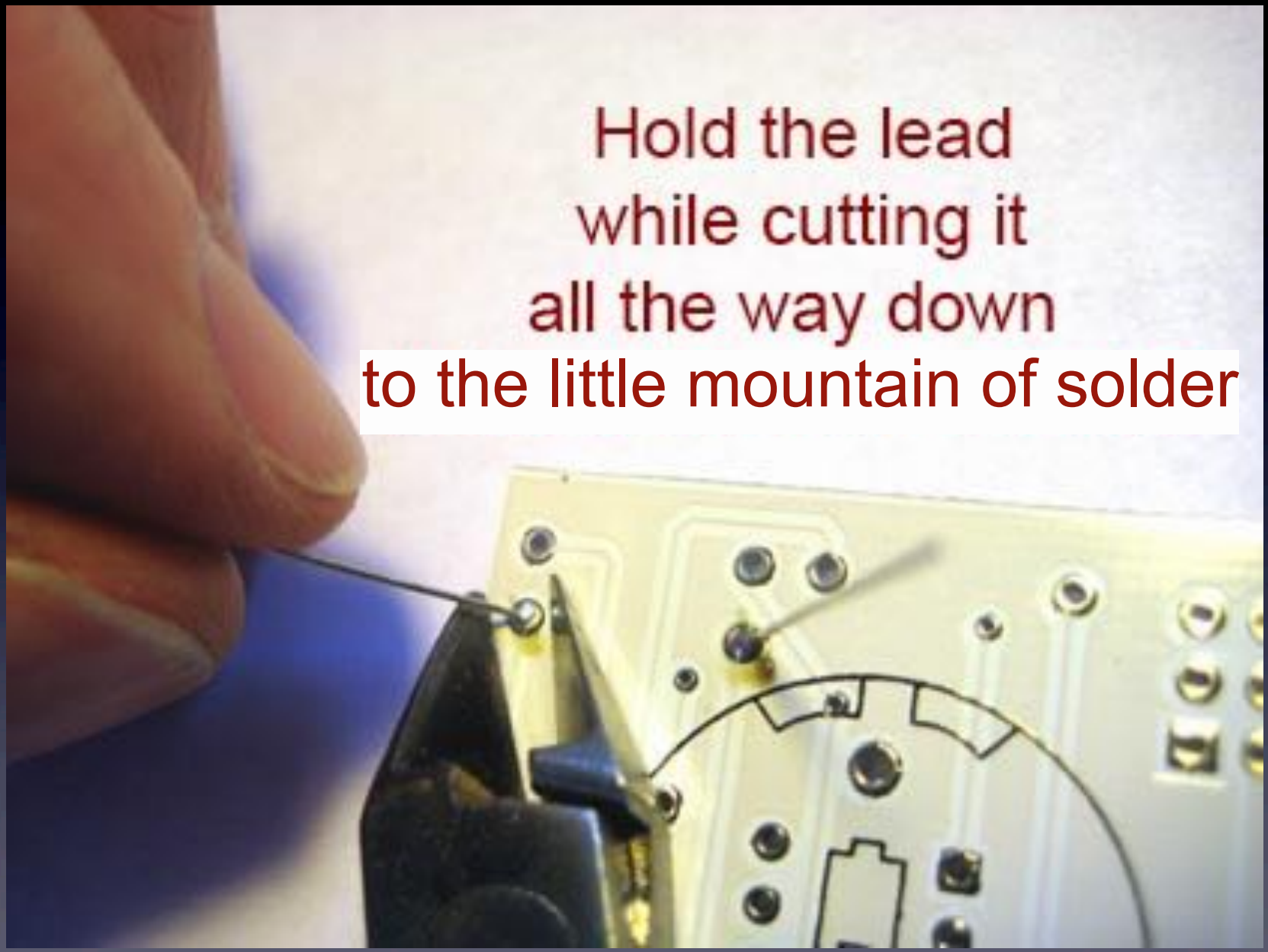
Here you can see two good solder connections

# Two good solder connections



- Little mountains (not flat)
- Pads totally covered in solder
- Can’t see the hole
- No connections to other pads

# Now cut the leads short




Cutting with the tip of the wire cutter gives you more control

# Safety Tip #3:

## Hold or cover the lead !

(or it will fly into your eye!)

*(They like doing that – so please hold or cover the lead when you cut.)*




No wires sticking out



# A closer look at good solder connections

Notice that:

- Each connection is a small mountain (not flat)
- You cannot see any pad (they're totally covered with solder)
- You cannot see the holes (they're totally covered with solder)
- No connections to other pads

# One part at a time

Till all the parts are soldered

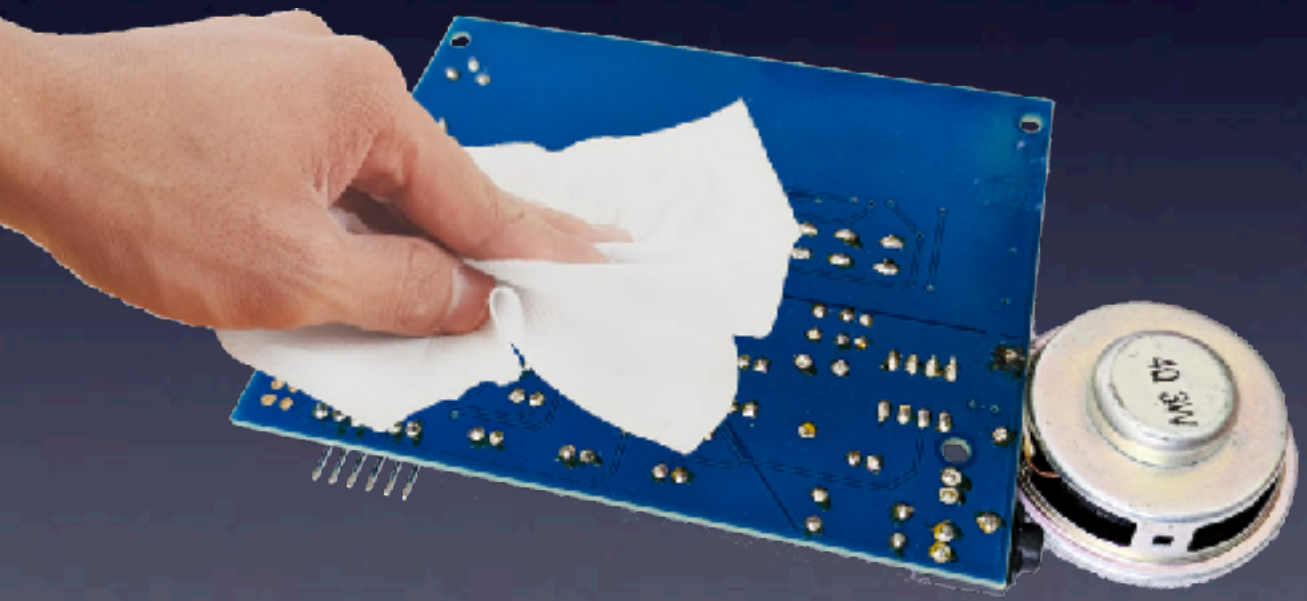



And it will look like this when you're done.

# If you used any *flux paste for re-working problems*



The bottom of the PCB will be sticky from the flux






***You can clean it with a cloth wet with Isopropyl Alcohol***

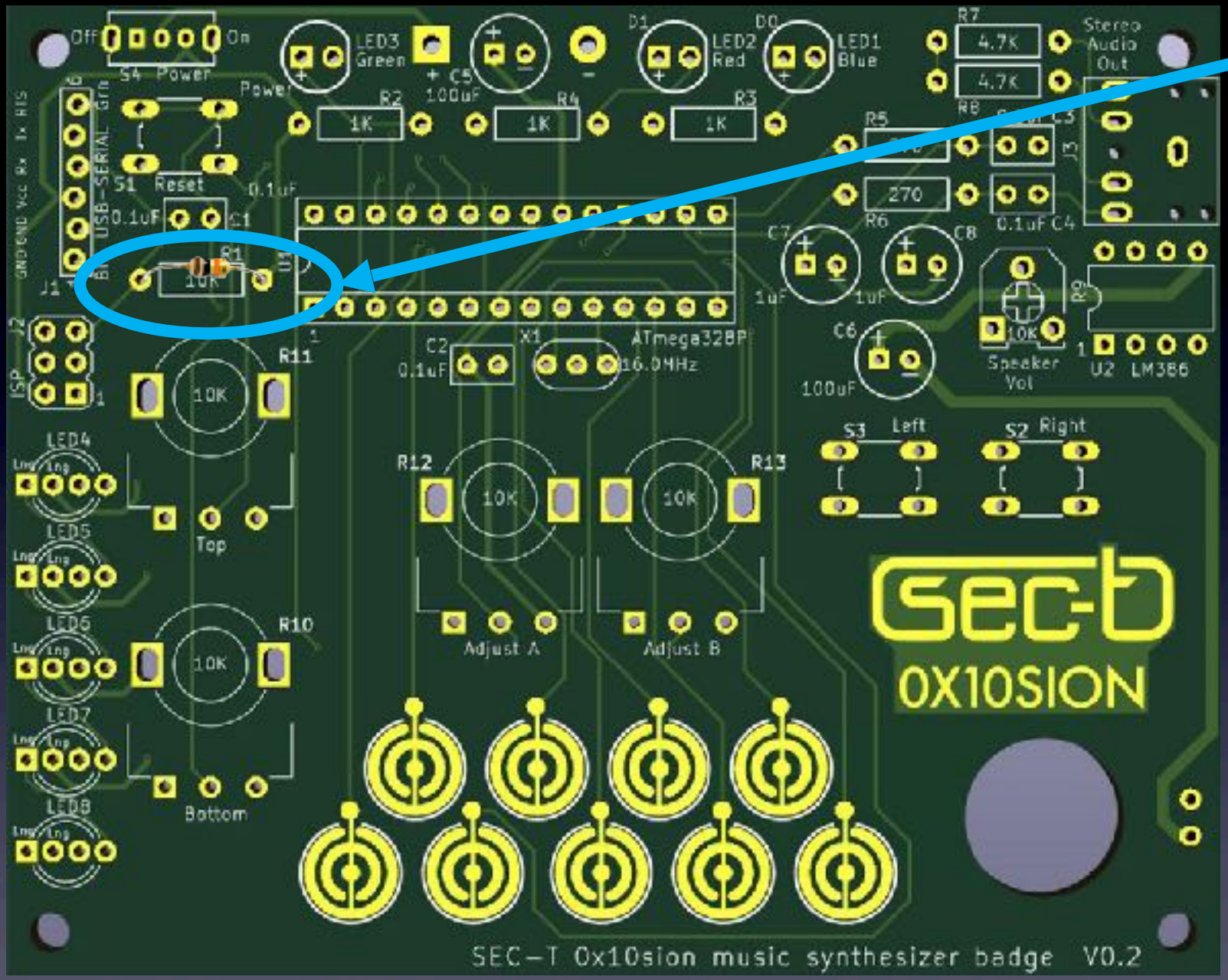
Then put in the batteries,

Turn it on,

And it works!

(Or you start debugging.)

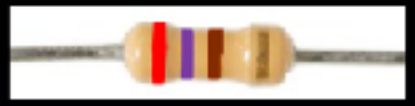
# Let’s start!



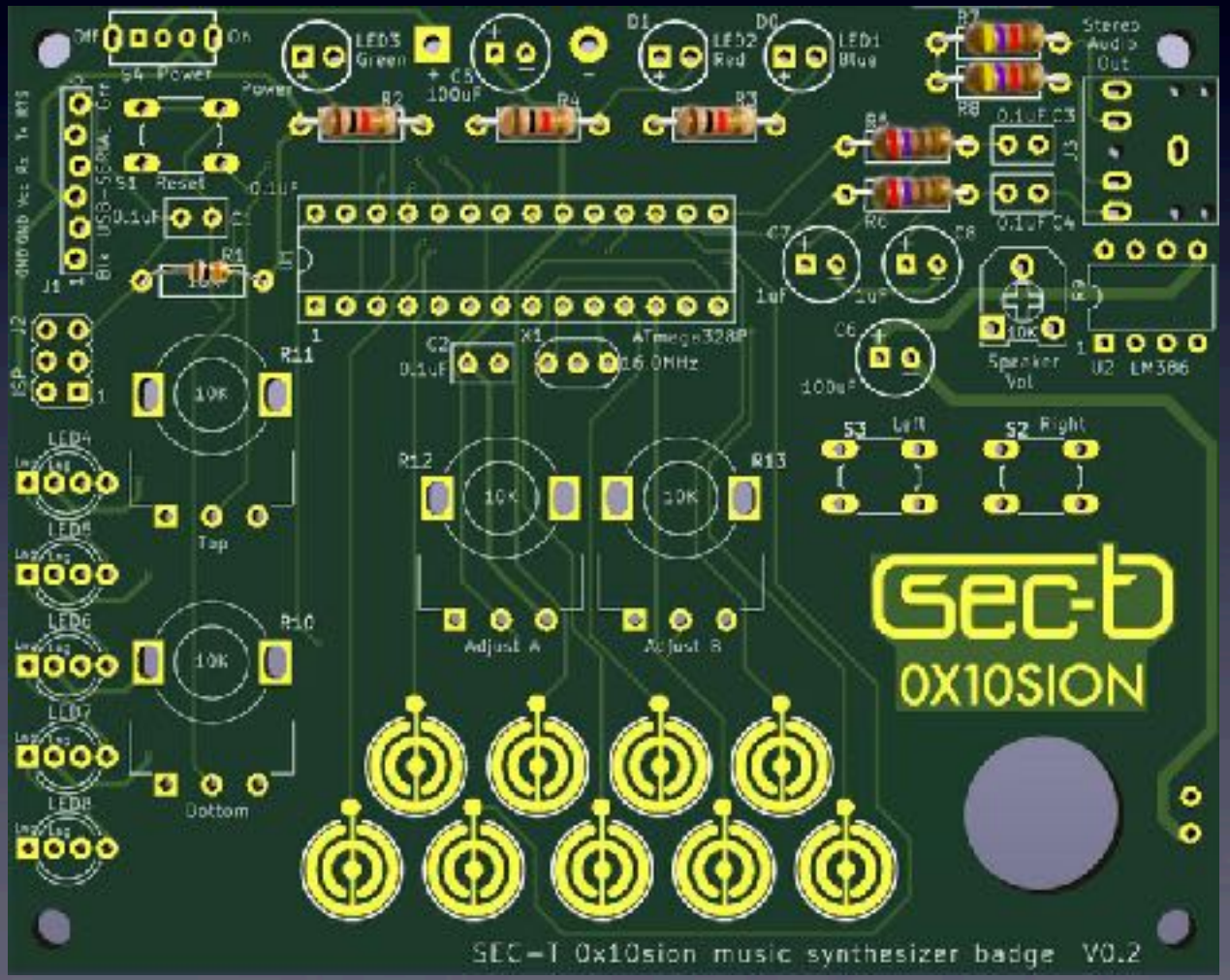

If you haven't done so already, solder R1: brown, black, orange

R1:  10K: (the small one) Brown, Black, Orange

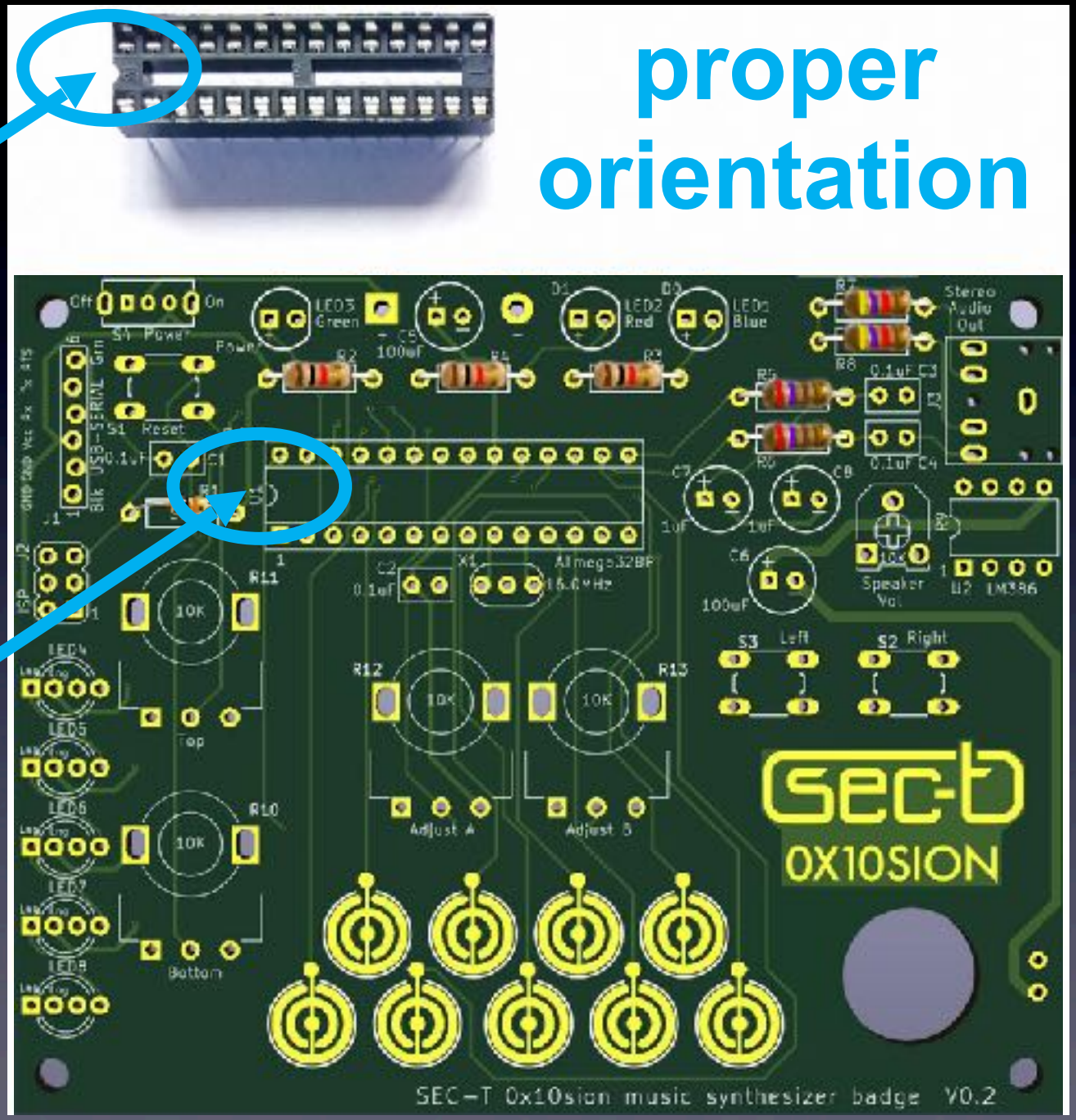
R2, R3, R4:  1K: Brown, Black, Red

R5, R6:  270: Red, Violet, Brown

R7, R8:  4.7K: Yellow, Violet, Red

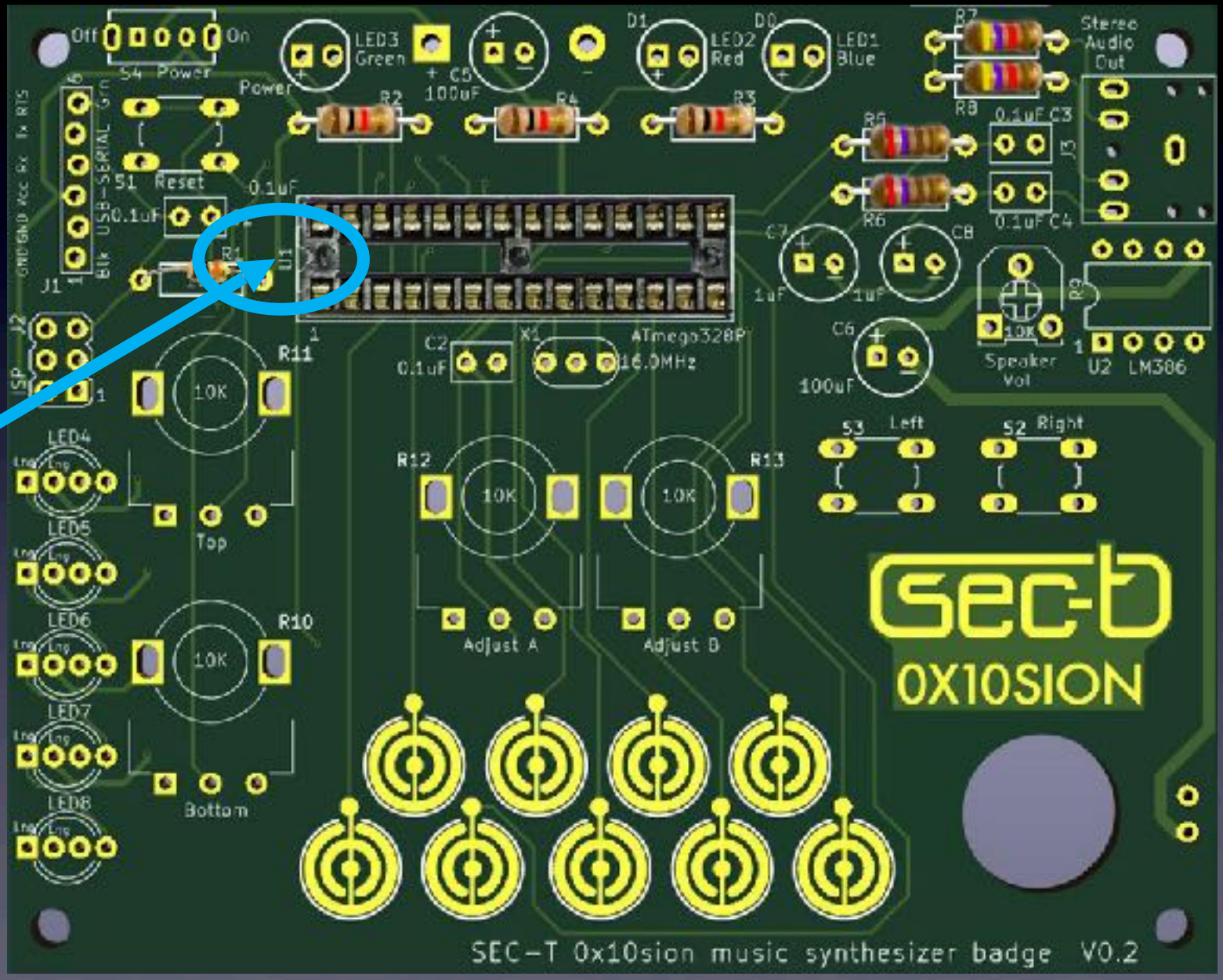


# U1: microcontroller socket

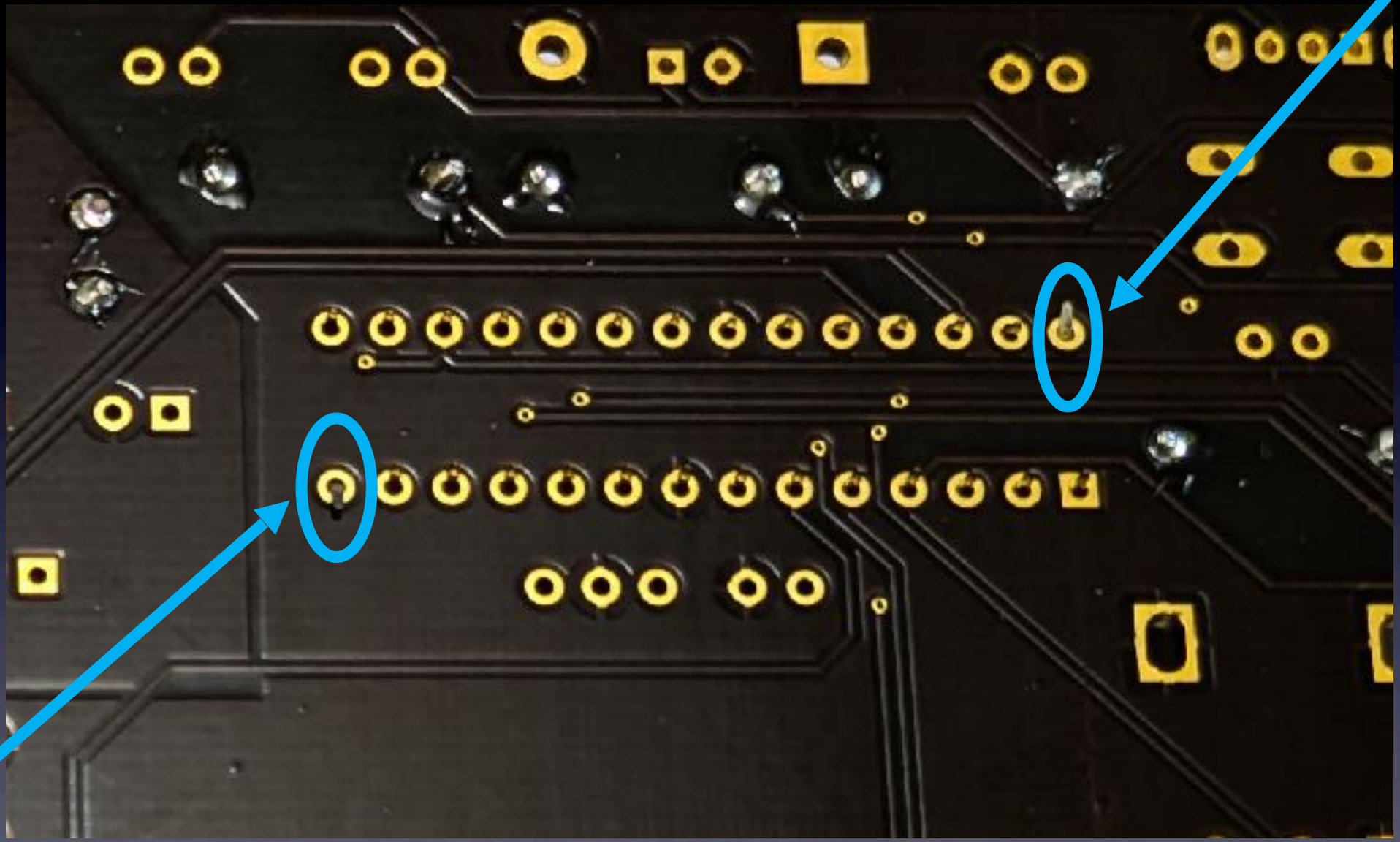


# U1: microcontroller socket

## inserted correctly

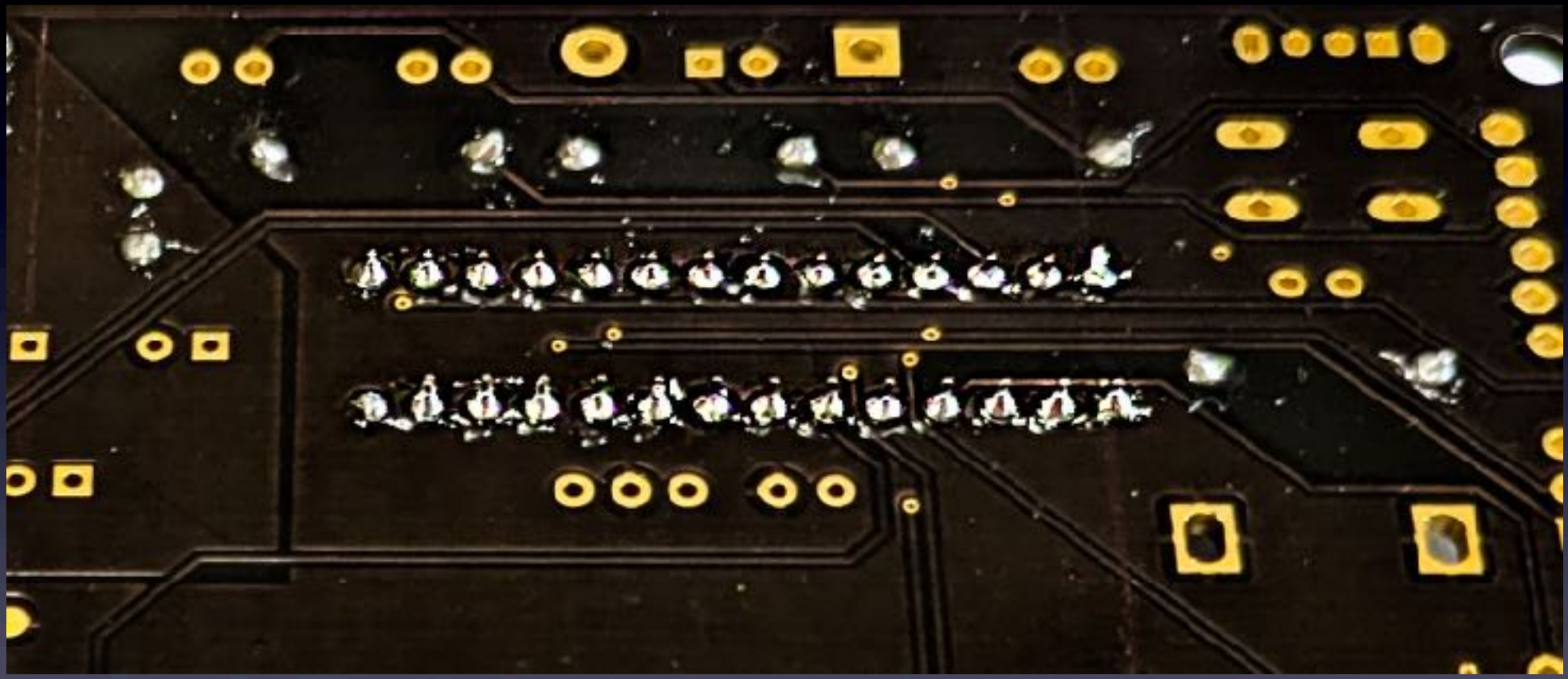


# U1: microcontroller socket



bend pins down on two corners,
and solder all 28 leads to the board

# U1:  microcontroller socket

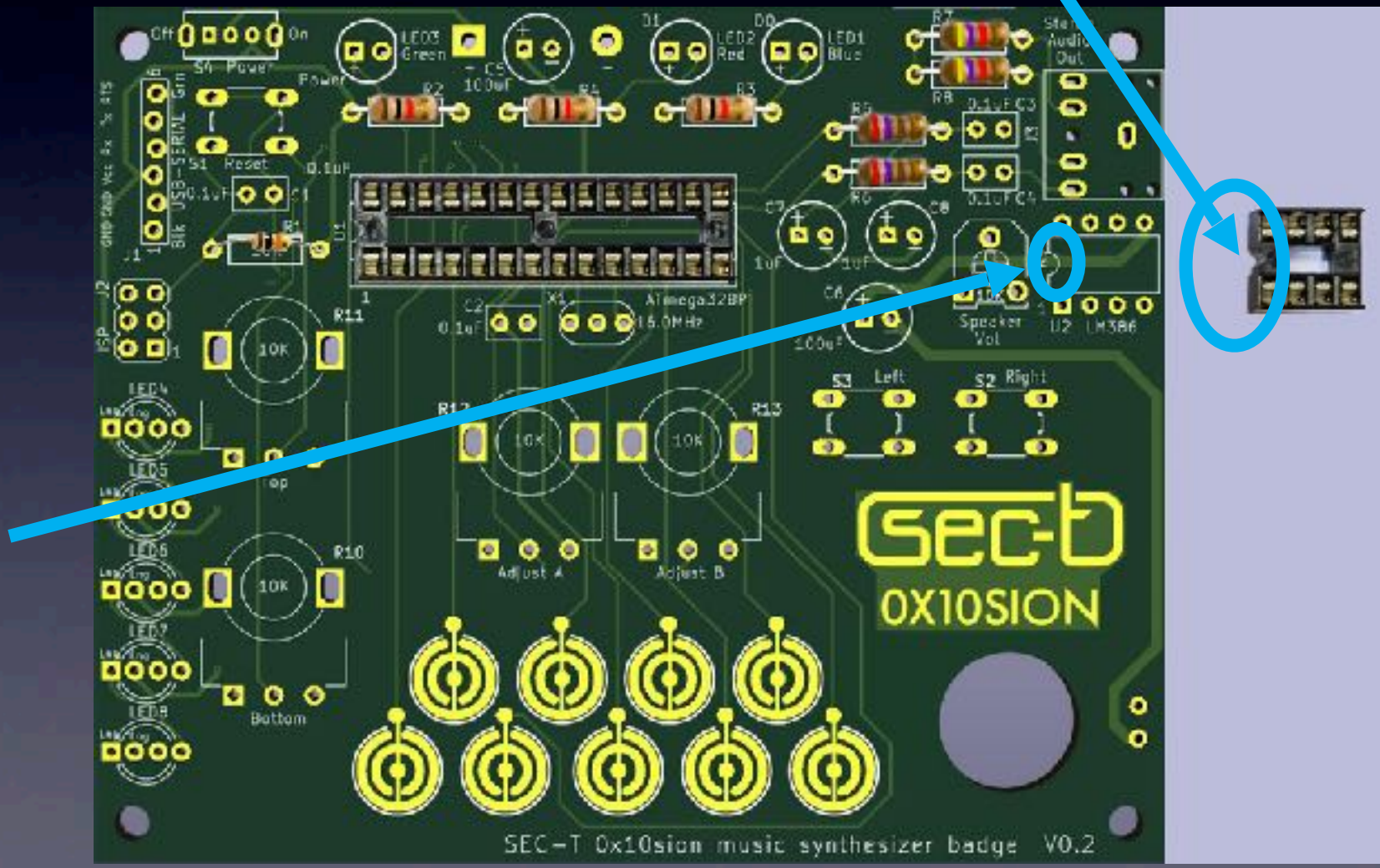


All 28 leads soldered to the board:

→ Notice that each pad is totally covered with solder. ←

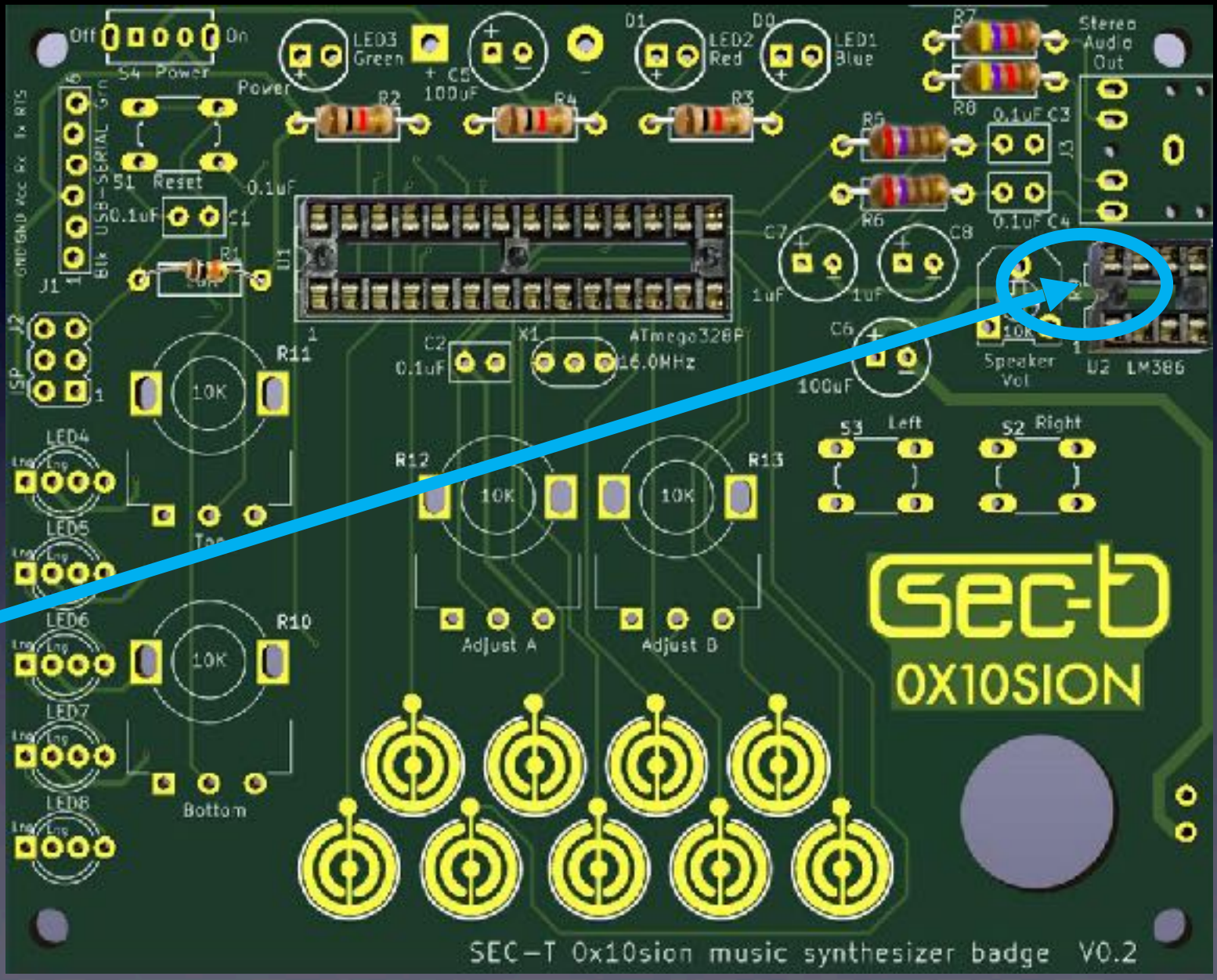
# U2: amplifier chip socket

proper orientation

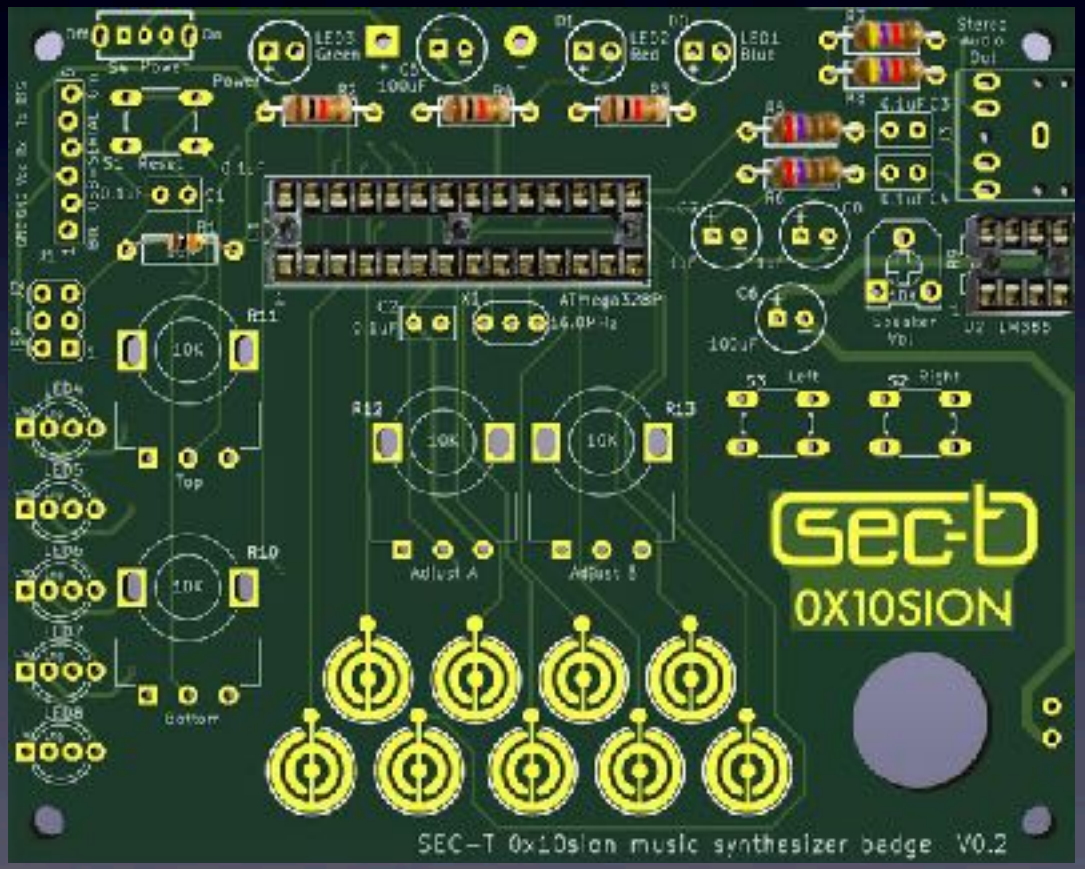


# U2: amplifier chip socket

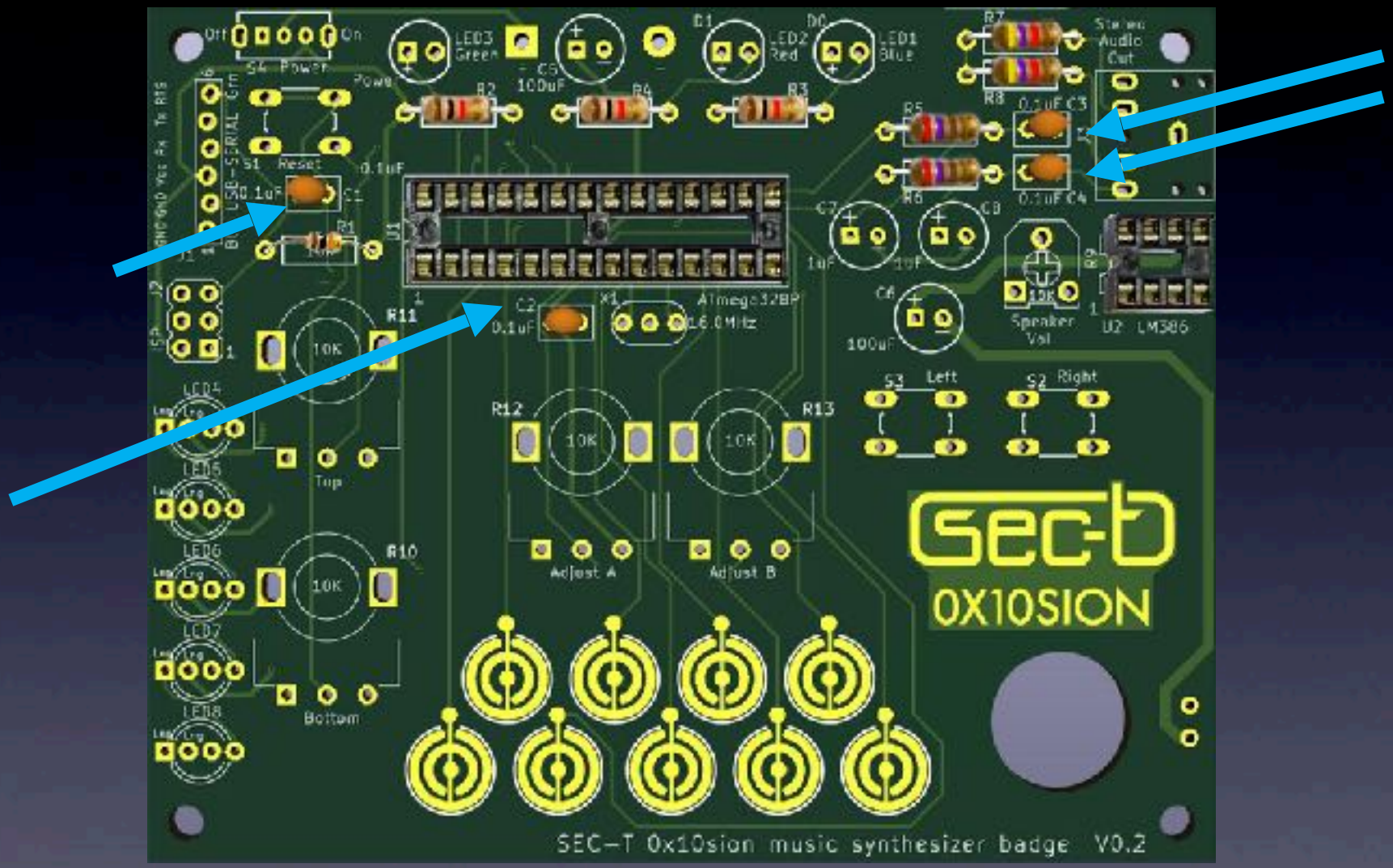
## inserted correctly



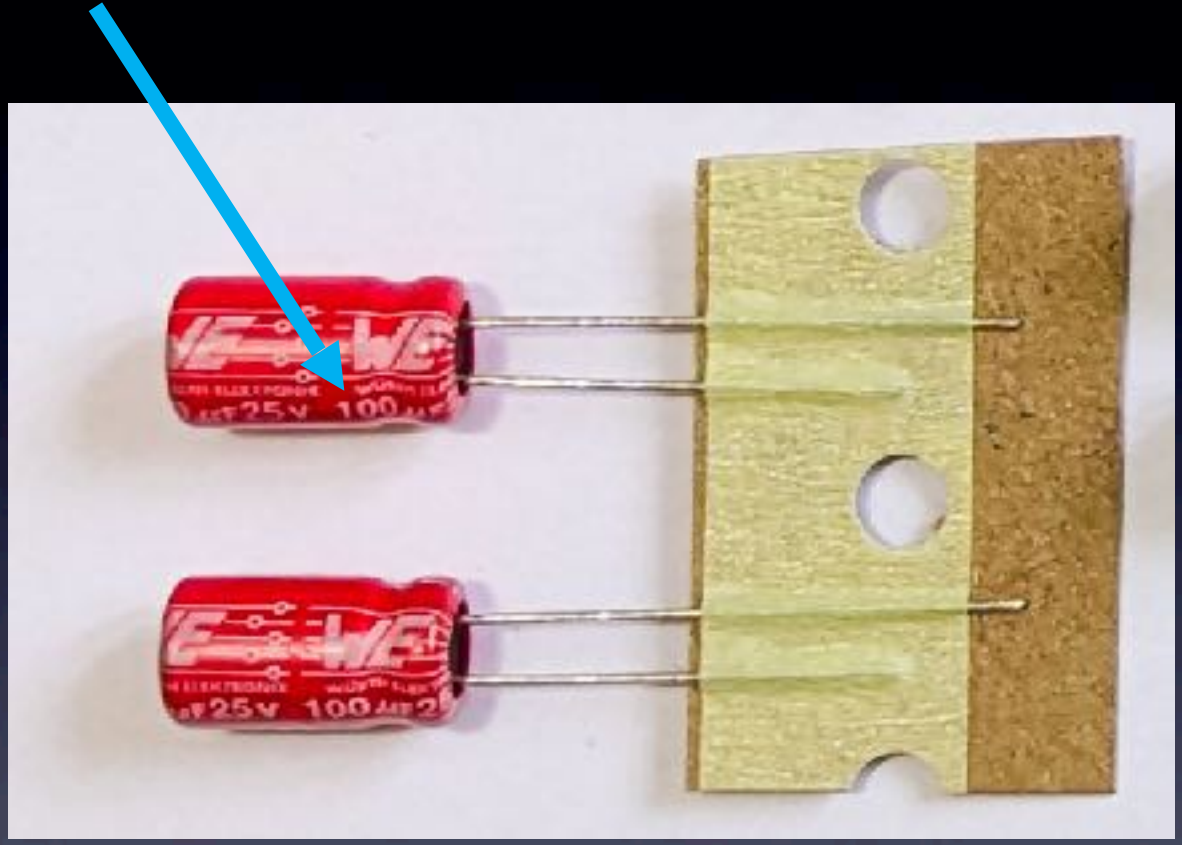
# U2: amplifier chip socket



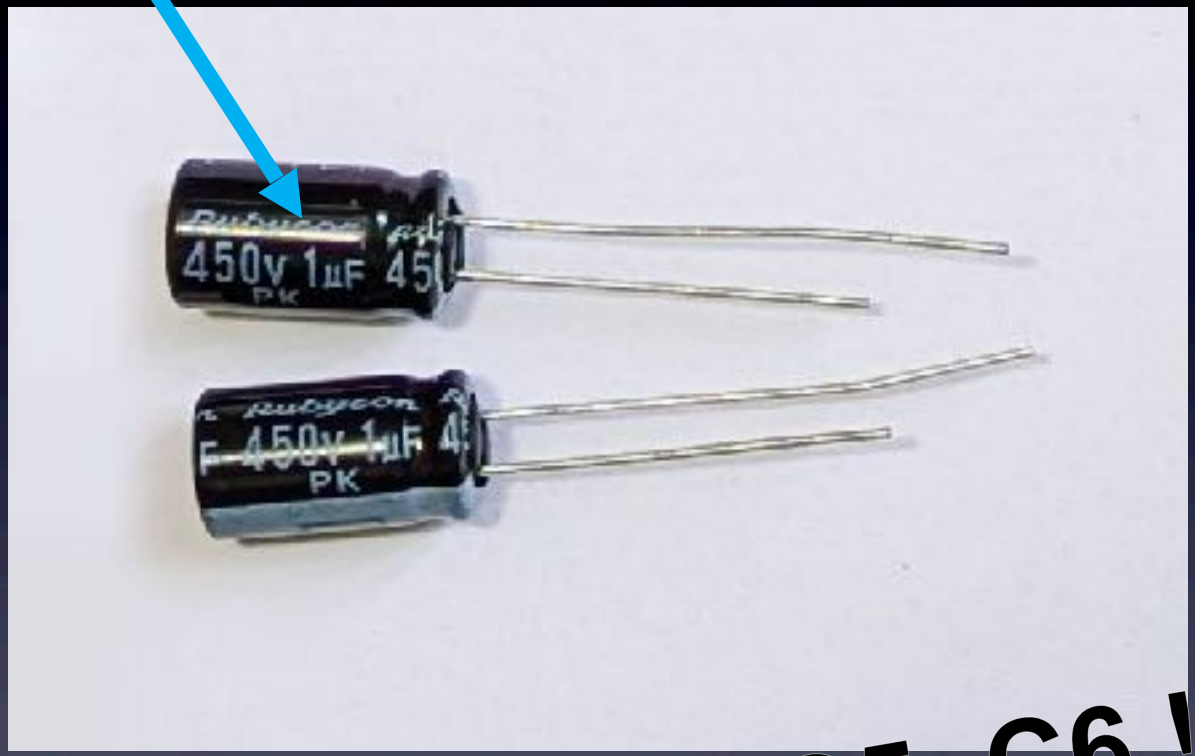

bend pins down on two corners,
and solder all 8 leads to the board


SEC-T 0x10sion music synthesizer badge V0.2
0X10SION
Atmega328P
16.0MHz
Speaker Vol
Adjust A
Adjust B
Top
Bottom
Left
Right
Reset
Power
Stereo Audio Out
LED1 Blue
LED2 Red
LED3 Green
U2 LM386

# C1, C2, C3, C4

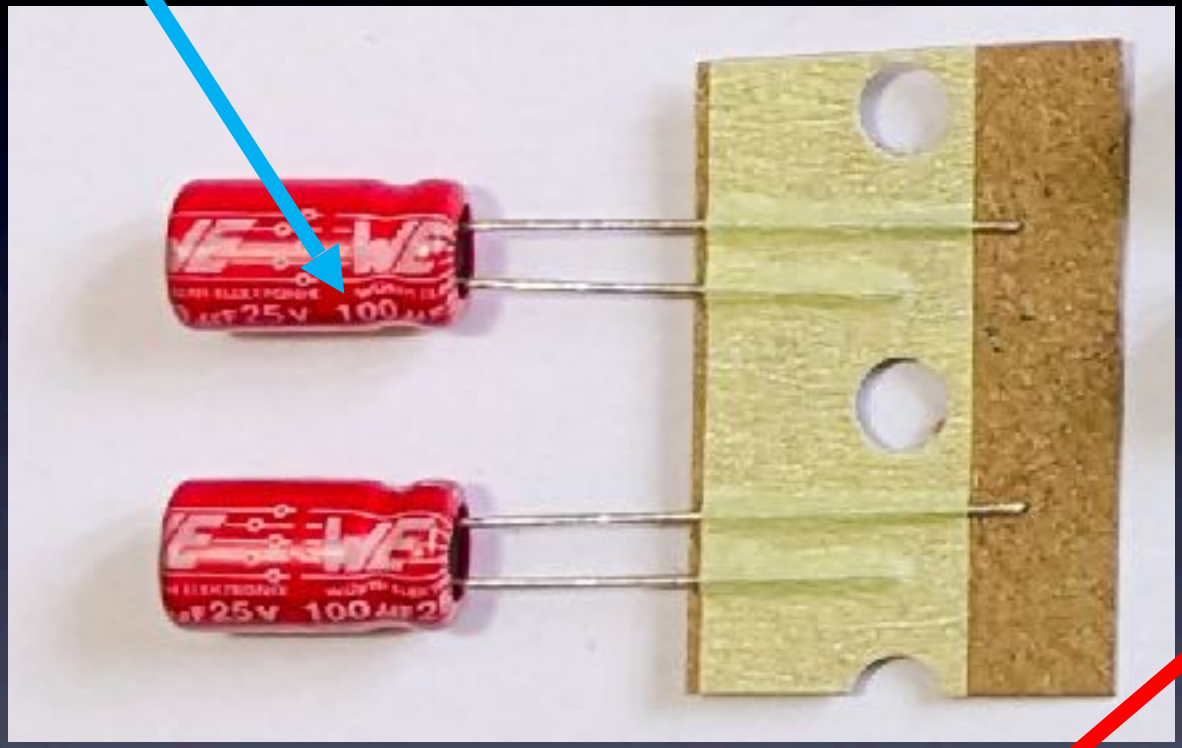



# C5, C6:  100uF

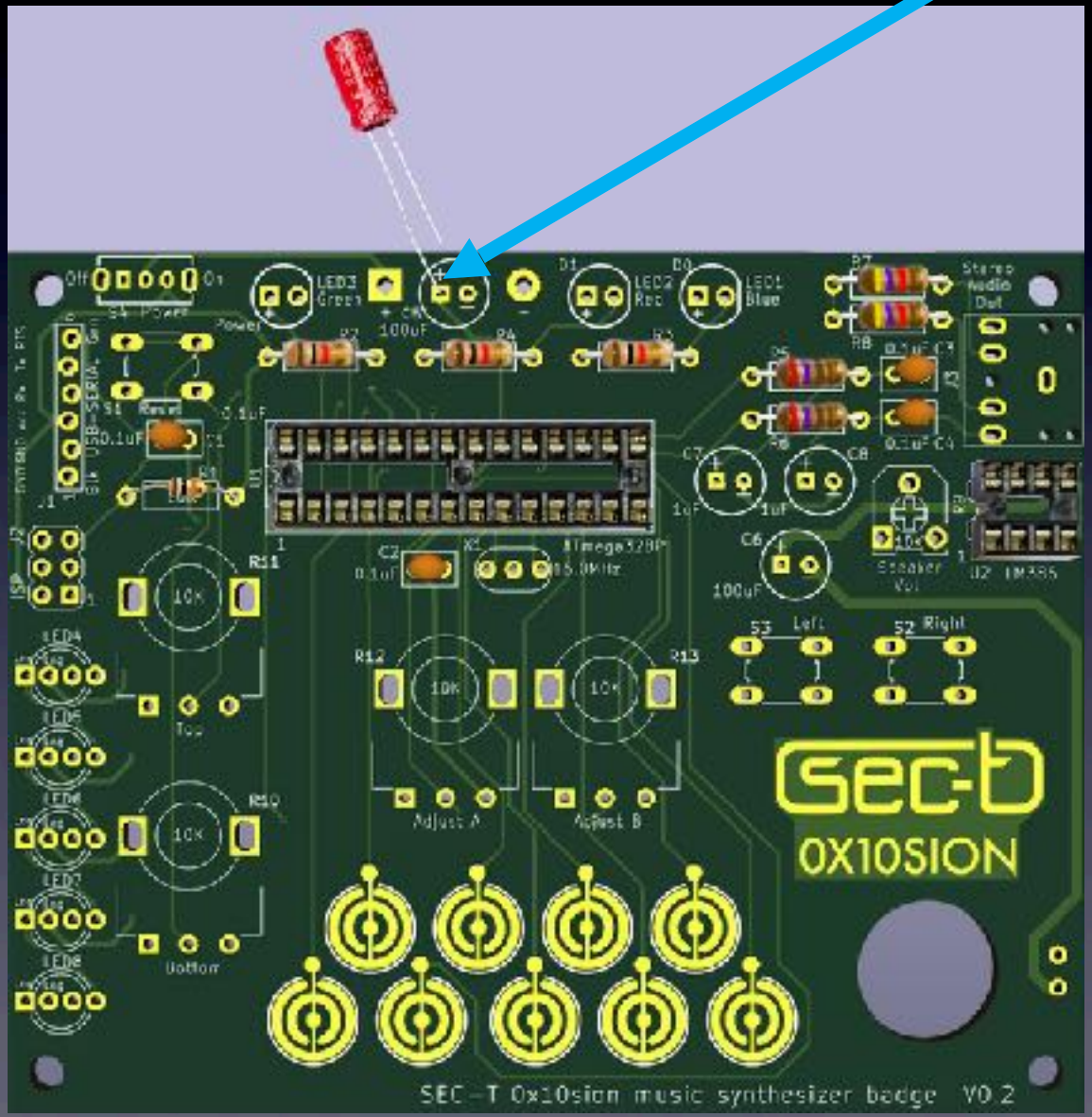



**Different than C5, C6 !**

**C7, C8: 1uF**

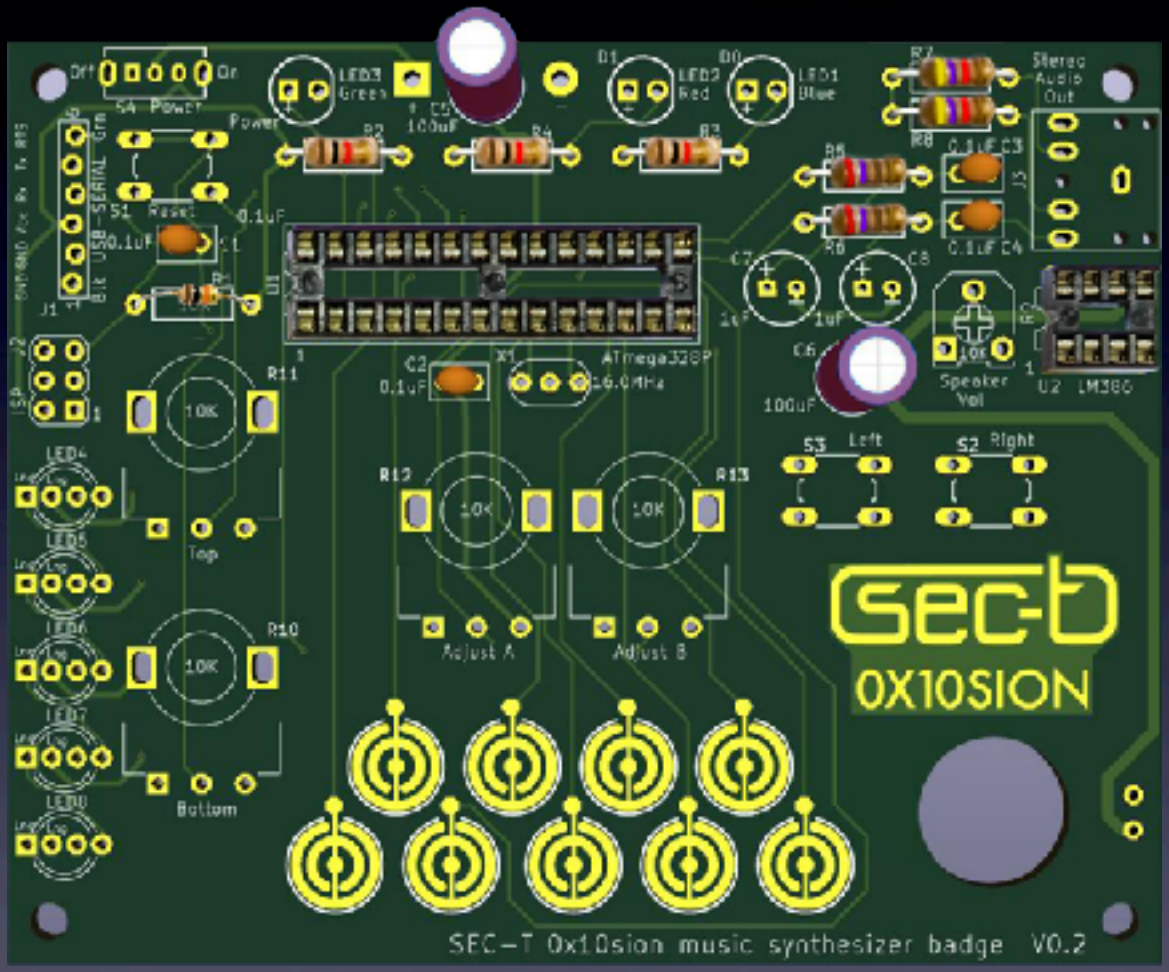



C5, C6: 100uF

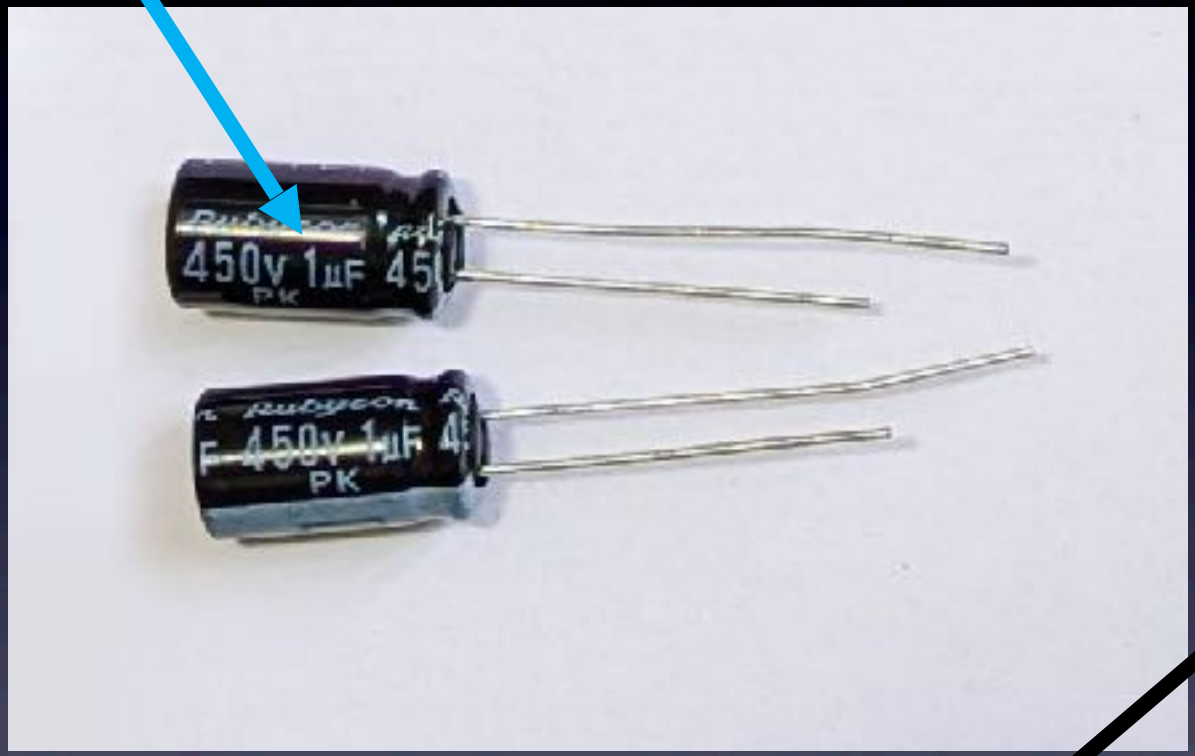



# C5, C6: Long Lead “+”

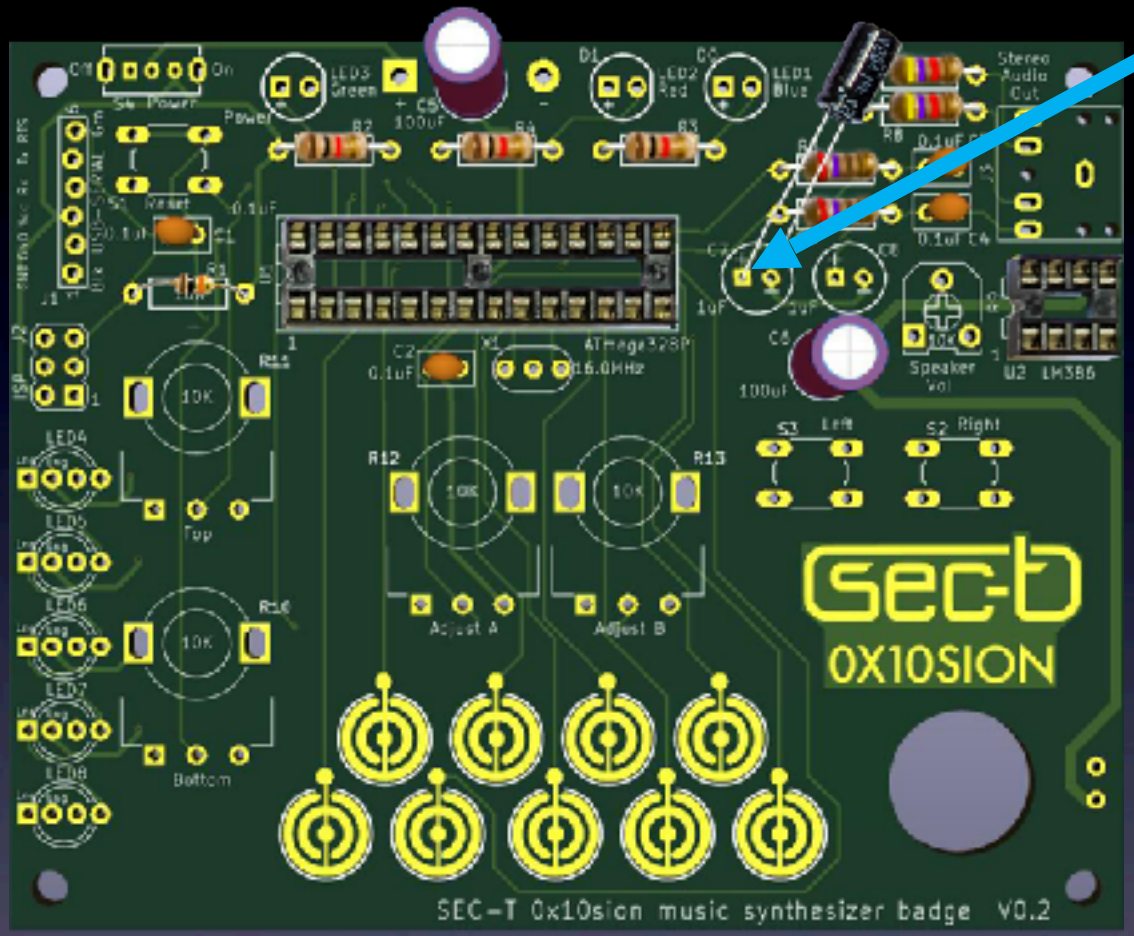
**Use 100uF !!**



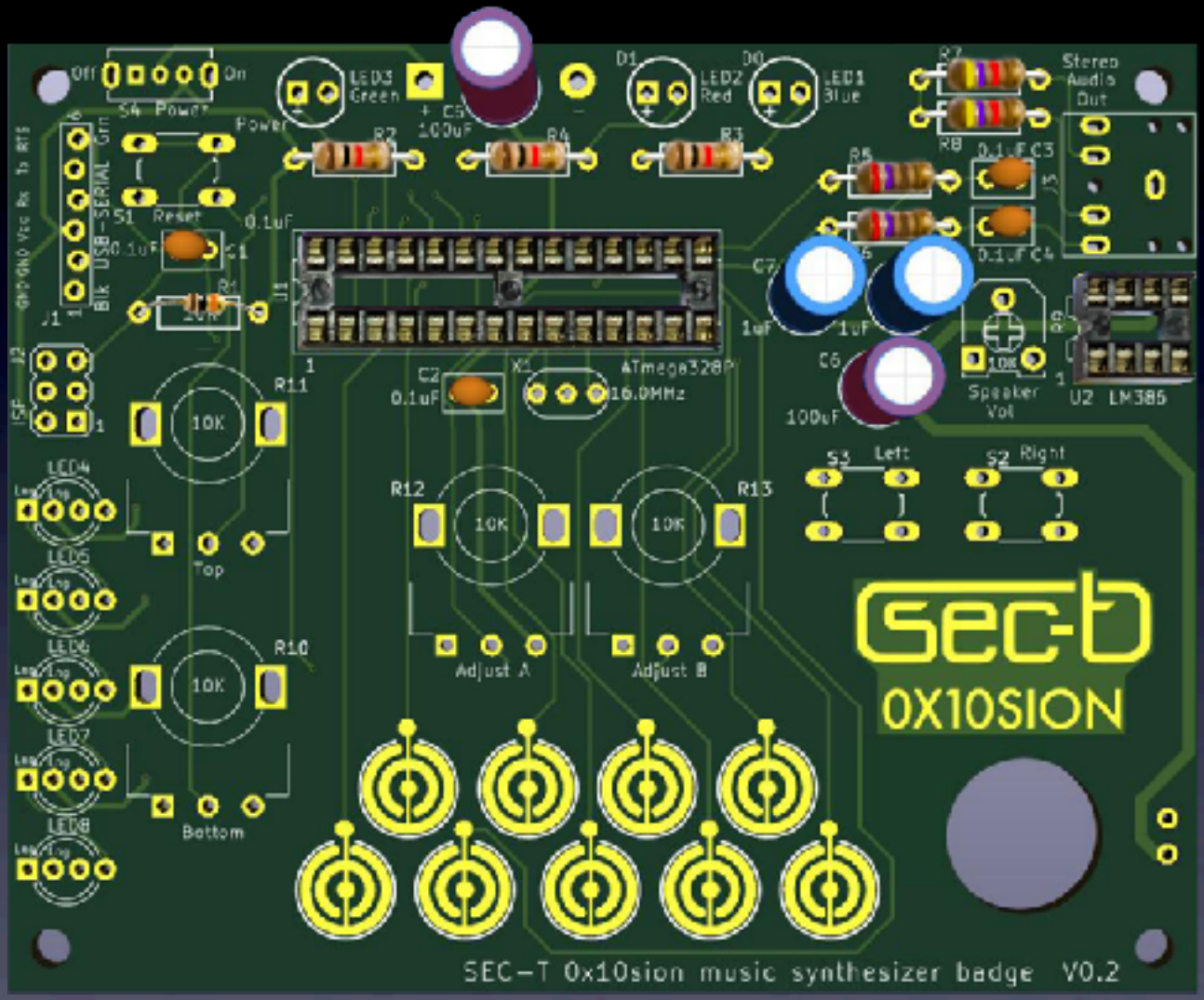

# C5, C6: 100uF – soldered to board




**C7, C8: 1uF**

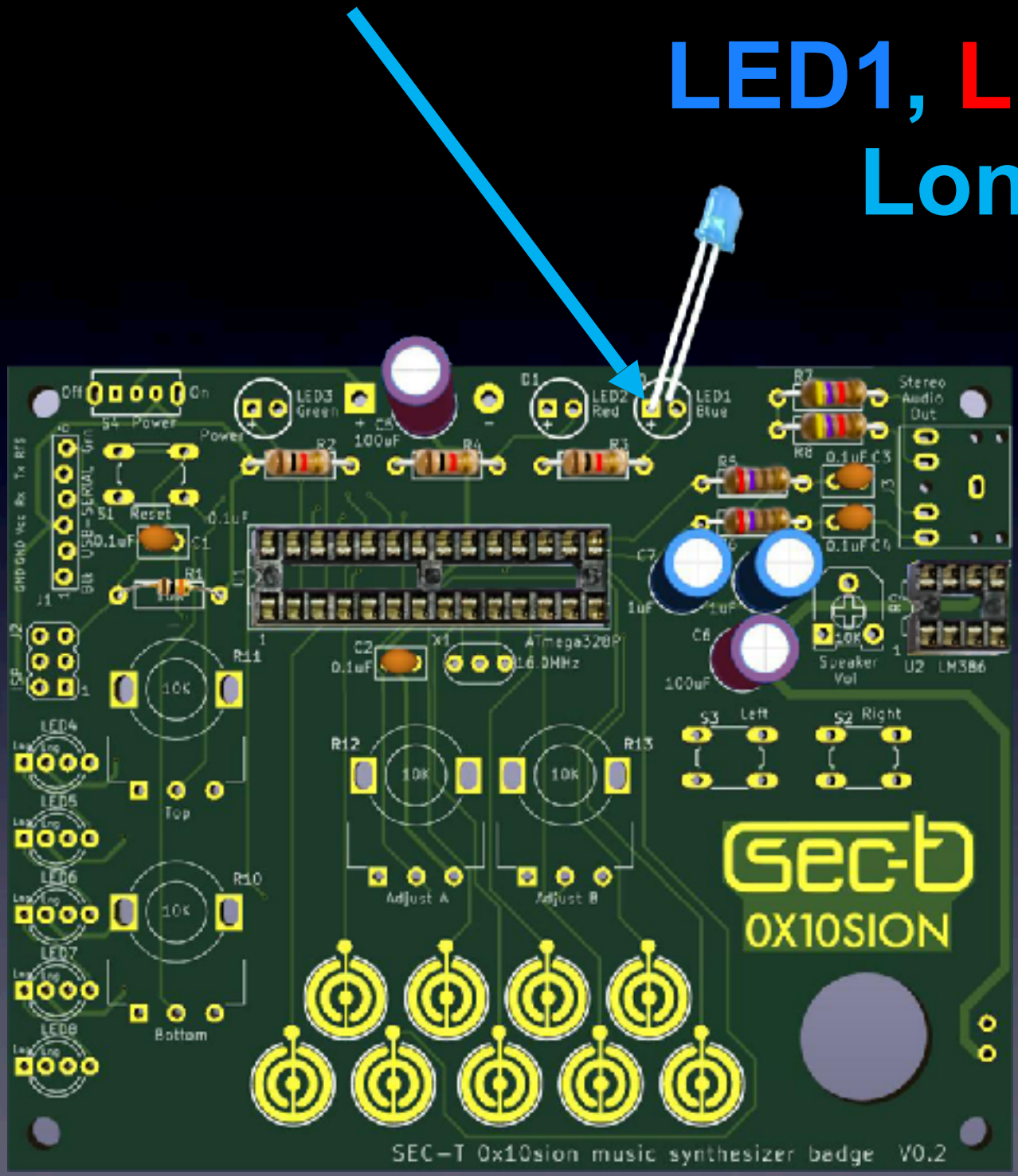


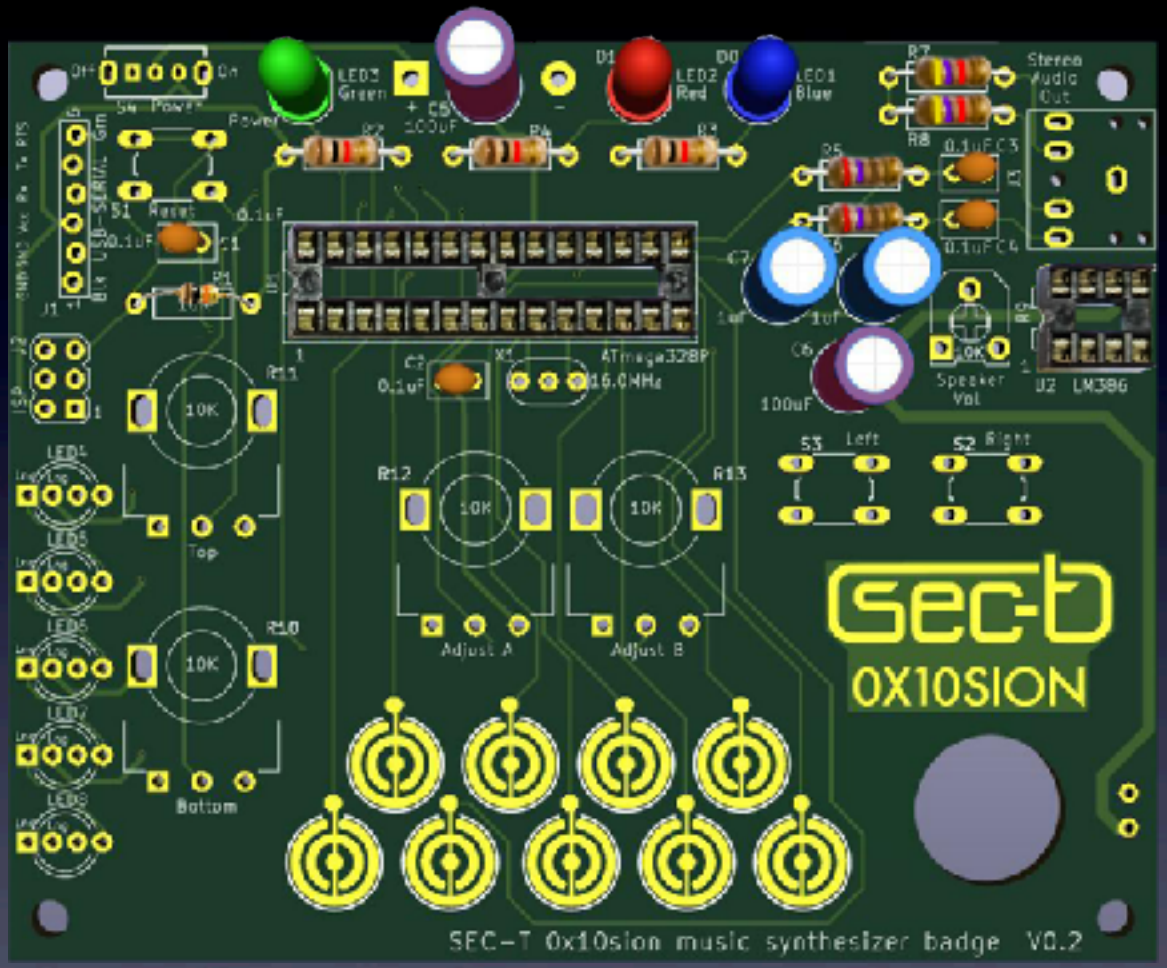

# C7, C8: Long Lead “+”




# C7, C8: 1uF – soldered to board

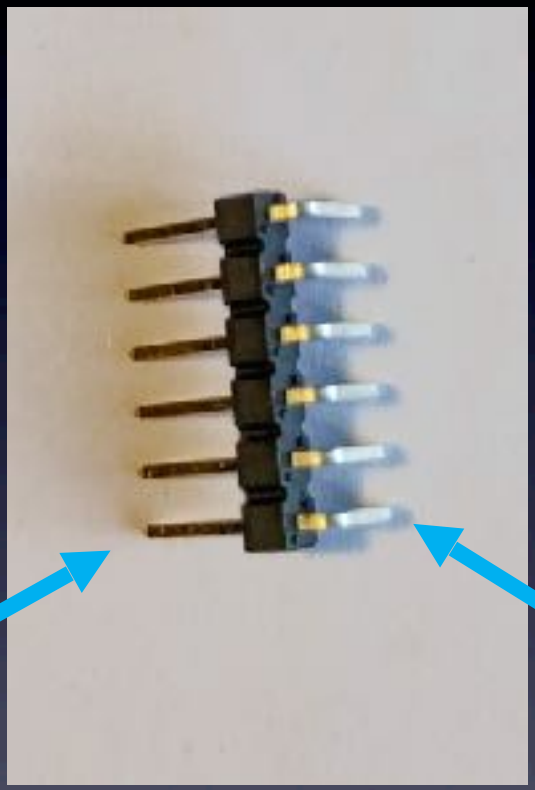
# LED1, LED2, LED3: Long Lead “+”






LED3, LED2, LED1

Green, Red, Blue – soldered to board



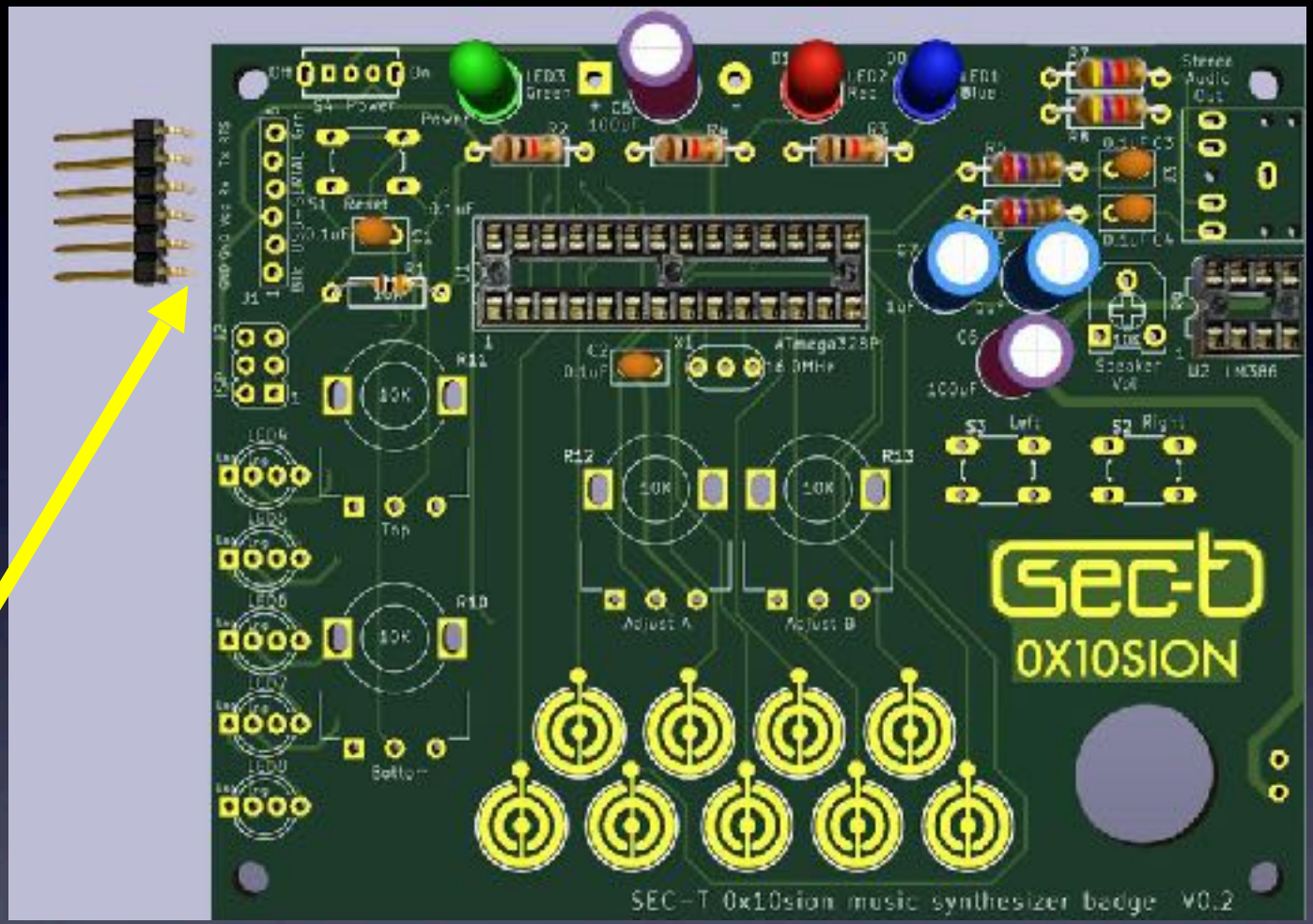
long leads

short leads

**J1**

# Short leads into board

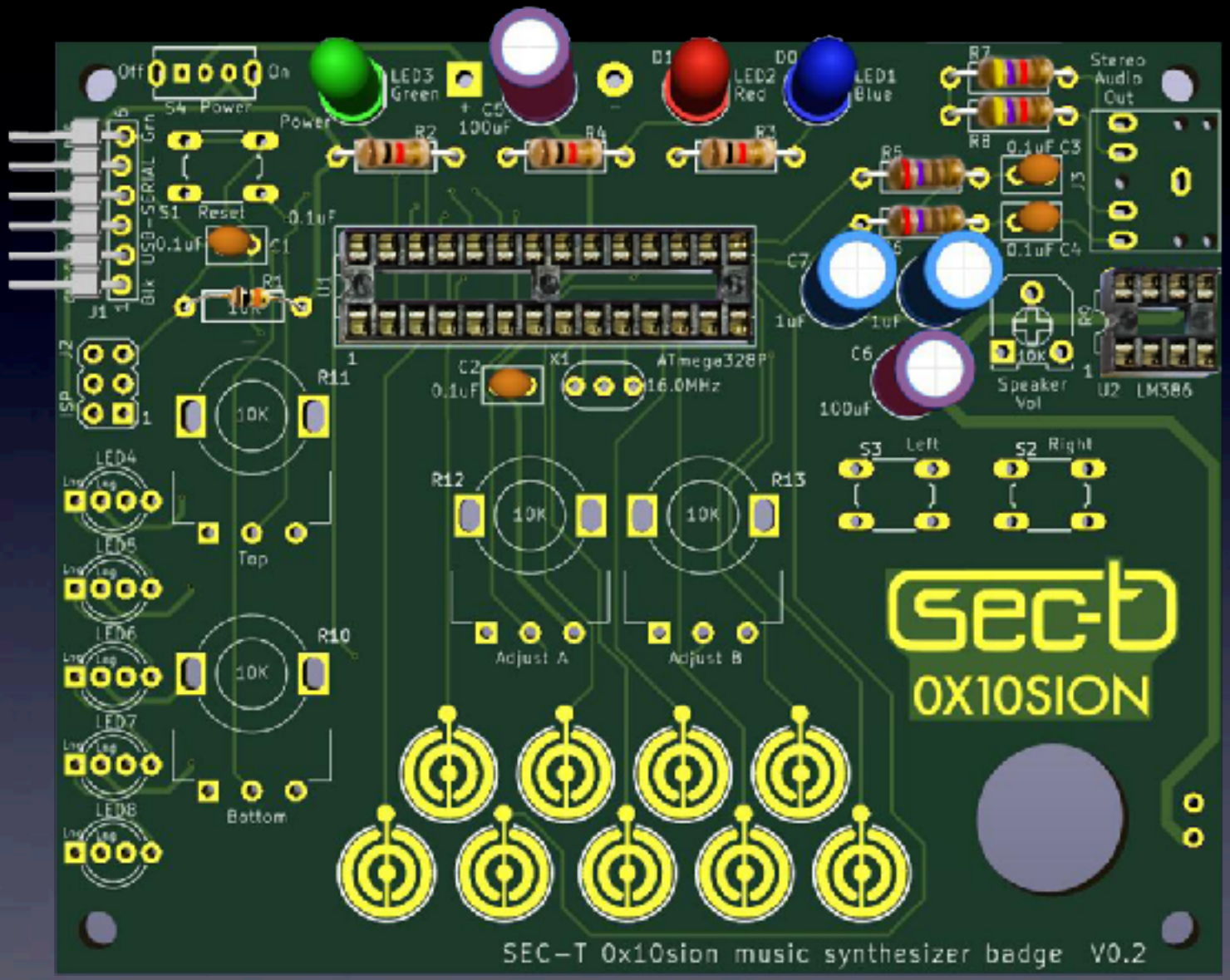
J1

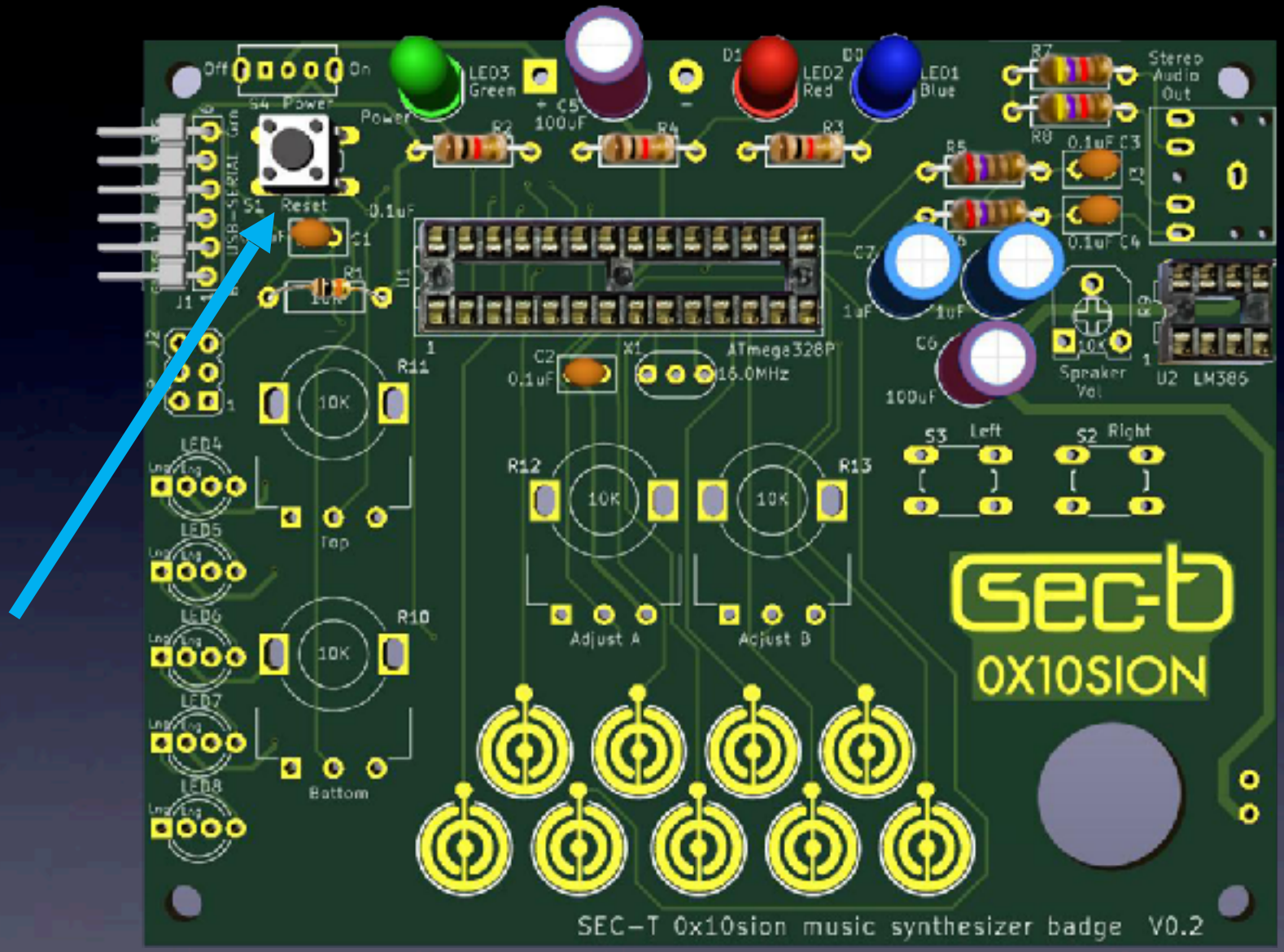



IMPORTANT:
short leads
go into the board

→ long leads sticking out from board

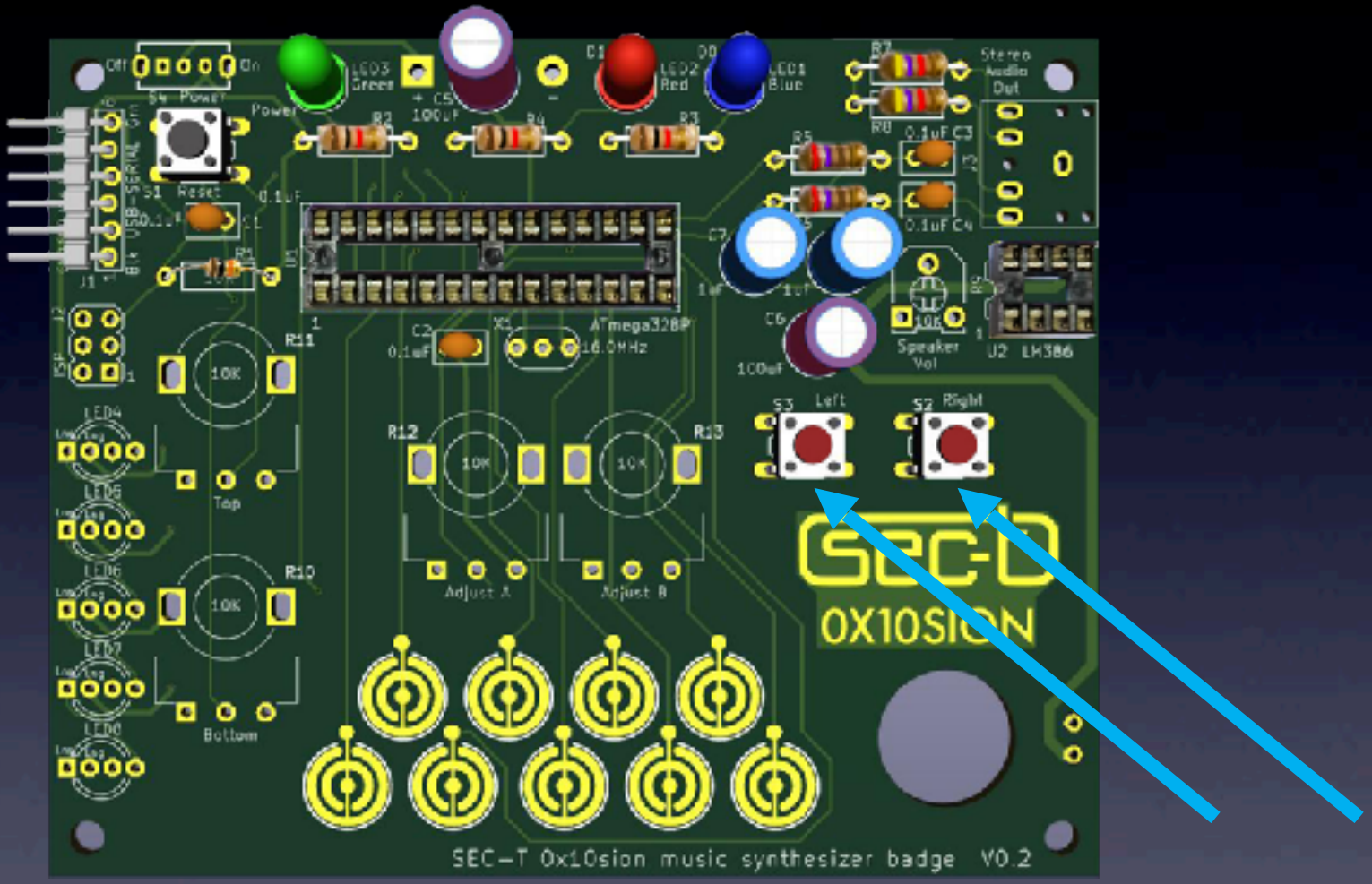
# J1

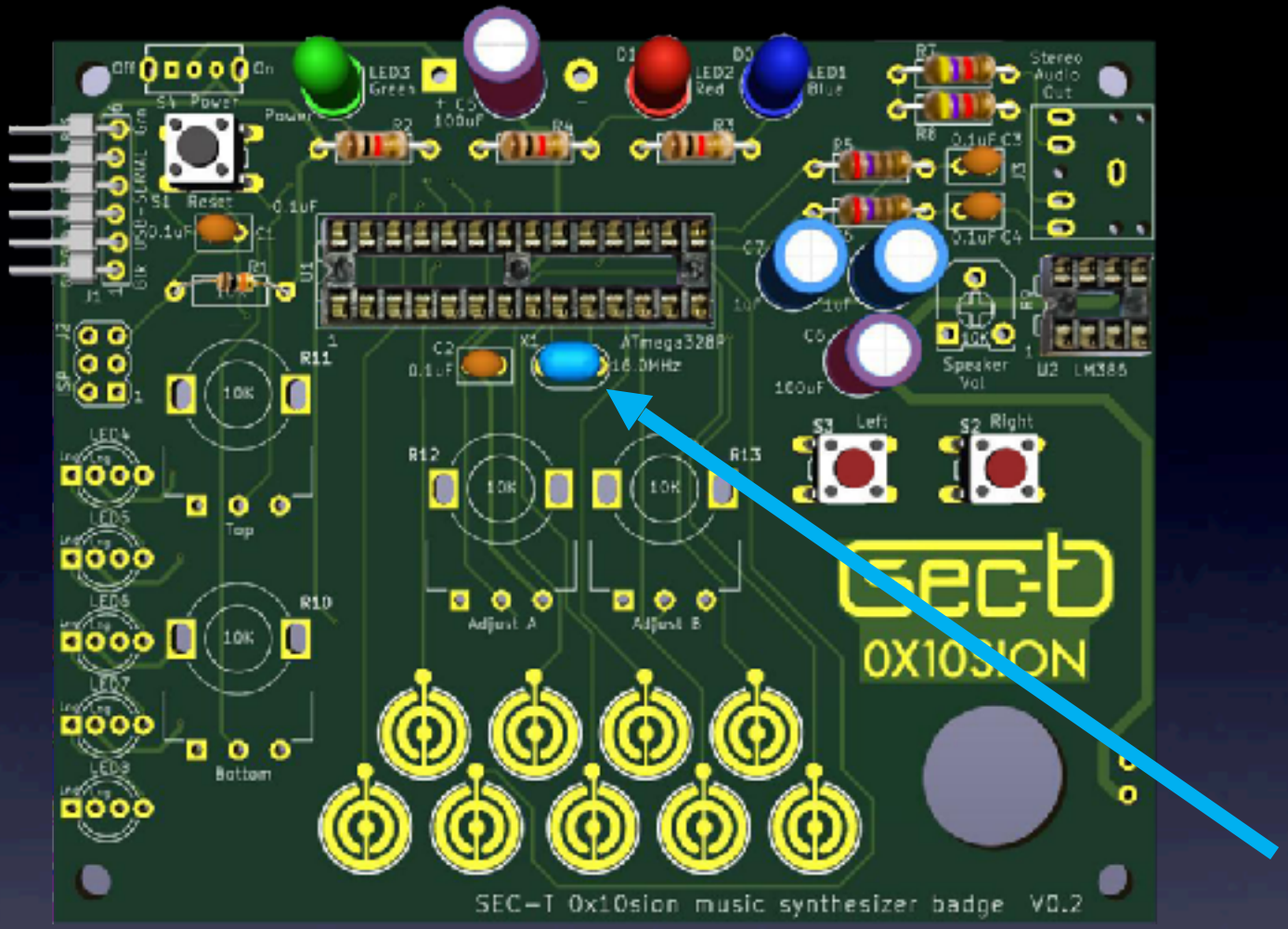

Off
On
S4 Power
Power
LED3
Green
LED2
Red
LED1
Blue
Stereo
Audio
Out
S1 Reset
0.1uF
J1
SER
Blk
Grn
J2
SP
R11
10K
Top
Bottom
R10
R12
R13
Adjust A
Adjust B
C2
X1
16.0MHz
ATmega328P
U1
C7
C6
1uF
100uF
Speaker
Vol
U2 LM386
S3 Left
S2 Right
J3
R7
R8
R5
R6
C3
C4
LED4
LED5
LED6
LED7
LED8
SEC-T
0X10SION
SEC-T 0x10sion music synthesizer badge V0.2




# S1: black Reset button

# S2, S3: Red buttons

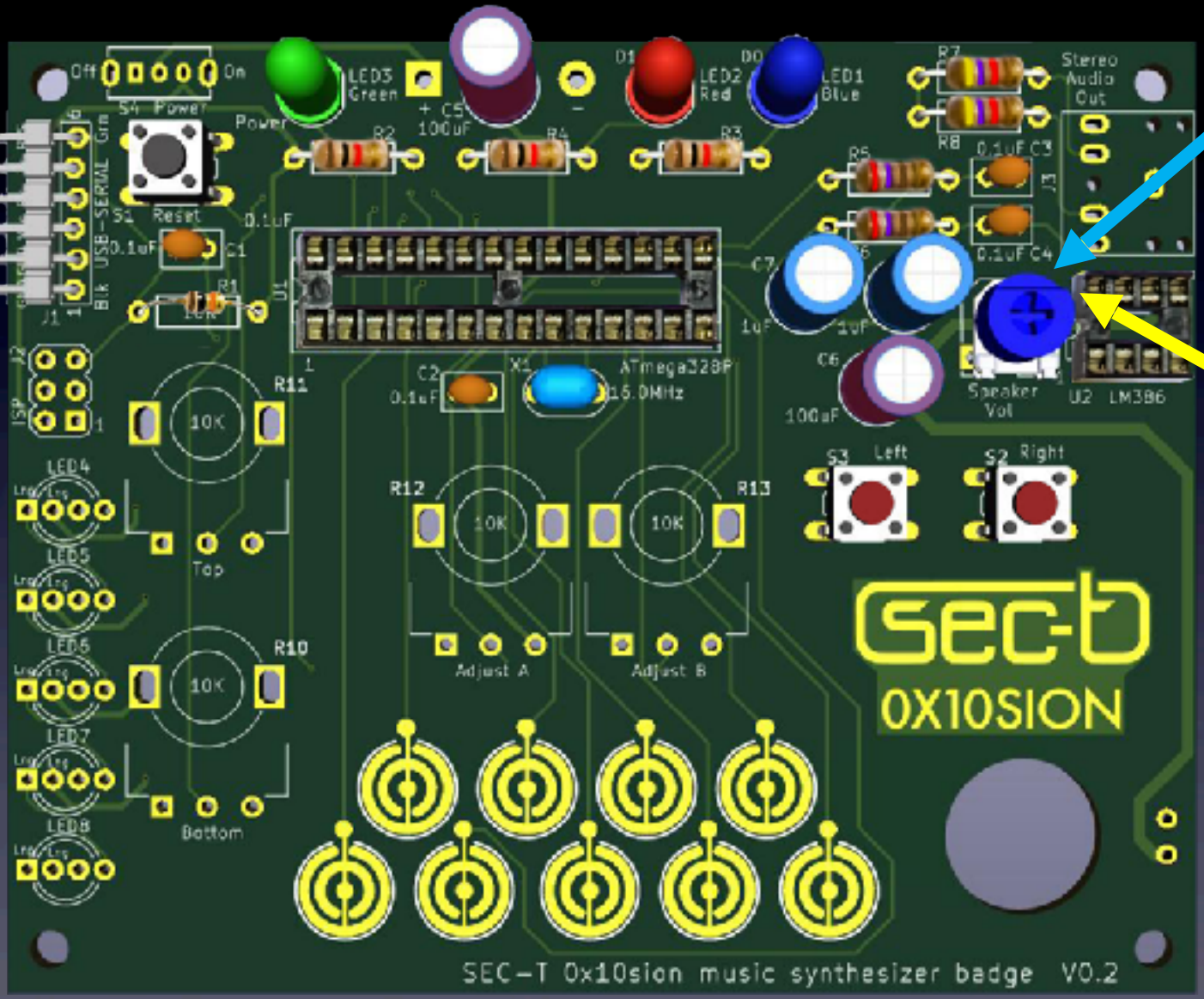





# X1

## The orientation of X1 does not matter.

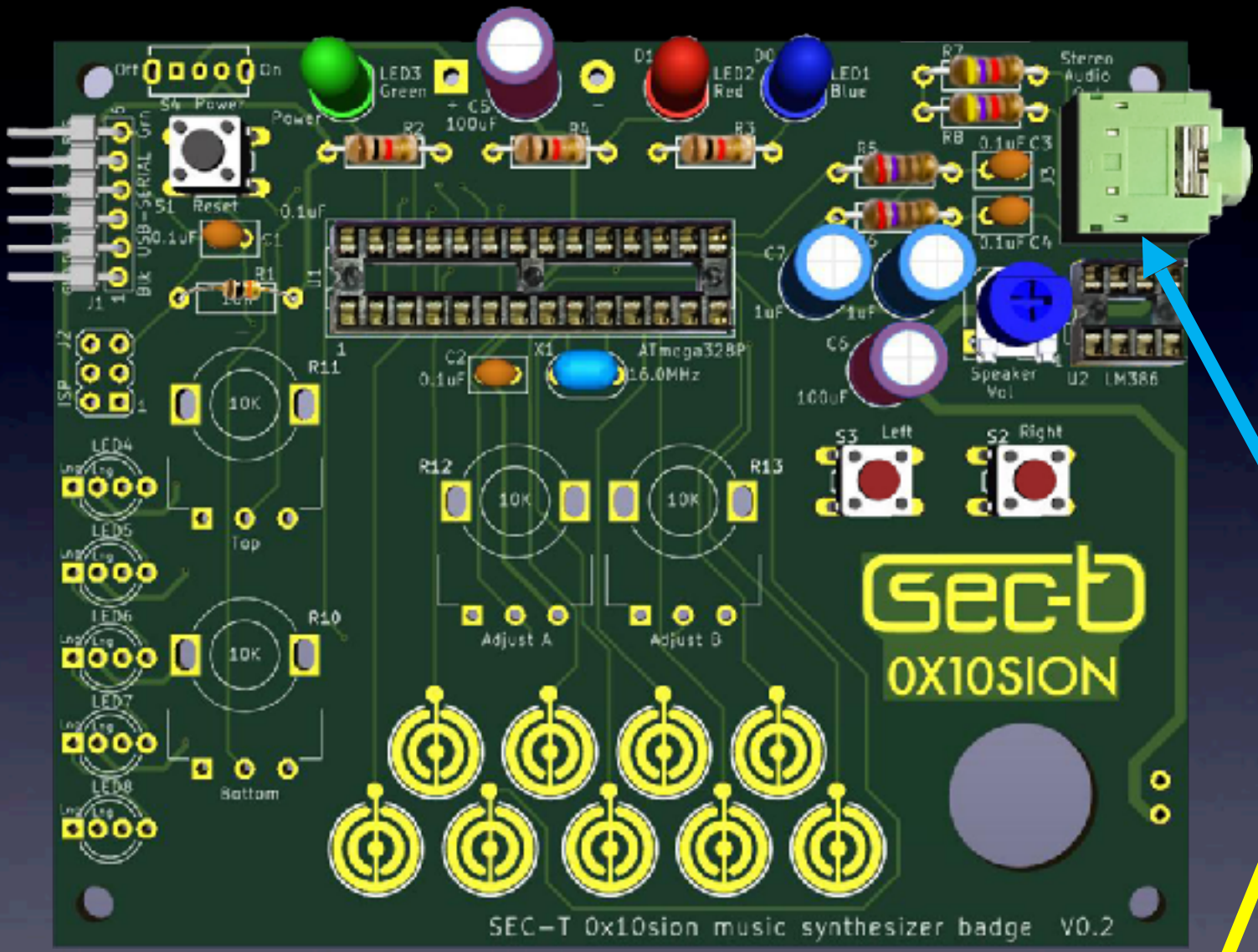
# R9: volume control




**If necessary, rotate the blue top so that it looks like this photo** *(rotated half-way)*

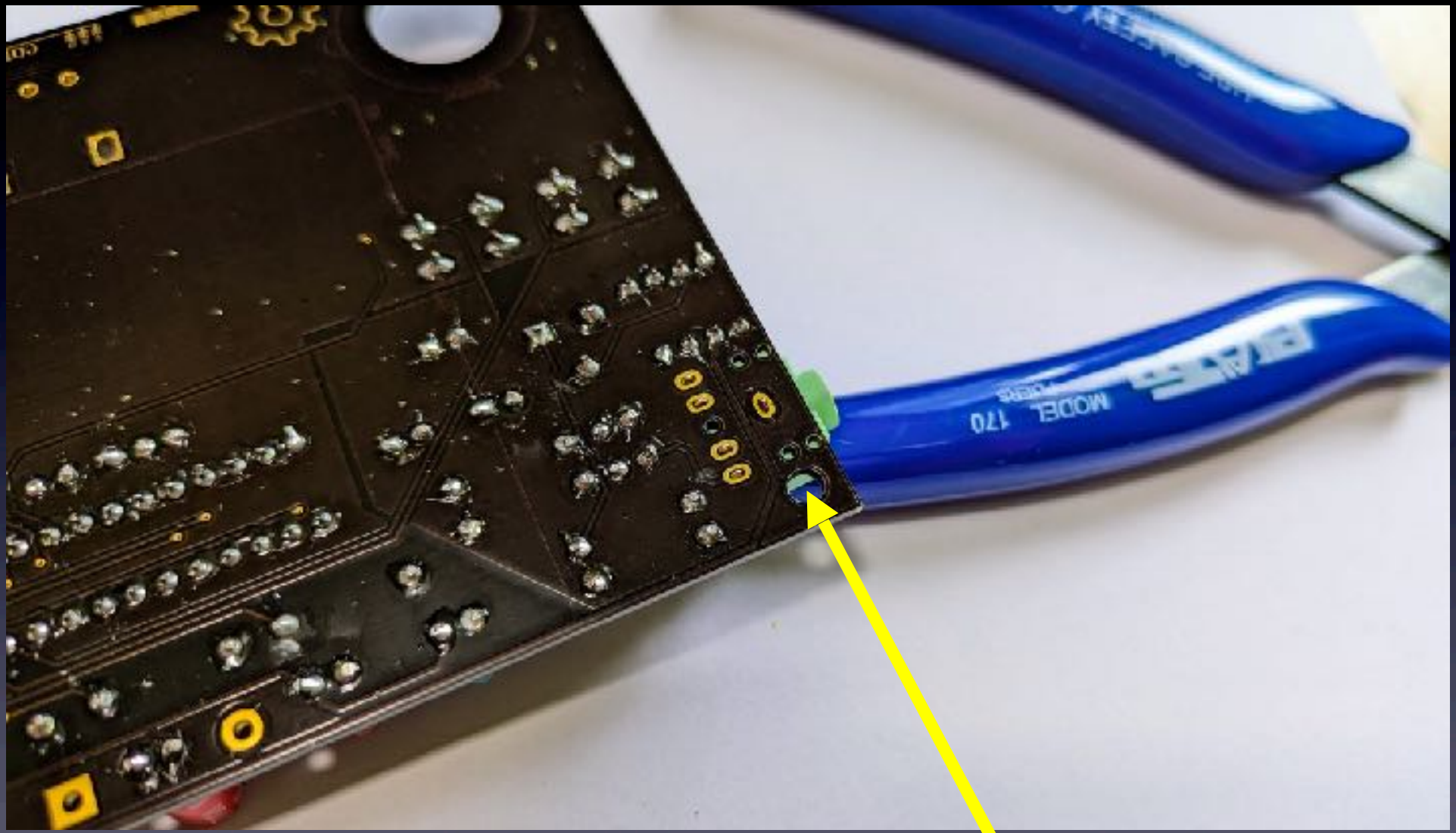
NOTE: R9 only controls the volume to the Speaker, not to Stereo Audio Out (J3)

# J3: Stereo Audio Output jack



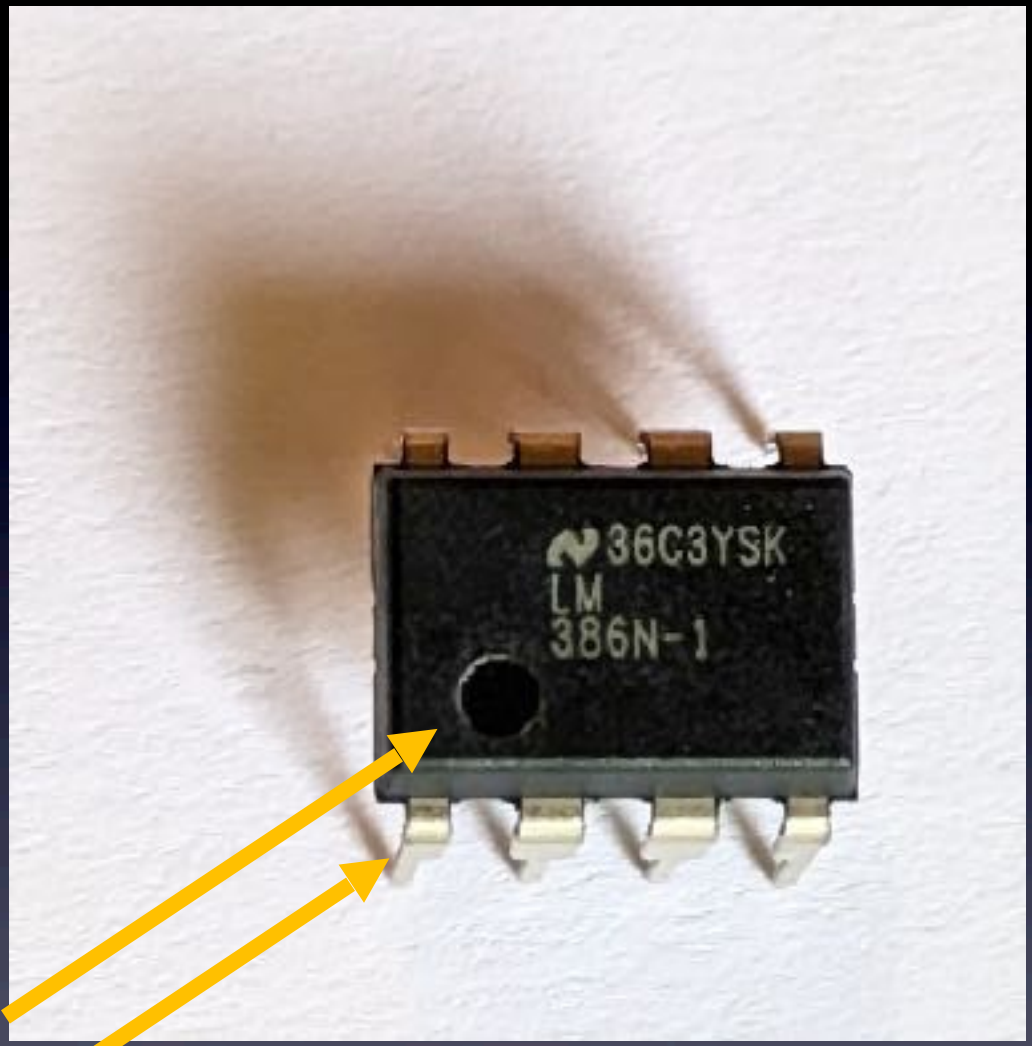

→ See next photo for how to solder this part…

# J3:  Stereo Audio Output jack



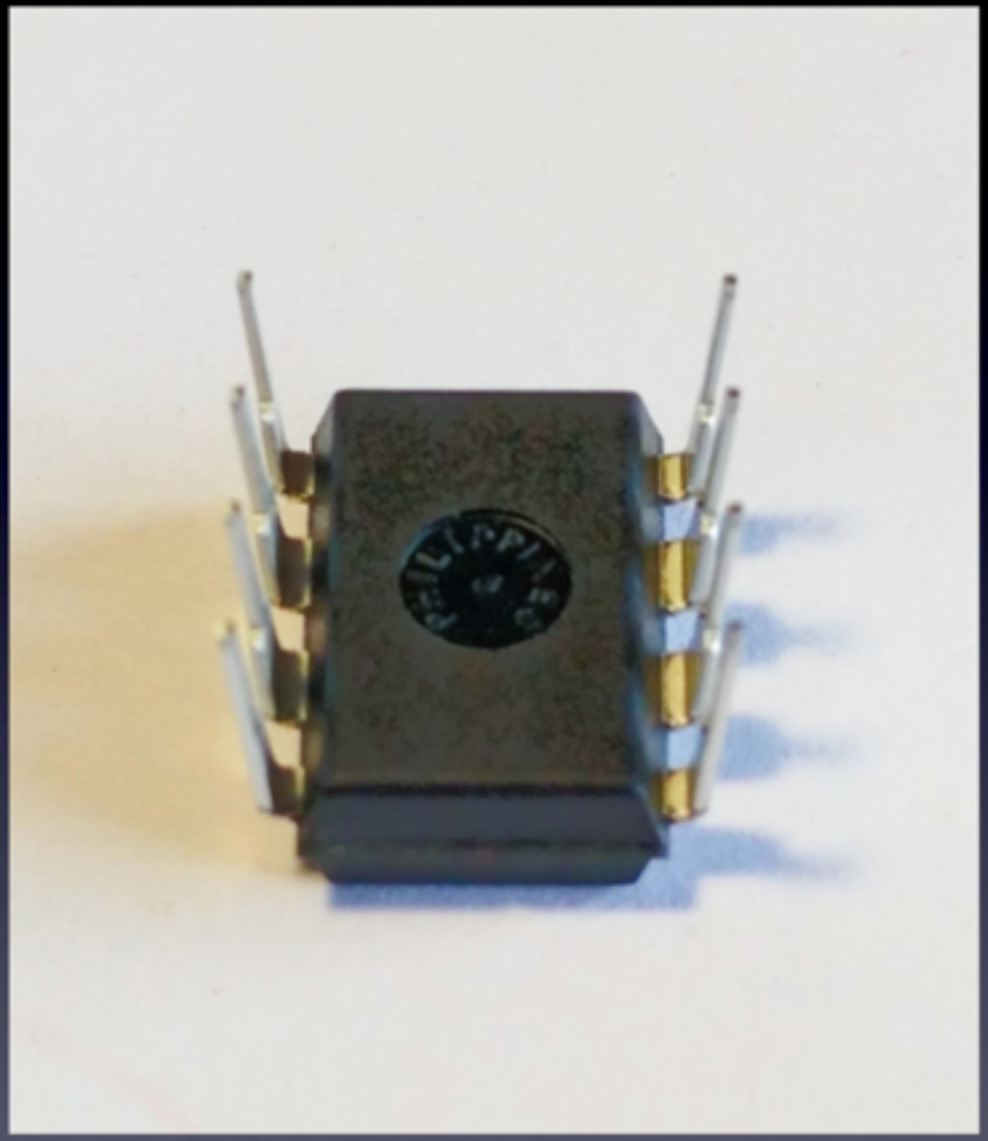
Resting on the wire cutters, J3 is held in place here, so you can solder all 5 leads.

# U2:  amplifier chip



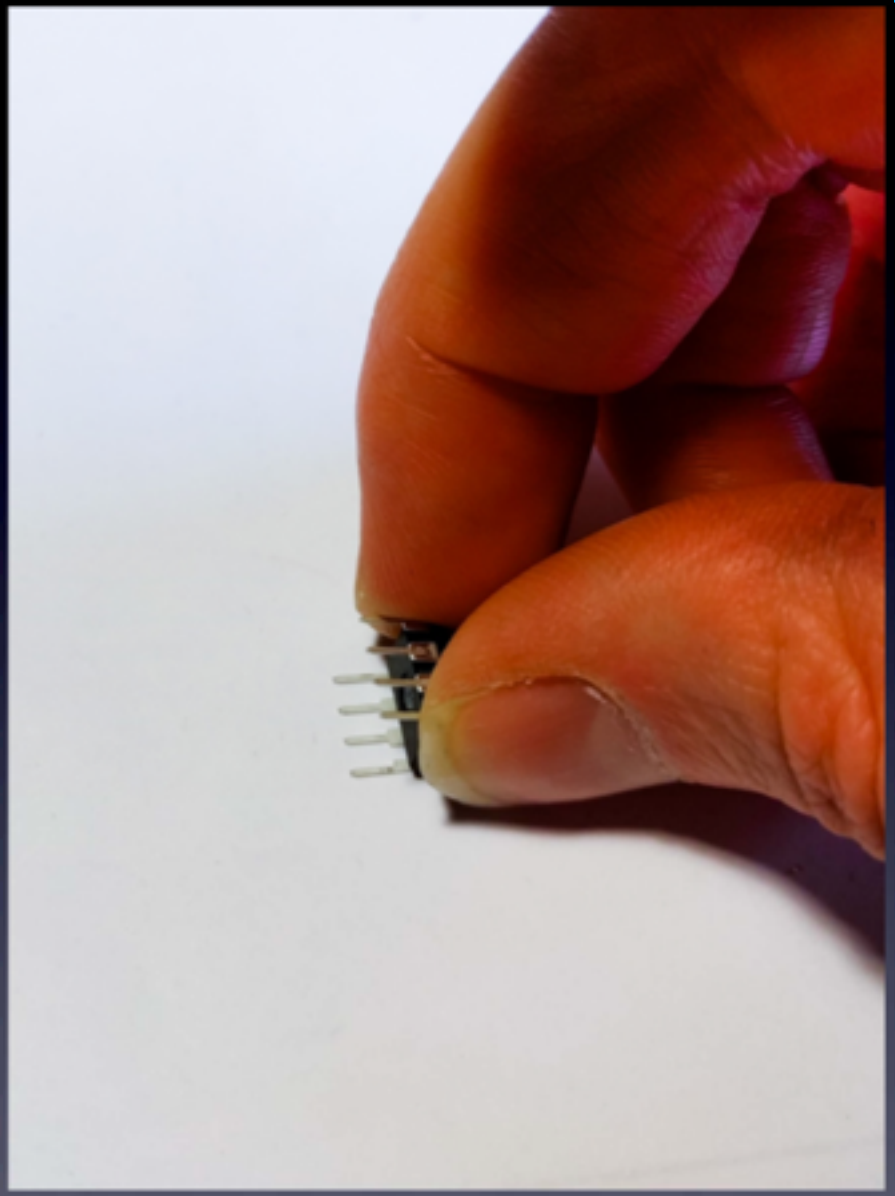

Indented black dot
Pin 1

# U2:  amplifier chip



**When chips are new,**
**their pins are bent out.**

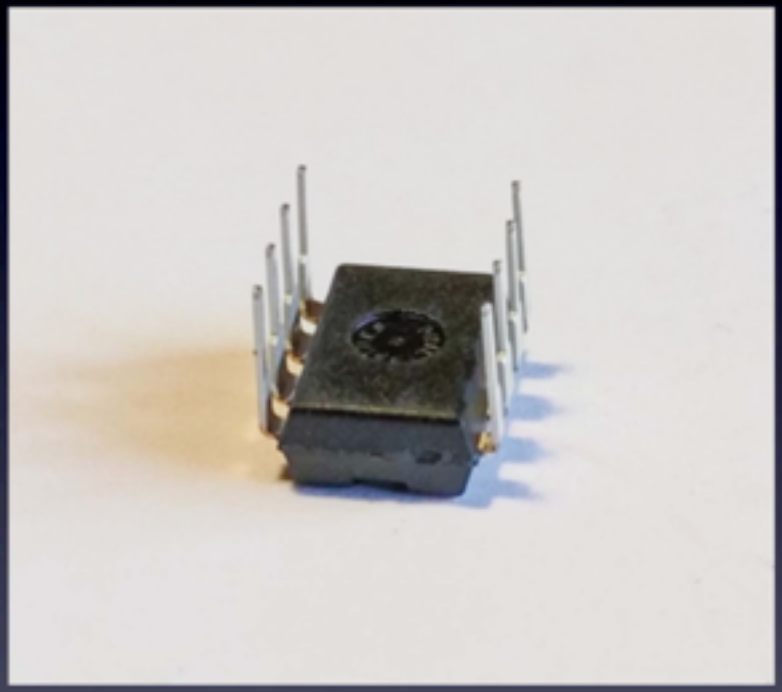
# U2:  amplifier chip



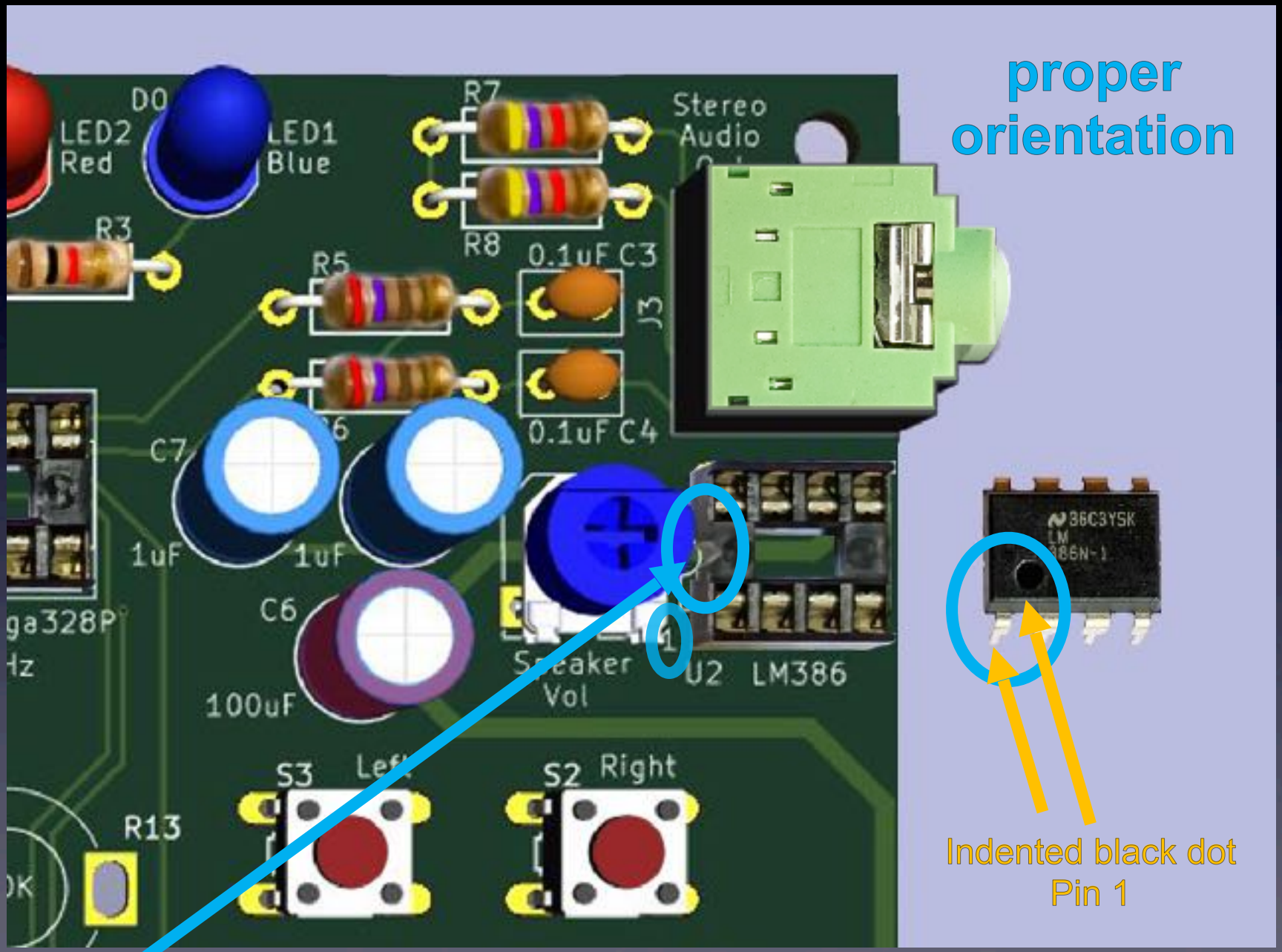
**We need the pins bent *straight and parallel*.**
**Use your work table to (gently) bend the leads.**

# U2: amplifier chip

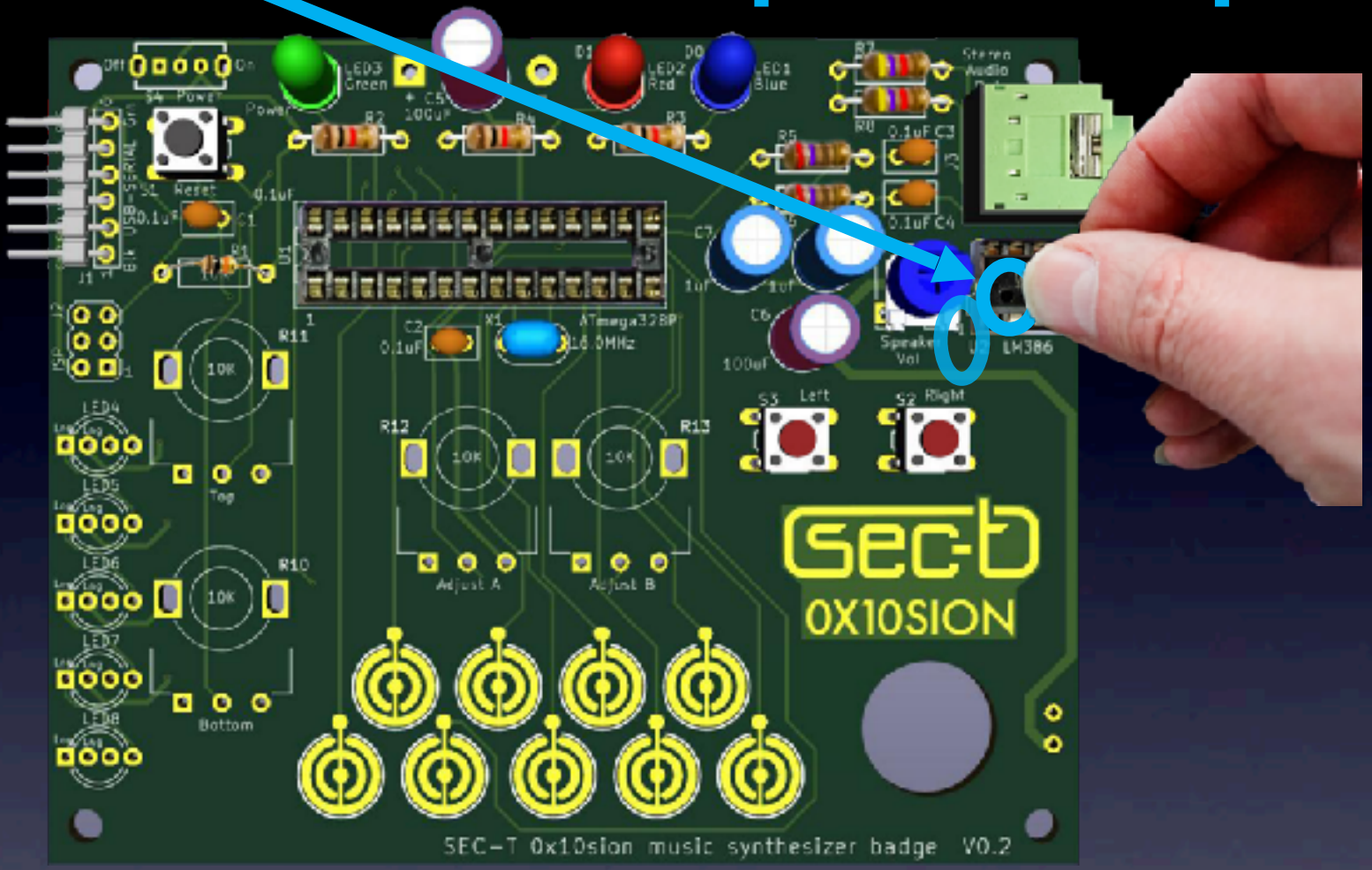
Gently
bend leads
so they're straight
and parallel



# U2: amplifier chip



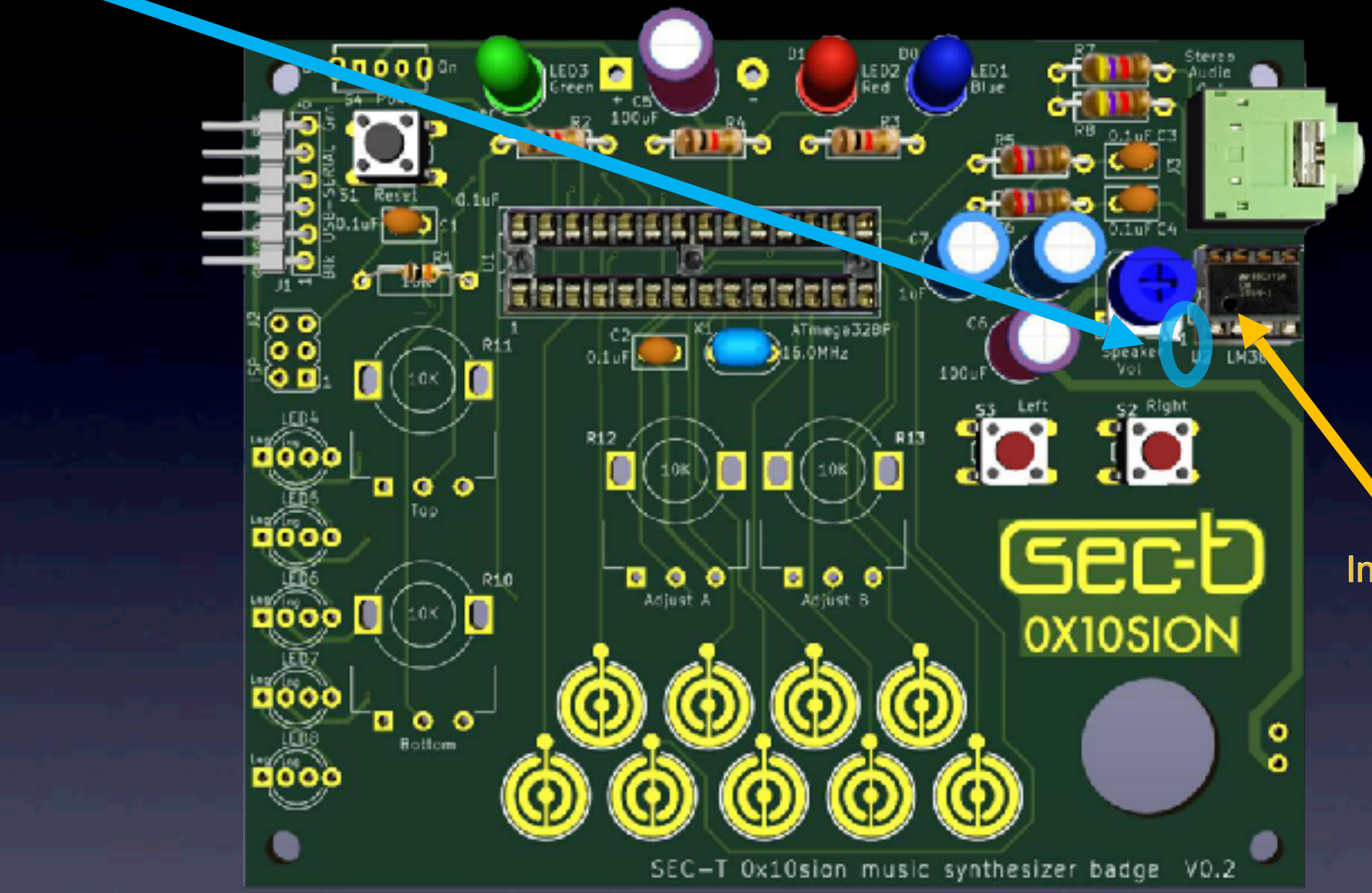
# U2: amplifier chip




Make sure all 8 pins rest in their holes in the socket

→ with the proper orientation

# U2: amplifier chip




Indented black dot
Pin 1

**Inspect all pins, and be sure each went into its hole in the socket – not bent.**

*If any pins are bent, (gently) pry out chip, straighten pins, and insert again.*

# U1: microcontroller



# U1: microcontroller

U1



When chips are new, their pins are bent out.

Note: Your kit's U1 chip may or may not have its pins already bent straight and parallel. If not, you need to bend them, as shown in the next picture.

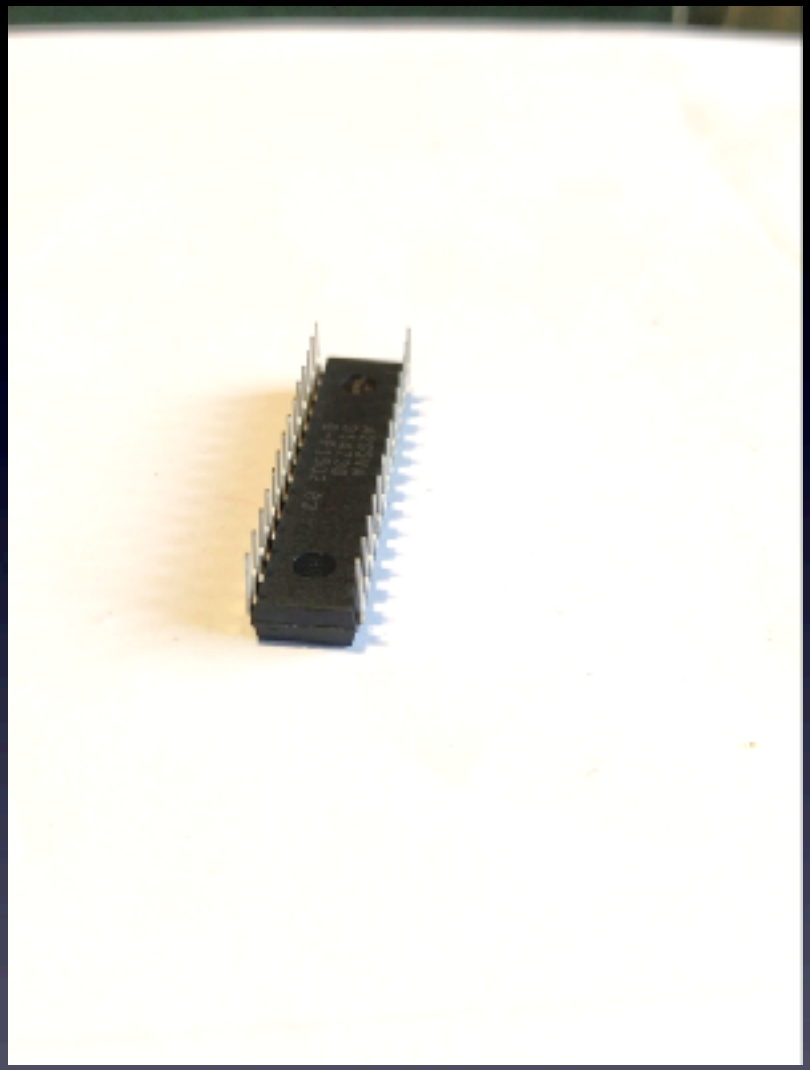
# U1: microcontroller

Note: Your kit's U1 chip may or may not have its pins already bent straight and parallel.
If not, you need to bend them, as shown in this picture.



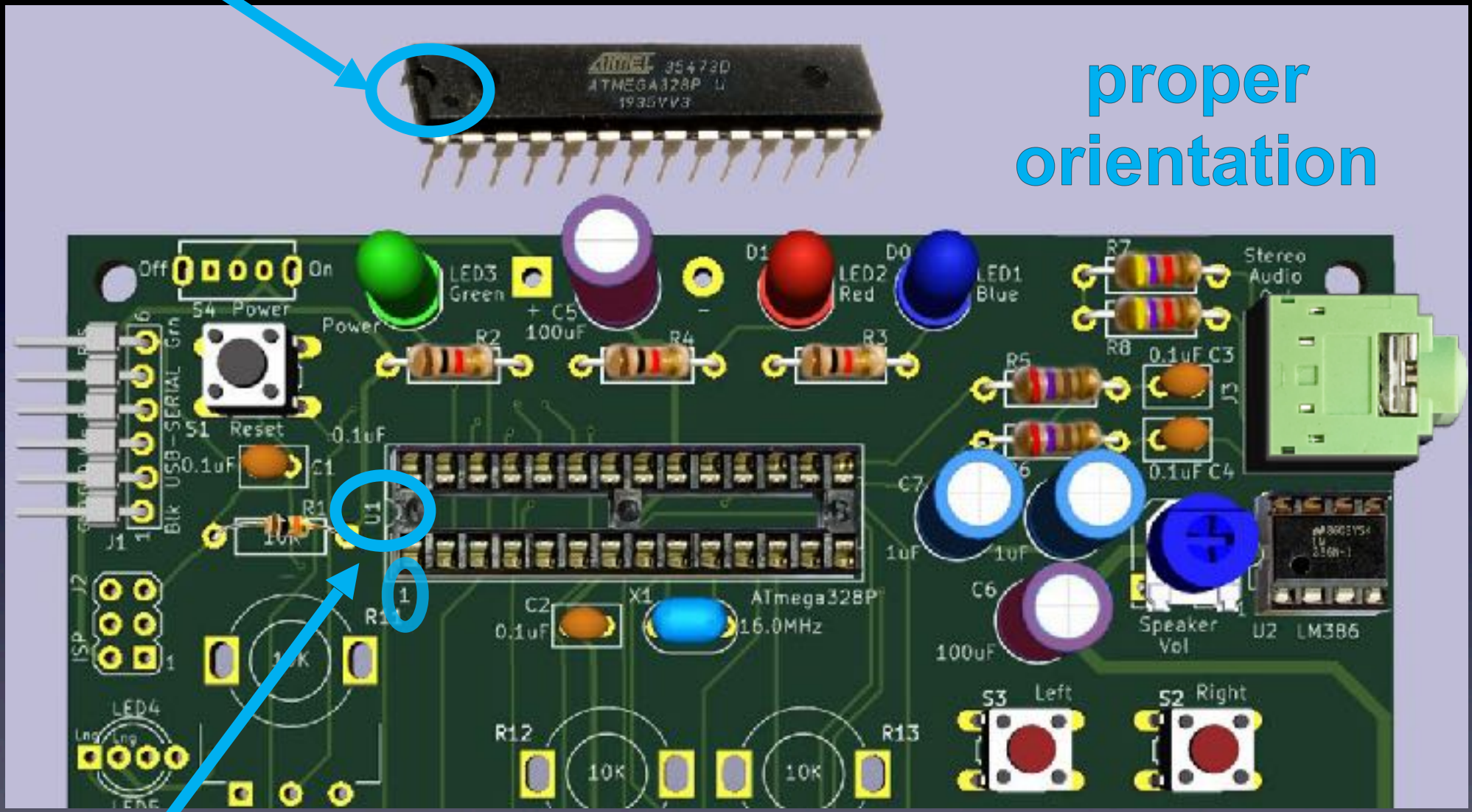
**We need the pins bent straight and parallel.**
**Use your work table to (gently) bend the leads.**

# U1:  microcontroller

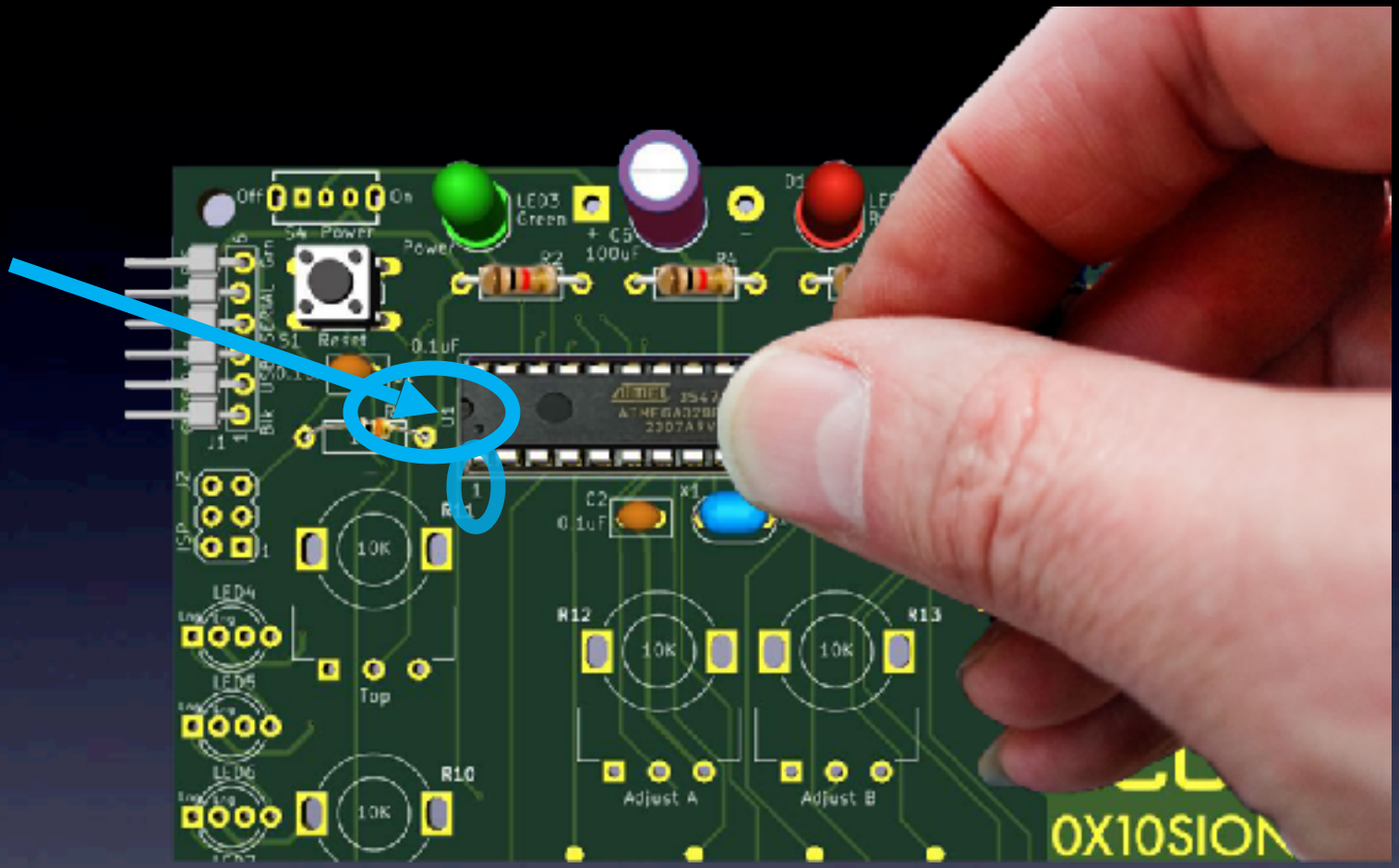


These pins must be straight and parallel

# U1: microcontroller



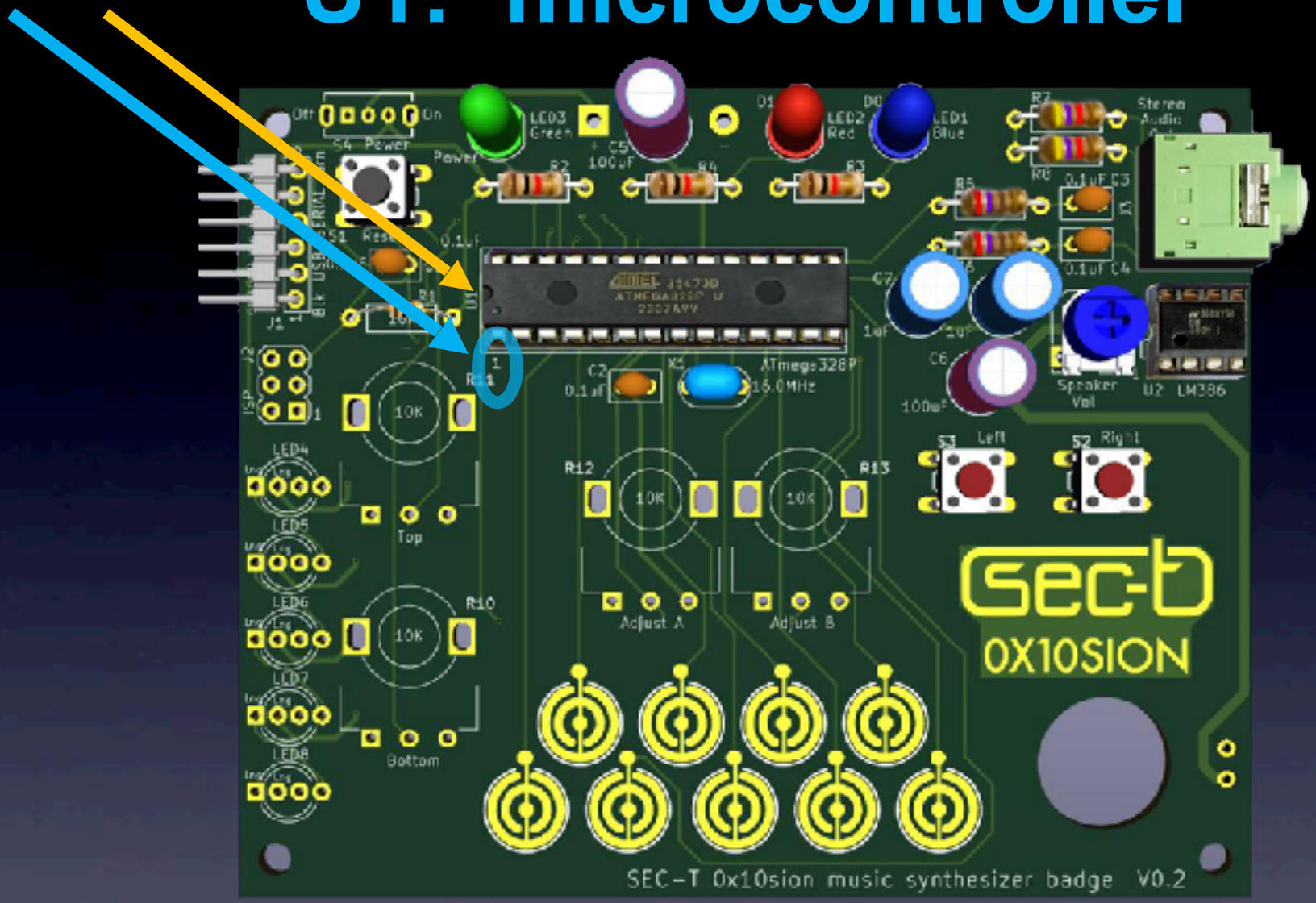
# U1: microcontroller




Make sure all 28 pins rest in their holes in the socket
→ with the proper orientation

**(This is actually way easier with 2 thumbs.)**

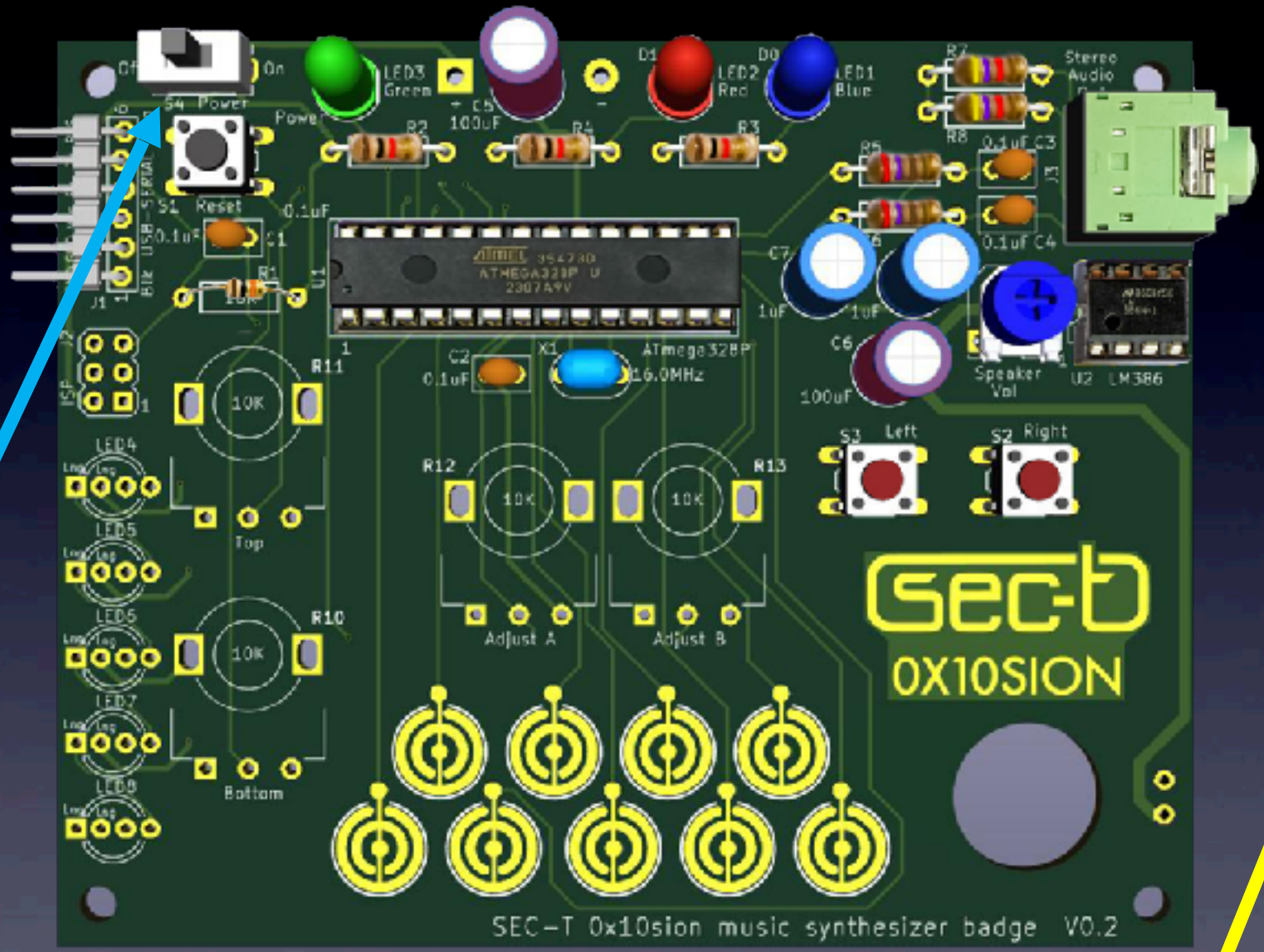
# U1: microcontroller




**Inspect all pins, and be sure each went into its hole in the socket – not bent.**

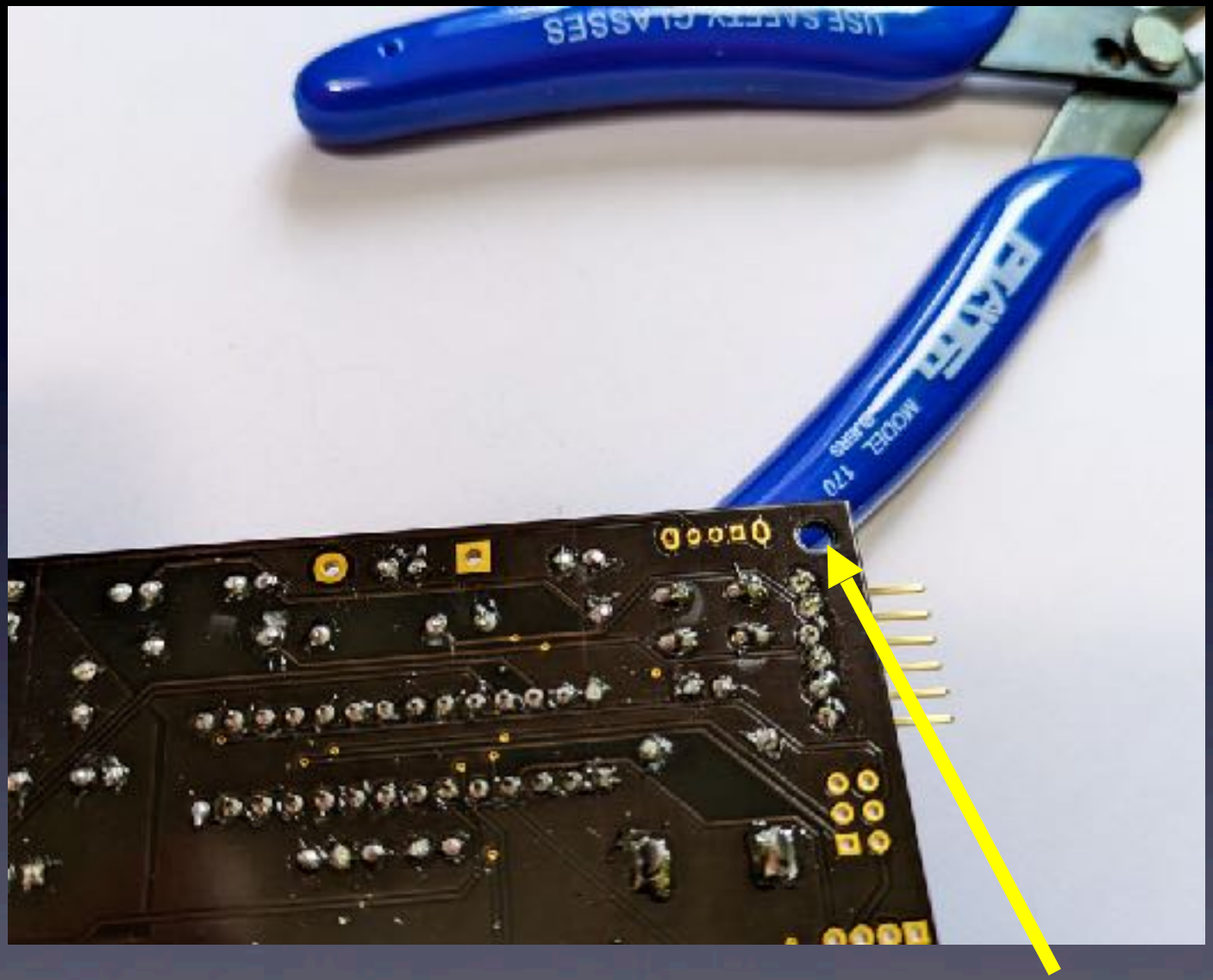
*If any pins are bent, (gently) pry out chip, straighten pins, and insert again.*

# J4: Power switch



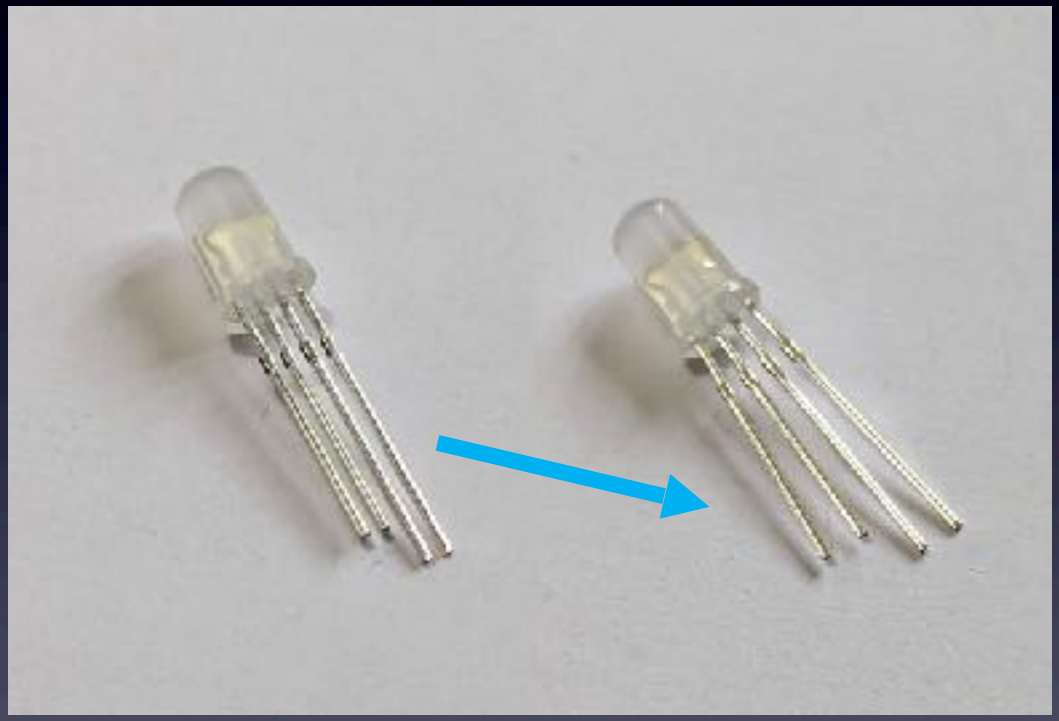

→ **See next photo for how to solder this part…**

# J4:  Power switch



Resting on the wire cutters, S4 is held in place here, so you can solder all 5 leads.

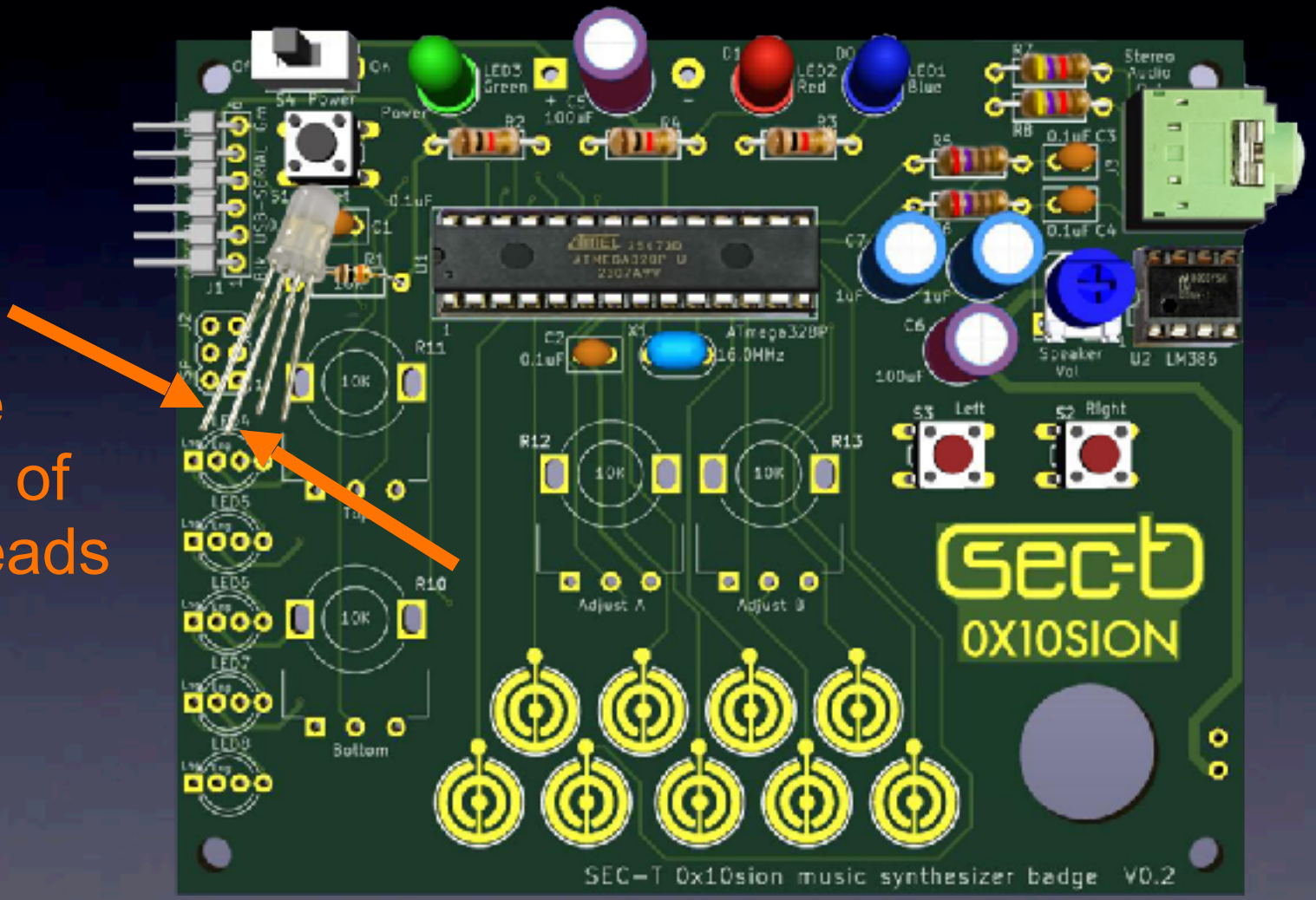
# LED4, LED5, LED6, LED7, LED8: RGB LEDs



Bend the leads out a little so they fit into their pads.

# LED4, LED5, LED6, LED7, LED8: RGB LEDs

Note the orientation of the 2 long leads

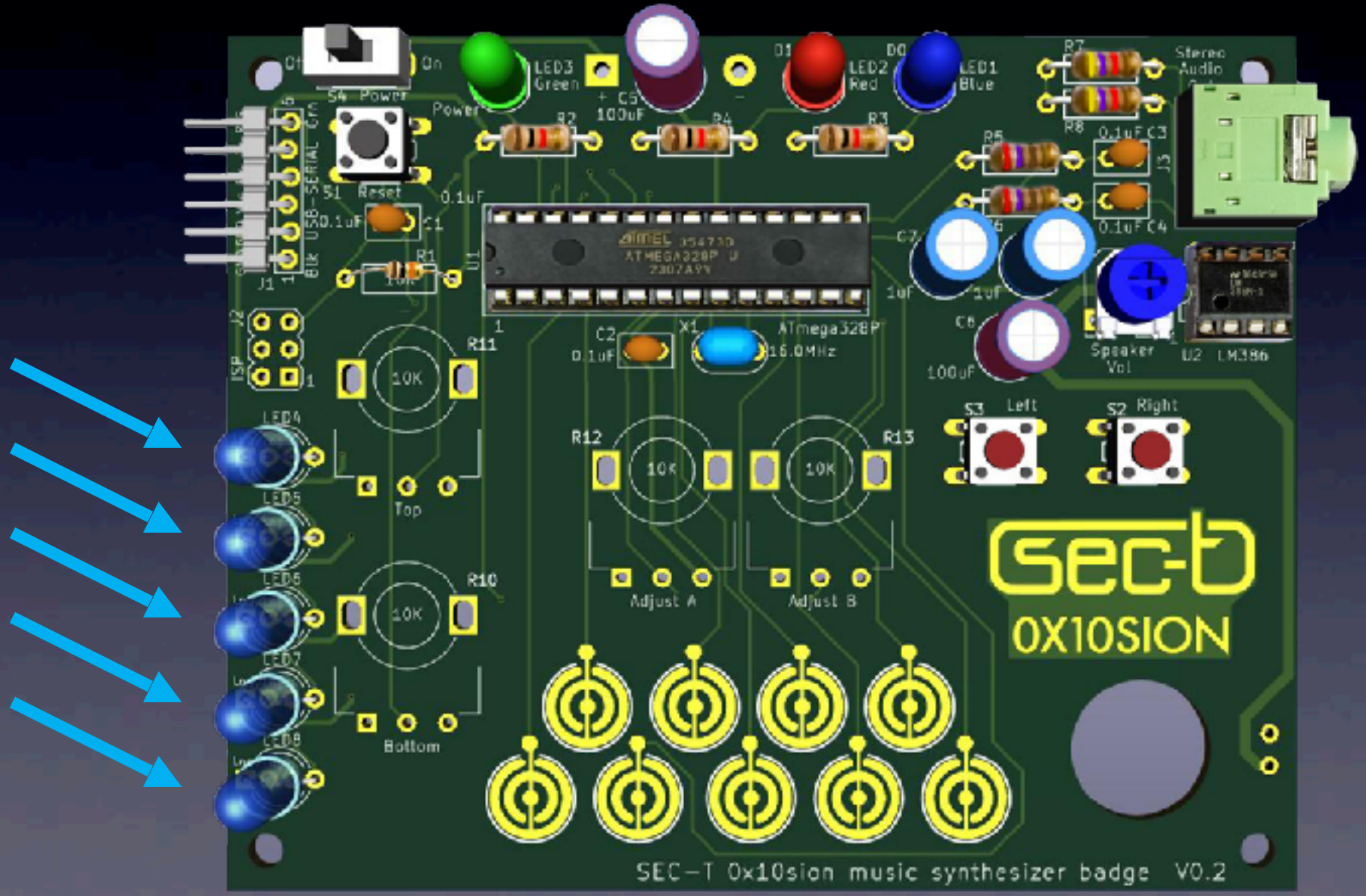


# LED4, LED5, LED6, LED7, LED8: RGB LEDs

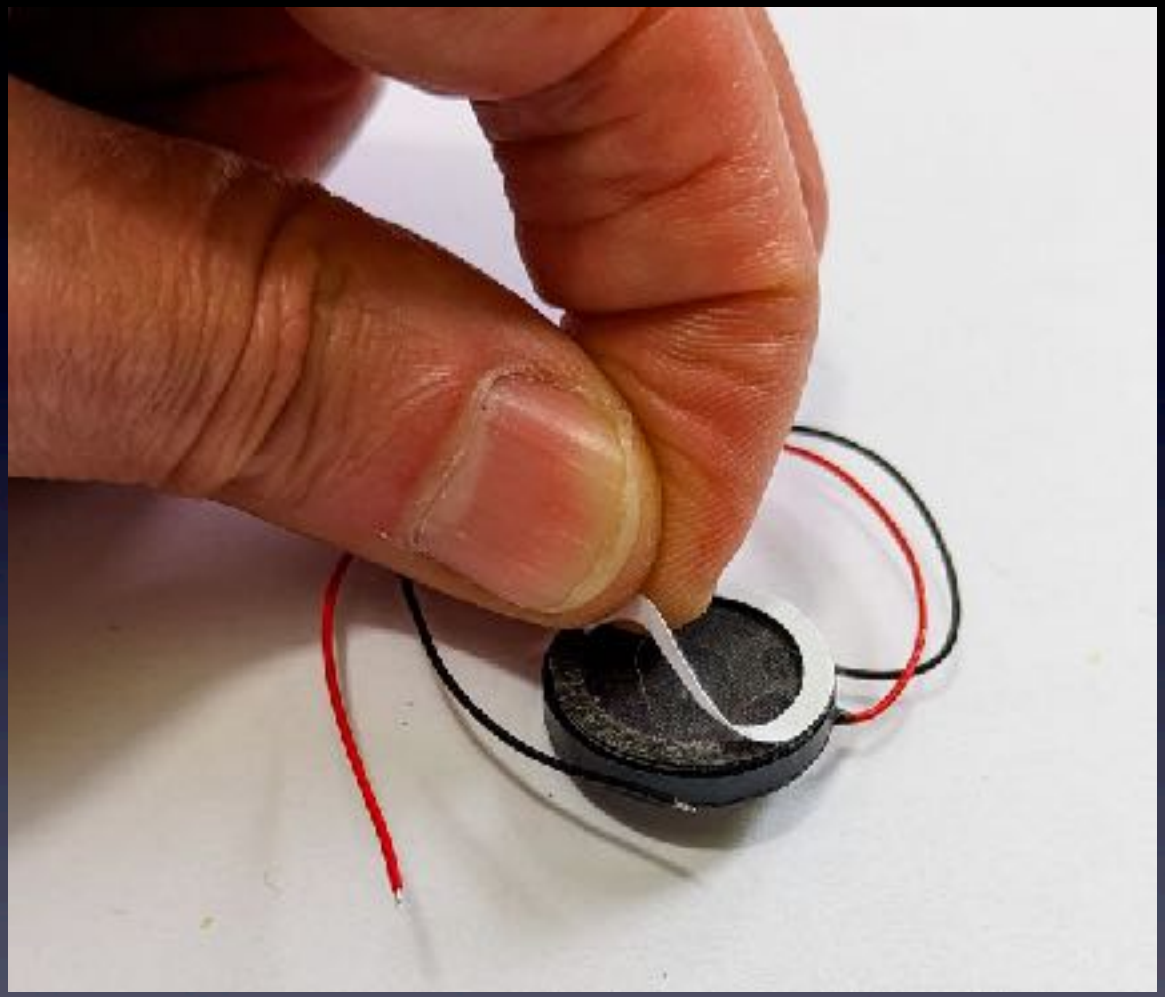


These LEDs will be inserted about this far into the board

# LED4, LED5, LED6, LED7, LED8: RGB LEDs

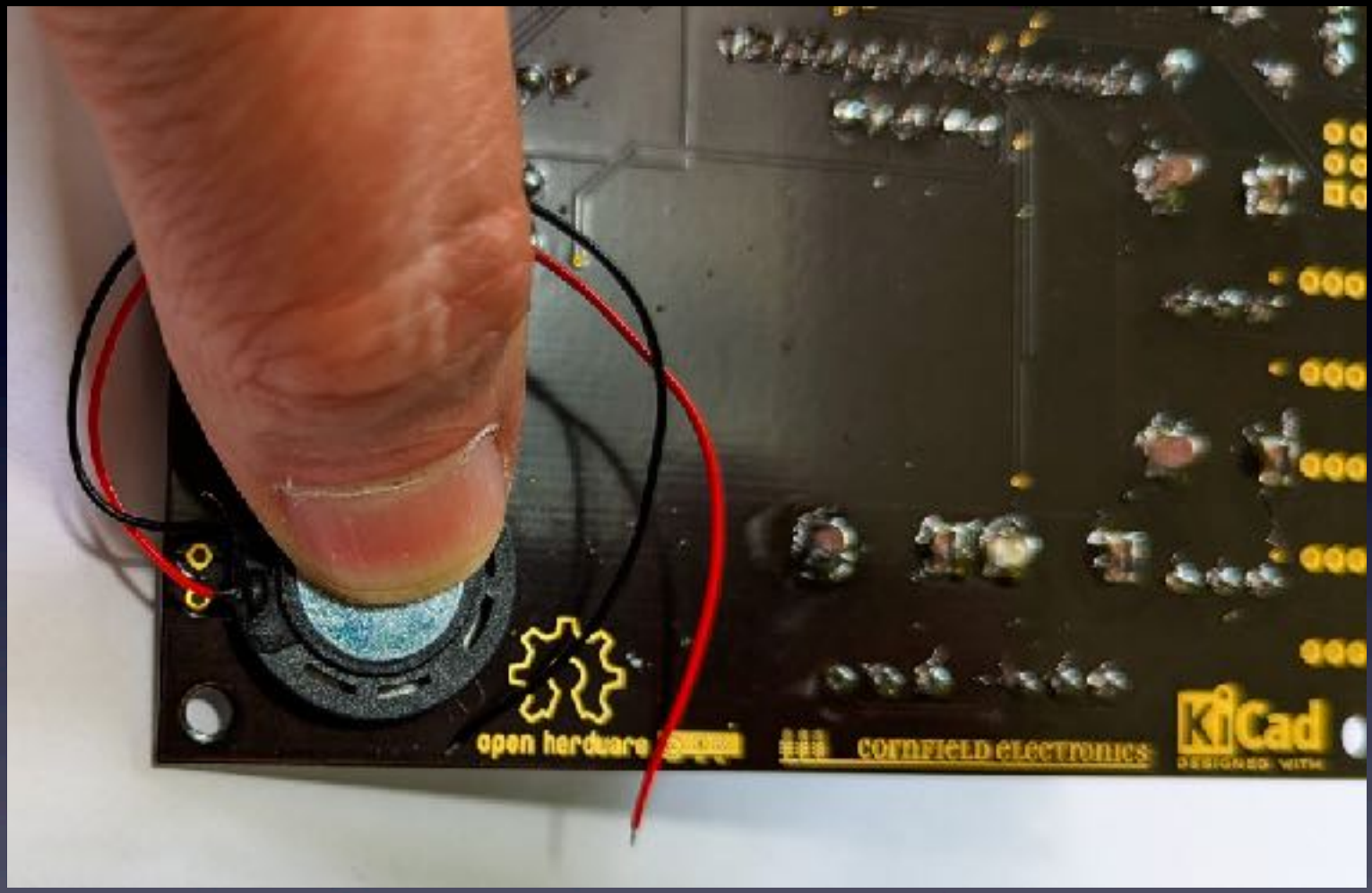


# Speaker



Pull off the protective covering to reveal the adhesive

# Speaker




Carefully center the speaker over its hole, and then push

# Speaker



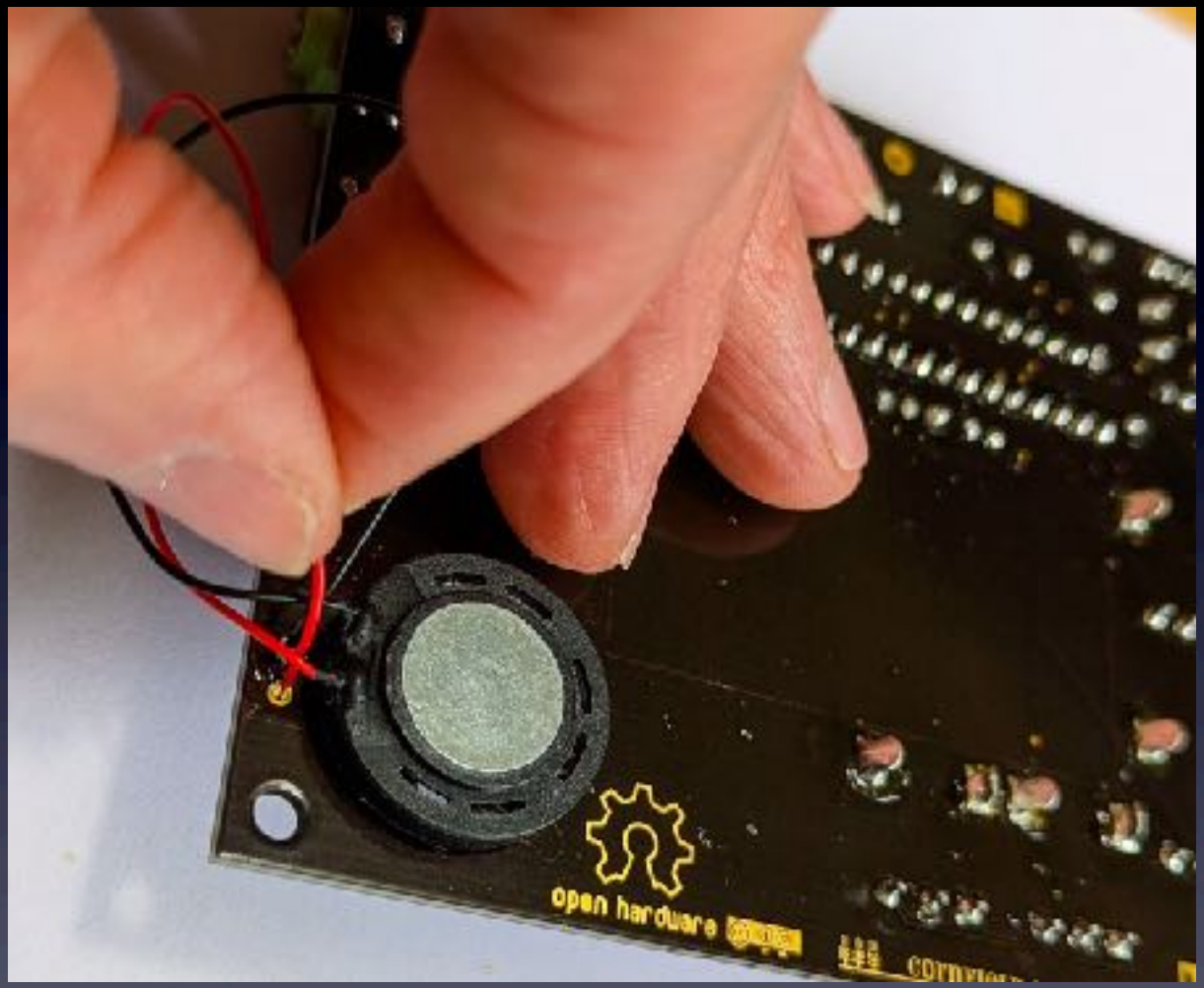

Here you can see the speaker centered under its hole

# Speaker



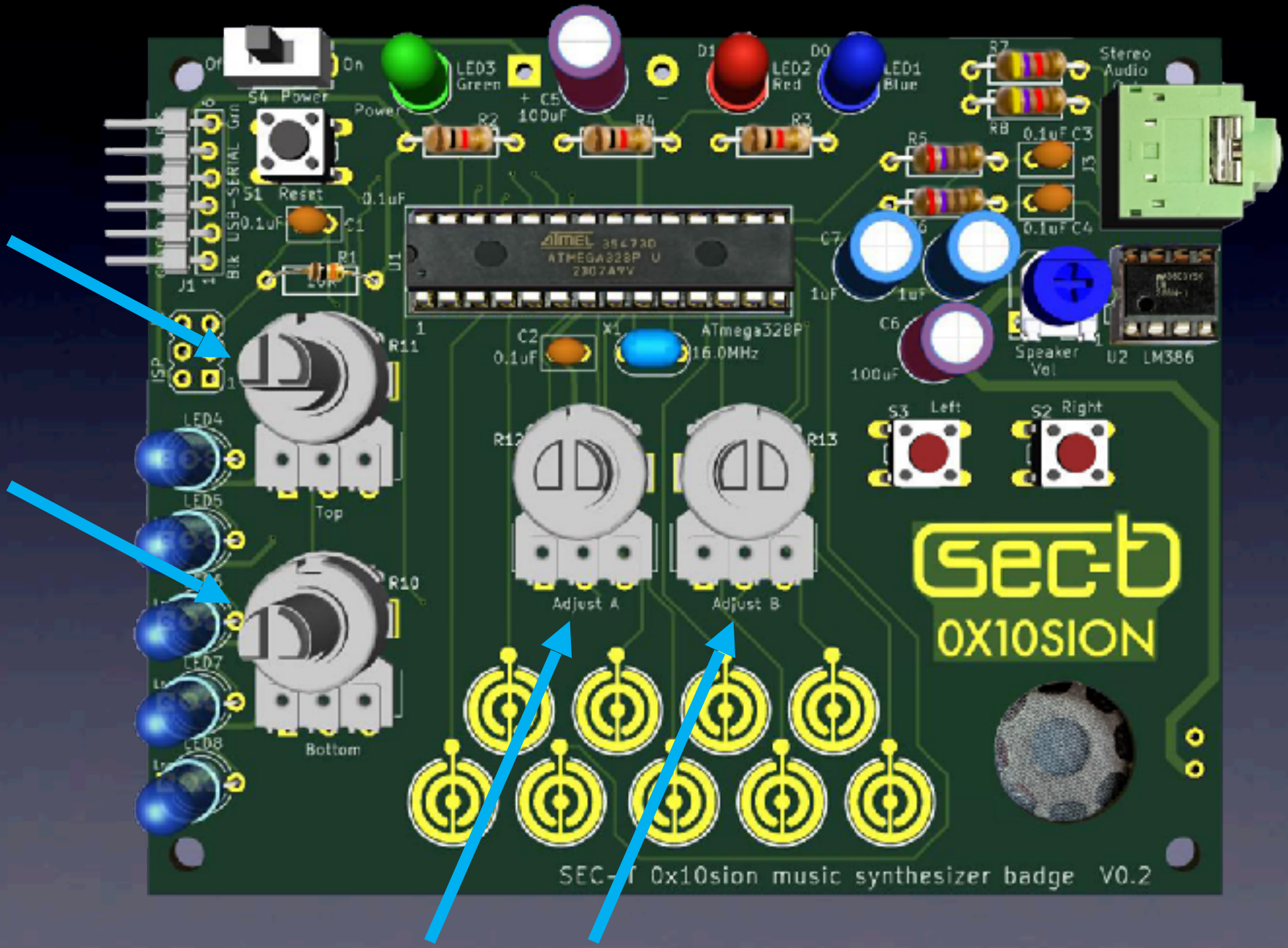
Place the black wire into its pad, and solder it

# Speaker

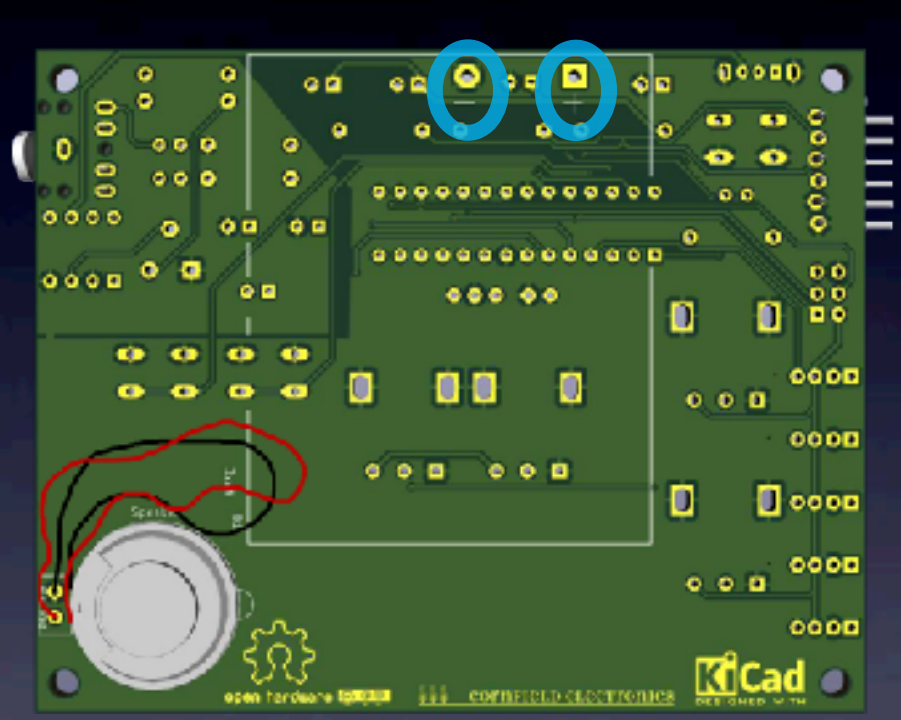


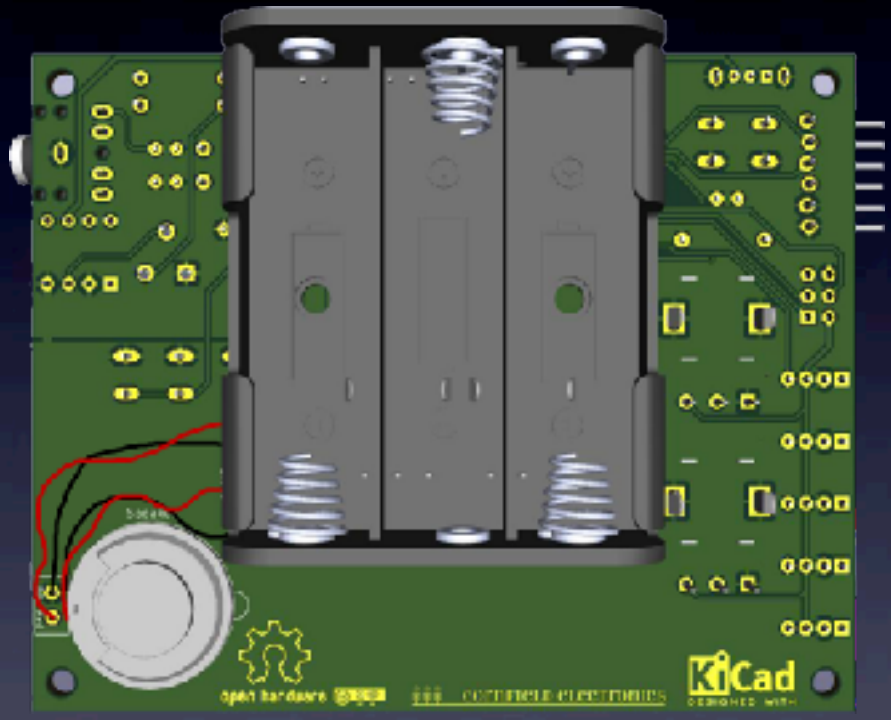

Place the red wire into its pad, and solder it

# R10, R11, R12, R13: potentiometers



# Battery pack



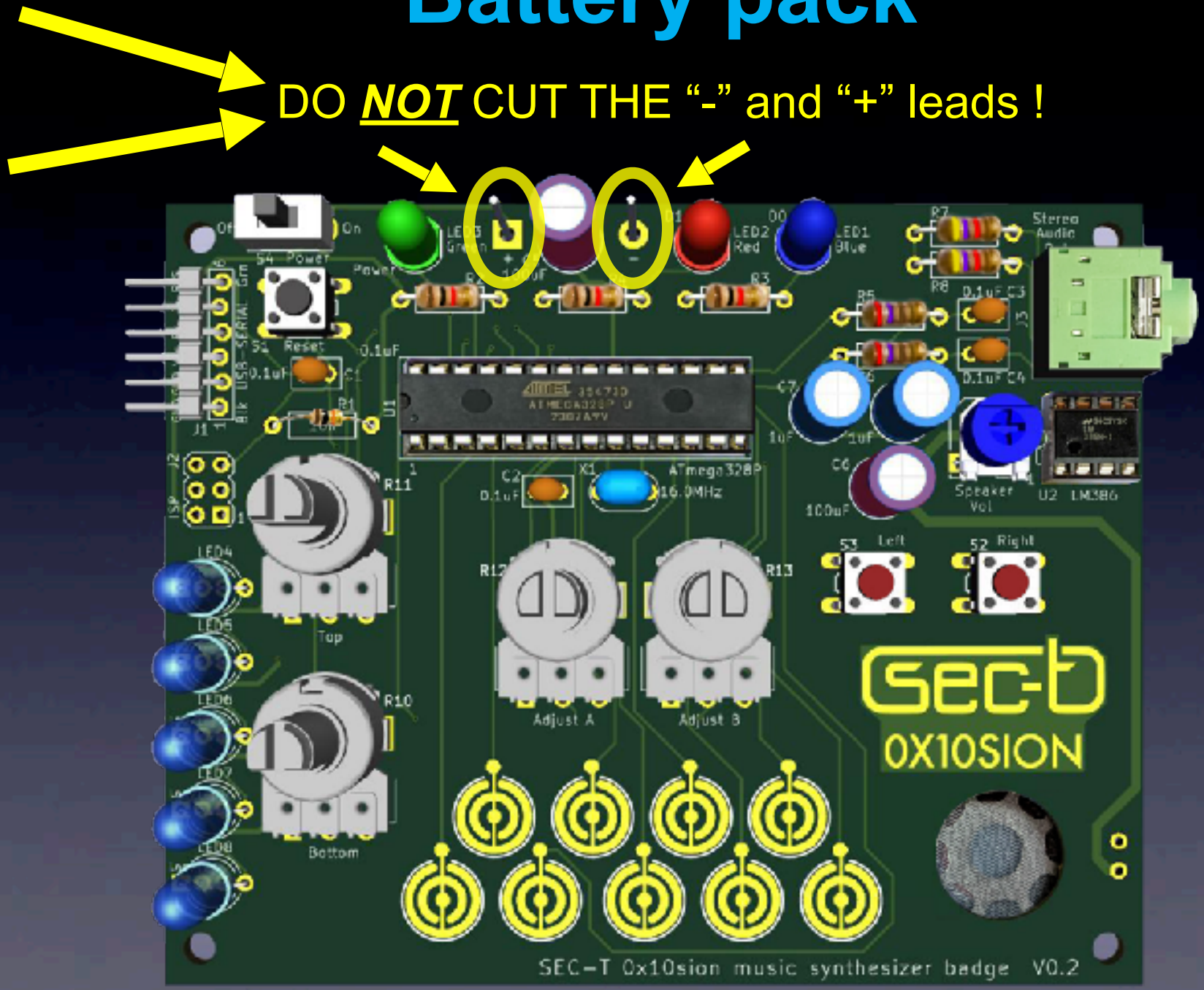


Place the Battery pack into place, with its leads in its “-” and “+” pads and solder the leads on the front

DO ***NOT*** CUT THE “-” and “+” leads !

# Battery pack

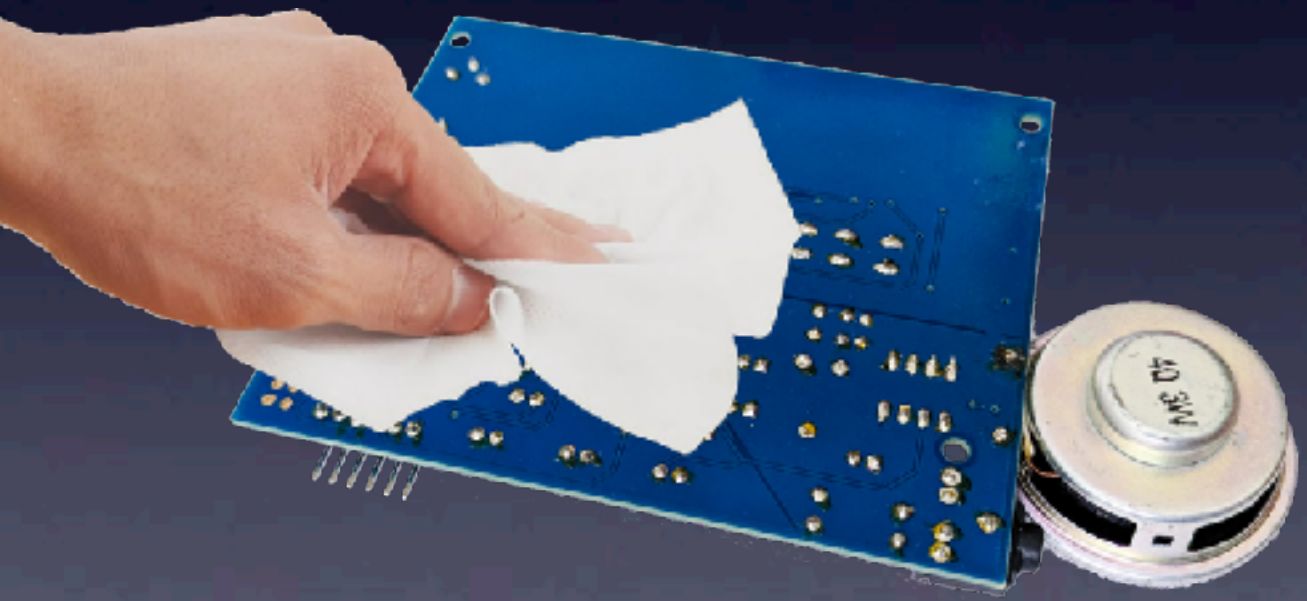
DO ***NOT*** CUT THE “-” and “+” leads !



## If you used any *flux paste for re-working problems*



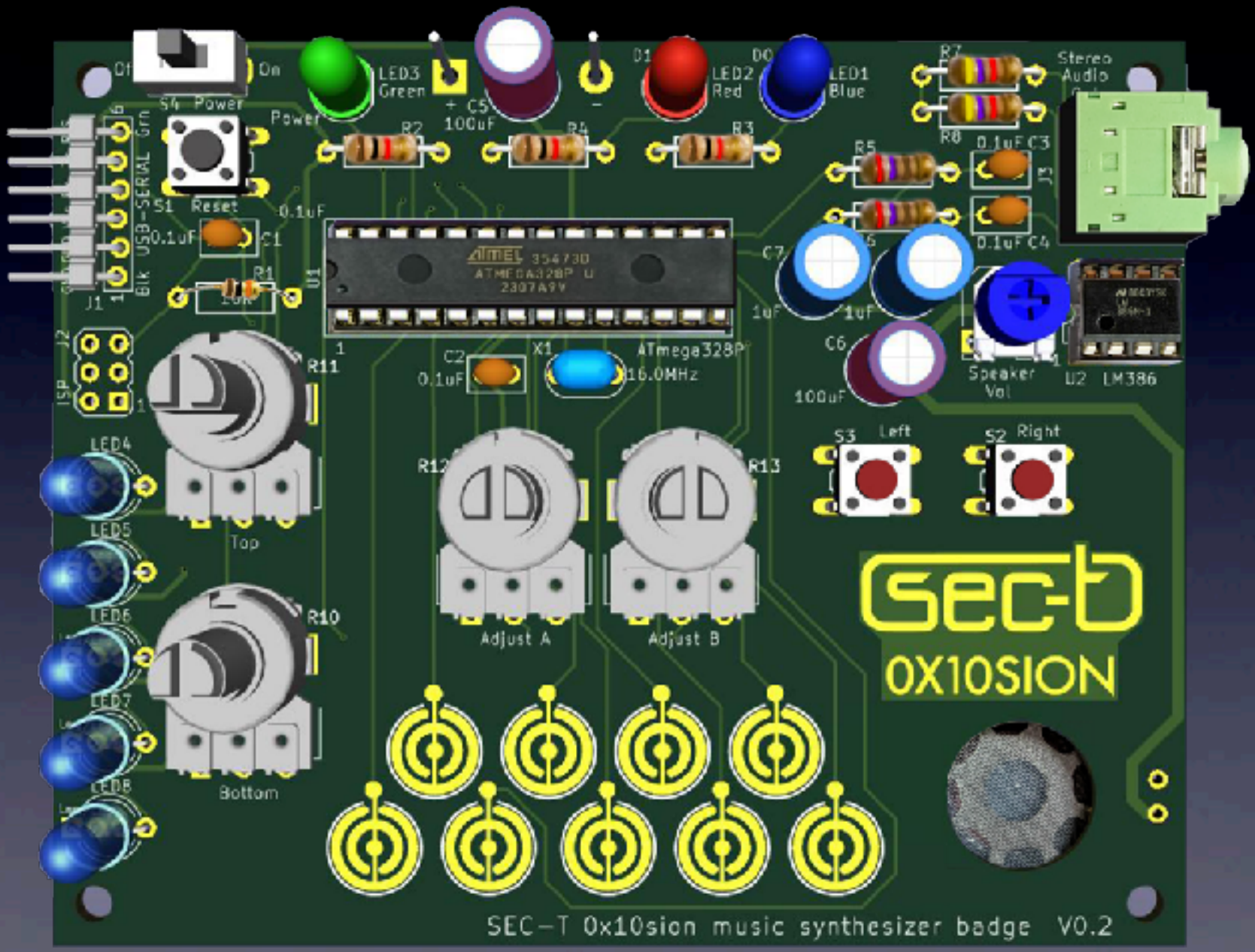
The bottom of the PCB will be sticky from the flux



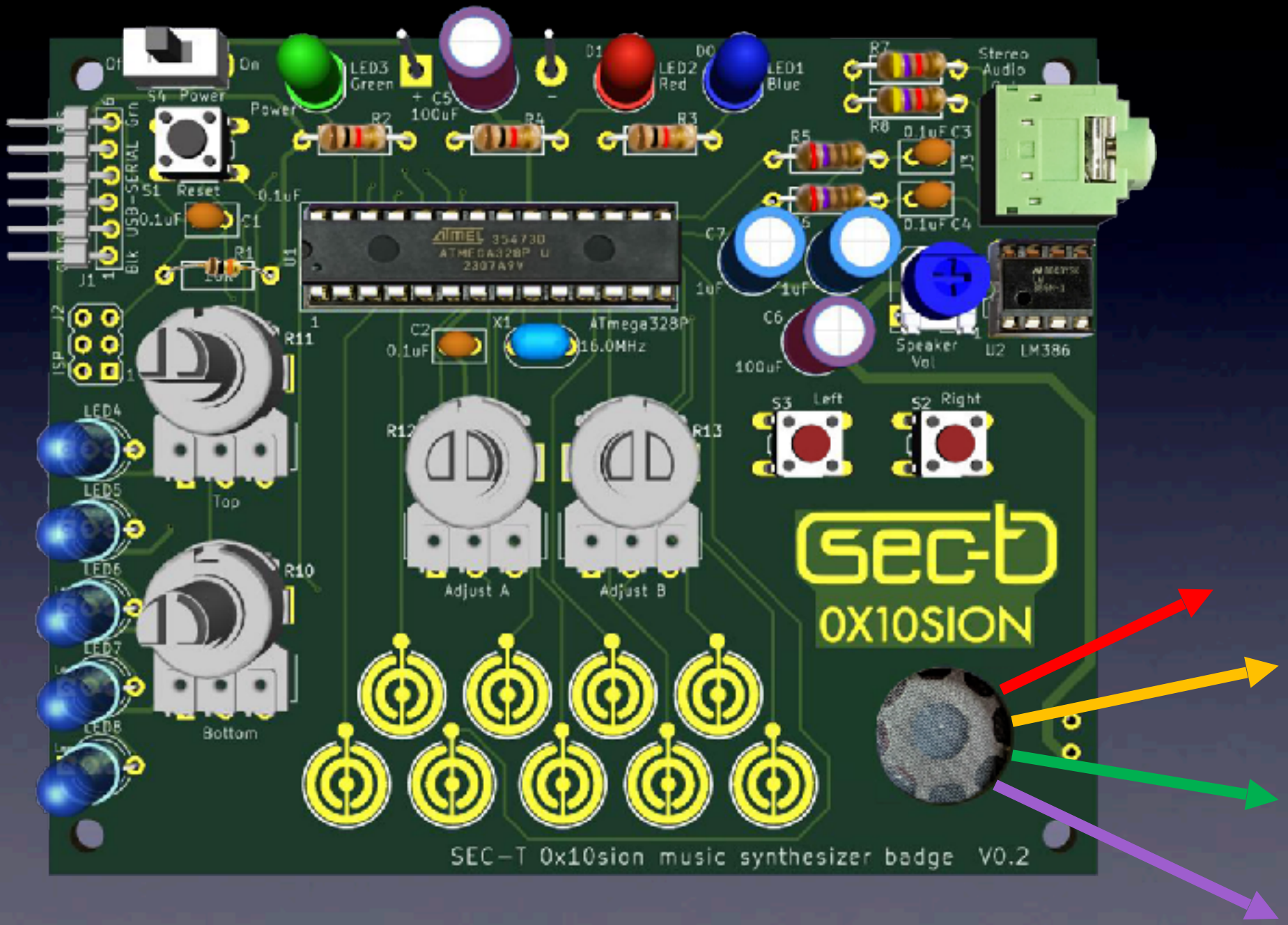



***You can clean it with a cloth wet with Isopropyl Alcohol***

# Done!


LED3 Green
LED2 Red
LED1 Blue
Power
Reset
Stereo Audio
USB-SERIAL
ATmega328P
16.0MHz
Speaker Vol
LM386
Top
Bottom
Adjust A
Adjust B
Left
Right
SEC-T
0X10SION
SEC-T 0x10sion music synthesizer badge V0.2

# Let’s make some noise!



Please Remember:

to

**Wash your hands**

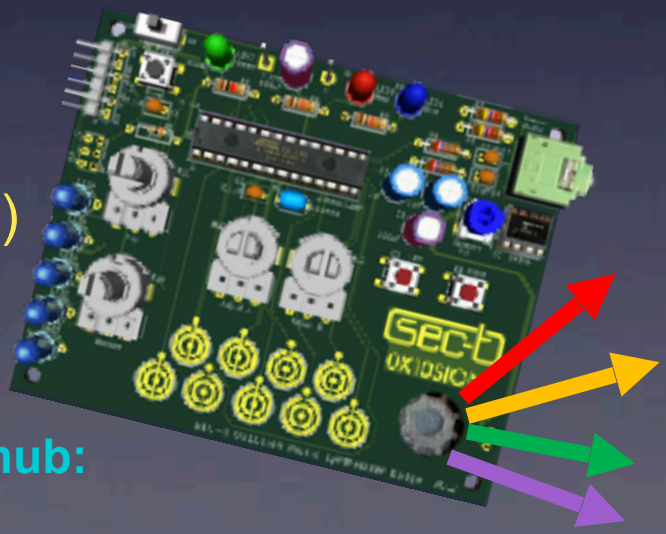
after soldering

# Let's make noise!

**Your SEC-T 0x10sion Synth Badge comes pre-programmed with a really cool synthesizer, called "Quadrant".**

"Quadrant" is like a looper, but with the ability to warp each time around the loop.

- the left and right red buttons change octaves
- double-click the right red button and then hit a bottom-row touch-key to change to different preset – there are 5 very different presets
- the Bottom knob controls the looping rate
- the Top knob controls warping (detuning) each time through the loop
- the Adjust A knob controls the wah speed
- the Adjust B knob controls the wah depth
- Try playing with these and see!
- Long-pressing the left and right red buttons changes other parameters that the knobs can control (more info on the project's Github)
- Experiment, and see what sounds you get!



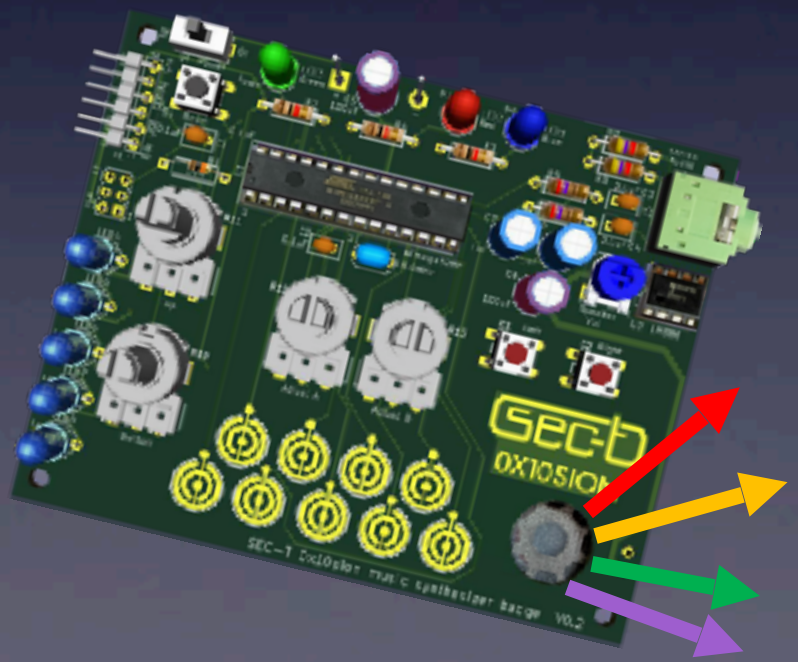
**Complete documentation is on the project's Github:**
https://github.com/SEC-T/badge-2024

# Let’s make noise!

**Your SEC-T 0x10sion Synth Badge comes pre-programmed with a really cool synthesizer, called “Quadrant”.**

**If you are happy playing with “Quadrant” then no need to re-program your SEC-T Synth Badge.**

**But if you want to program other synths into your SEC-T Synth Badge, the next pages show you how…**



# Re-programming the Synth Badge

**The SEC-T 0x10sion Music Synthesizer Badge**
**was created from the ArduTouch Music Synthesizer project**

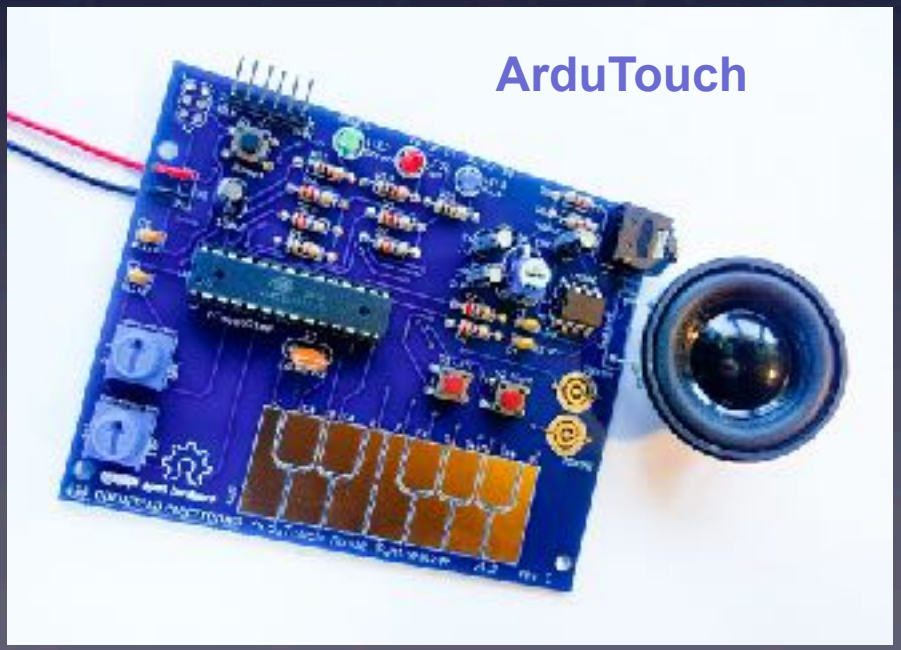
**The following pages show how to re-program the ArduTouch synthesizer**

*Re-Programming the Synth Badge board*
*is the same as*
*re-programming the ArduTouch board*




Synth Badge

***Or***

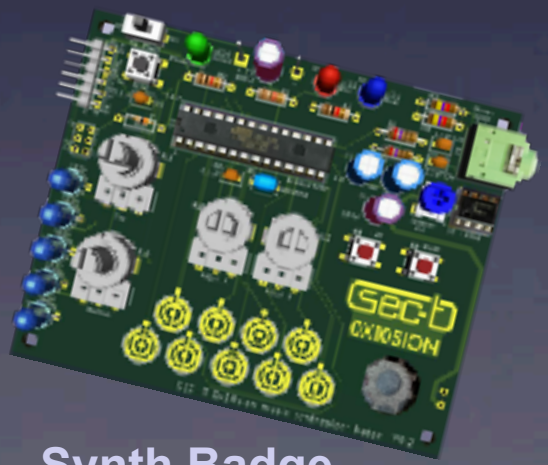


# Re-programming the Synth Badge

**We have written several way cool synthesizers for the ArduTouch!**
**Each is unique, and each way different than the others.**

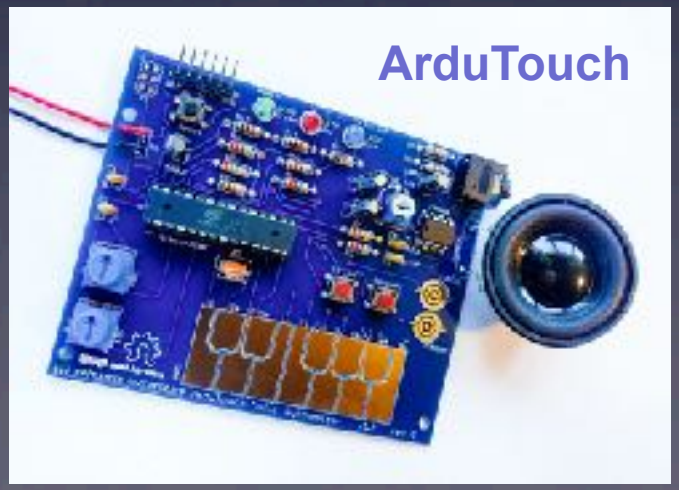
**To program in a new synth in your Synth Badge, you will need:**

- the Arduino software
  <http://arduino.cc>
- a USB-Serial adapter cable (such as an FTDI, or equivalent)
  with its driver installed
- the SEC-T 0x10sion Synth Badge Arduino library
  <https://github.com/SEC-T/badge-2024/tree/main/Firmware/Arduino%20Library>
- An ArduTouch synth sketch
  <http://cornfieldelectronics.com/cfe/projects.php#ardutouch>




Synth Badge

*Or*



# Re-programming the Synth Badge

**We have written several way cool synthesizers for the ArduTouch!**
**Each is unique, and each way different than the others.**

**To program in a new synth in your Synth Badge, you will need:**

- the Arduino software
  <http://arduino.cc>
- a USB-Serial adapter cable (such as an FTDI, or equivalent)
  with its driver installed
- the SEC-T 0x10sion Synth Badge Arduino library
  <https://github.com/SEC-T/badge-2024/tree/main/Firmware/Arduino%20Library>
- An ArduTouch synth sketch
  <http://cornfieldelectronics.com/cfe/projects.php#ardutouch>

**The following pages show you how to do the above, in detail.**

# Arduino

**Arduino is a very powerful tool!**
**But it is very easy to use.**
**It was designed for total beginners to use successfully.**

**I won't give a complete tutorial here – just some basics.**
**For more info, there are many good Arduino tutorials online.**
**A good place to start is:**
**<https://www.arduino.cc/en/Tutorial/HomePage>**



# Arduino

**First:**

Download and install the Arduino software

< http://arduino.cc >

Any version is OK



# Re-programming the Synth Badge

**Second:**

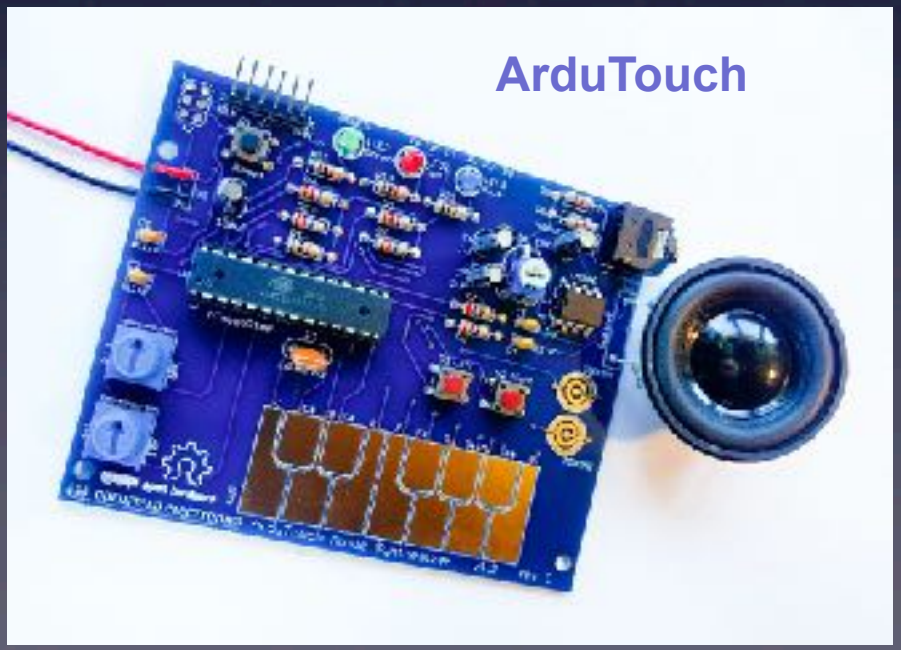
Download and install the SEC-T 0x10sion ArduTouch Arduino library
<https://github.com/SEC-T/badge-2024/tree/main/Firmware/Arduino%20Library>

**(details on this soon)**




Synth Badge

*Or*



# Re-programming the Synth Badge

**Third:**

Download ArduTouch synth sketches
<http://cornfieldelectronics.com/cfe/projects.php#ardutouch>

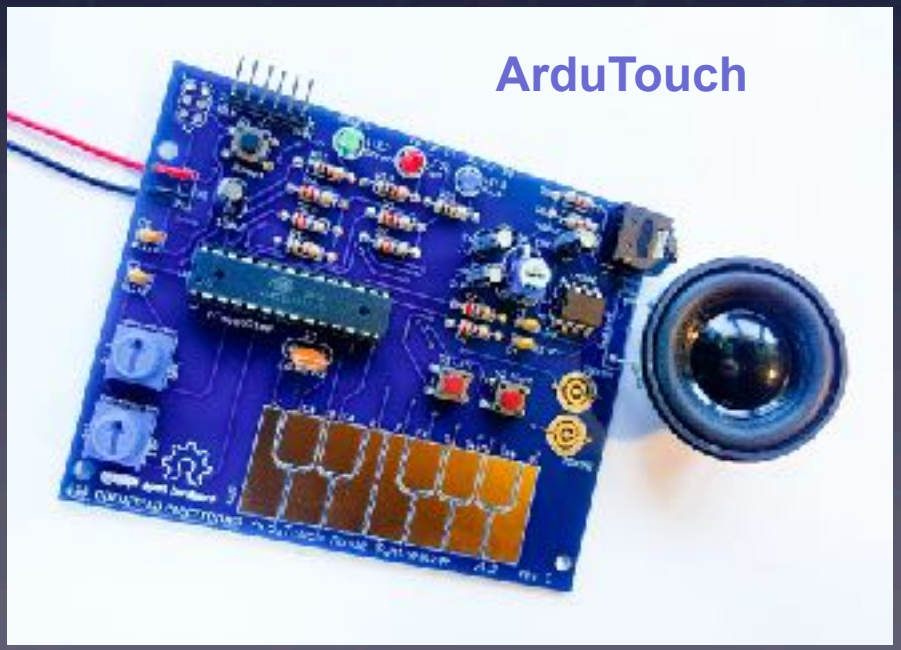
Store them on your computer anywhere you like.

**(details on this soon)**




Synth Badge

*Or*

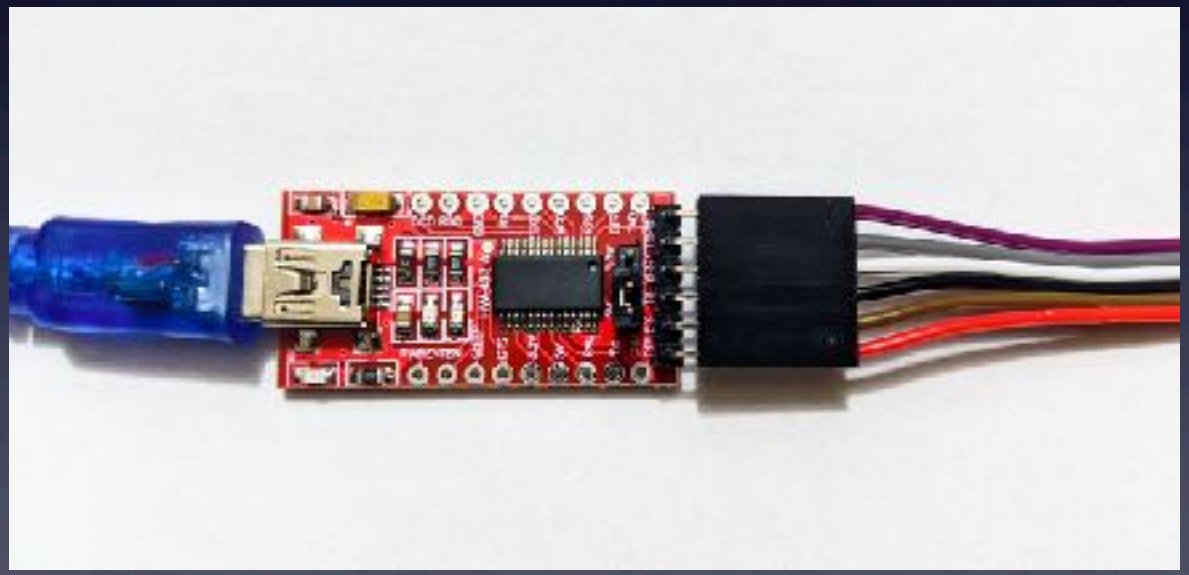


# Connecting your Synth Badge to your computer

**USB-Serial adapter cable**

**Ones available from Cornfield Electronics look like this:**

<https://cornfieldelectronics.com/cfe/products/buy.php?productId=usbcable>

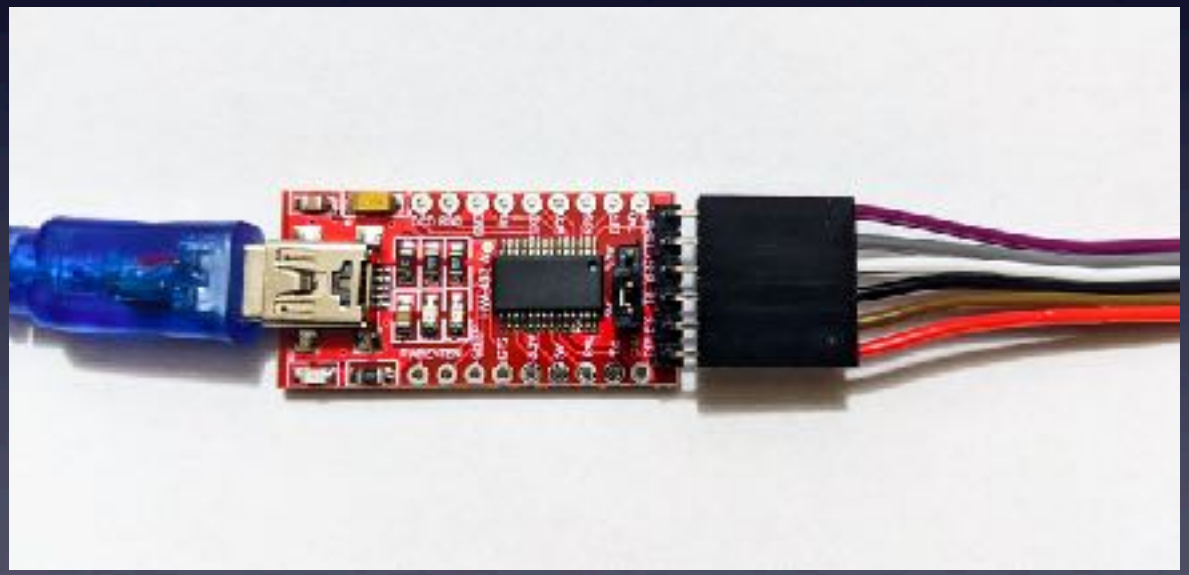


# Connecting your Synth Badge to your computer

**USB-Serial adapter cable**

**Ones available from Cornfield Electronics look like this:**

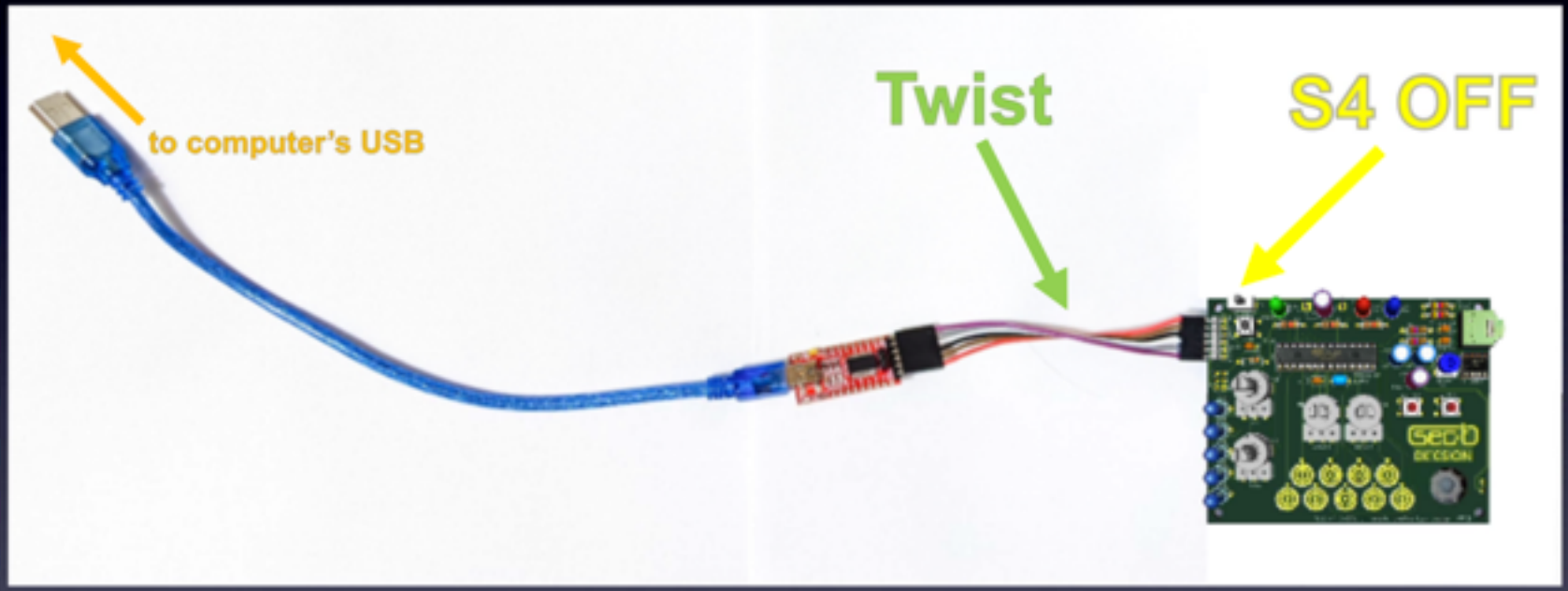
<https://cornfieldelectronics.com/cfe/products/buy.php?productId=usbcable>



**You may need to download and install a driver for your Operating System (Windows, MacOS, or Linux):**

<https://ftdichip.com/drivers/vcp-drivers/>

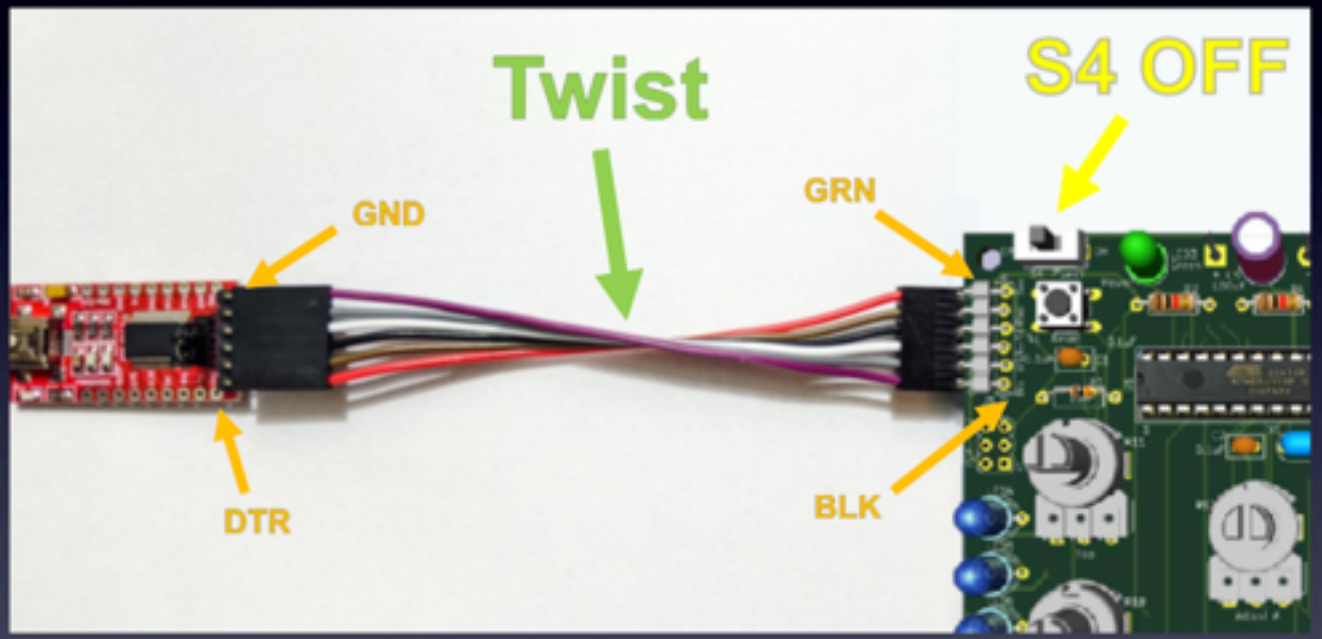
# Connecting your Synth Badge to your computer




**<u>IMPORTANT</u>:**
**Make sure the Power switch S4 on your Synth Badge is OFF**

# Connecting your Synth Badge to your computer

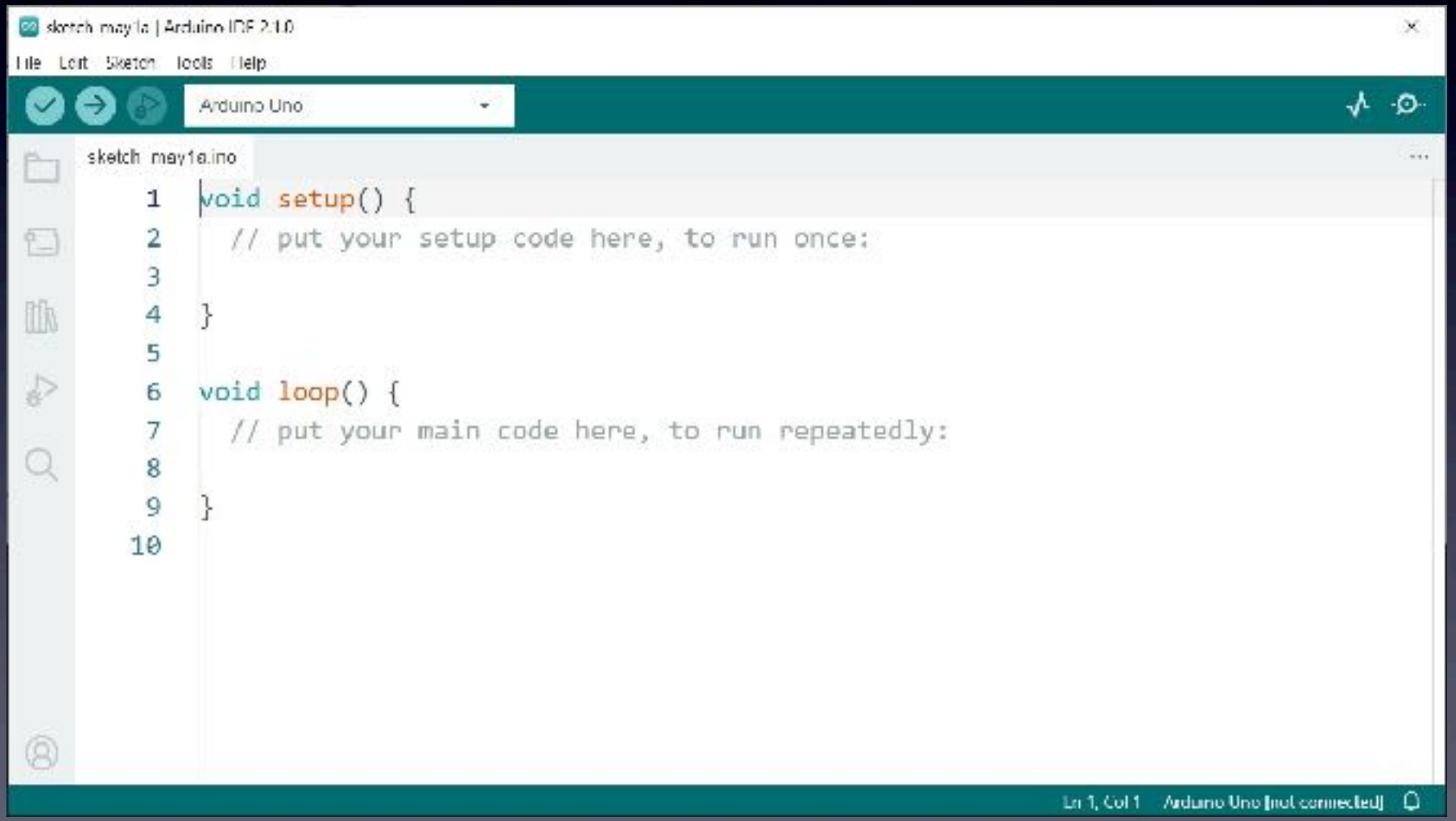
**This shows a few more details:**




**<u>IMPORTANT</u>:**
**Make sure the Power switch S4 on your Synth Badge is OFF**

# Arduino

**After you download and install the Arduino software start it, and you will see a screen that looks like this:**

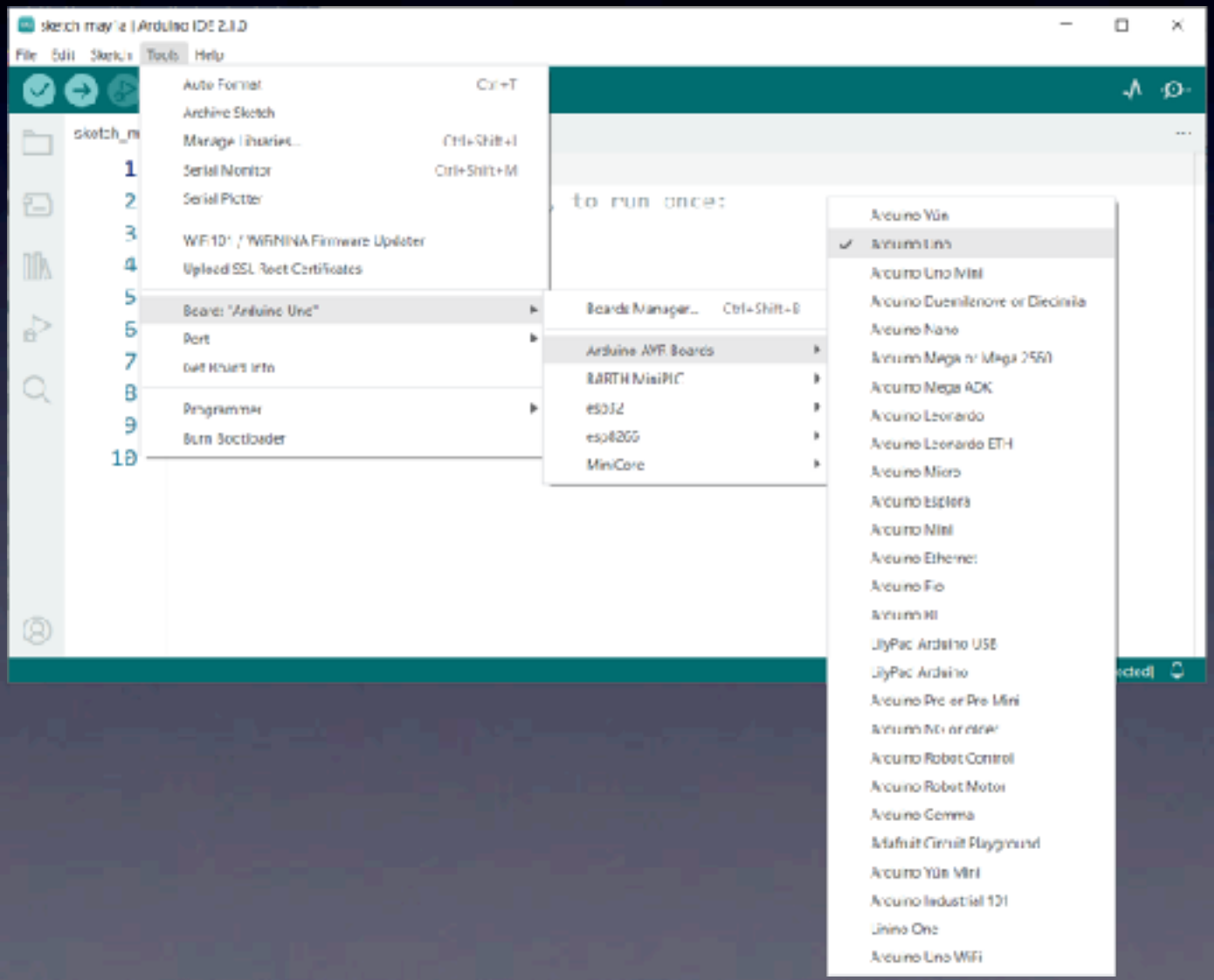


# Arduino

**The first time you start your Arduino software you need to do three things to set things up**

**(1)**
**Choose “Uno” as the Board**

**(Your Synth Badge board acts just like an Arduino Uno board)**

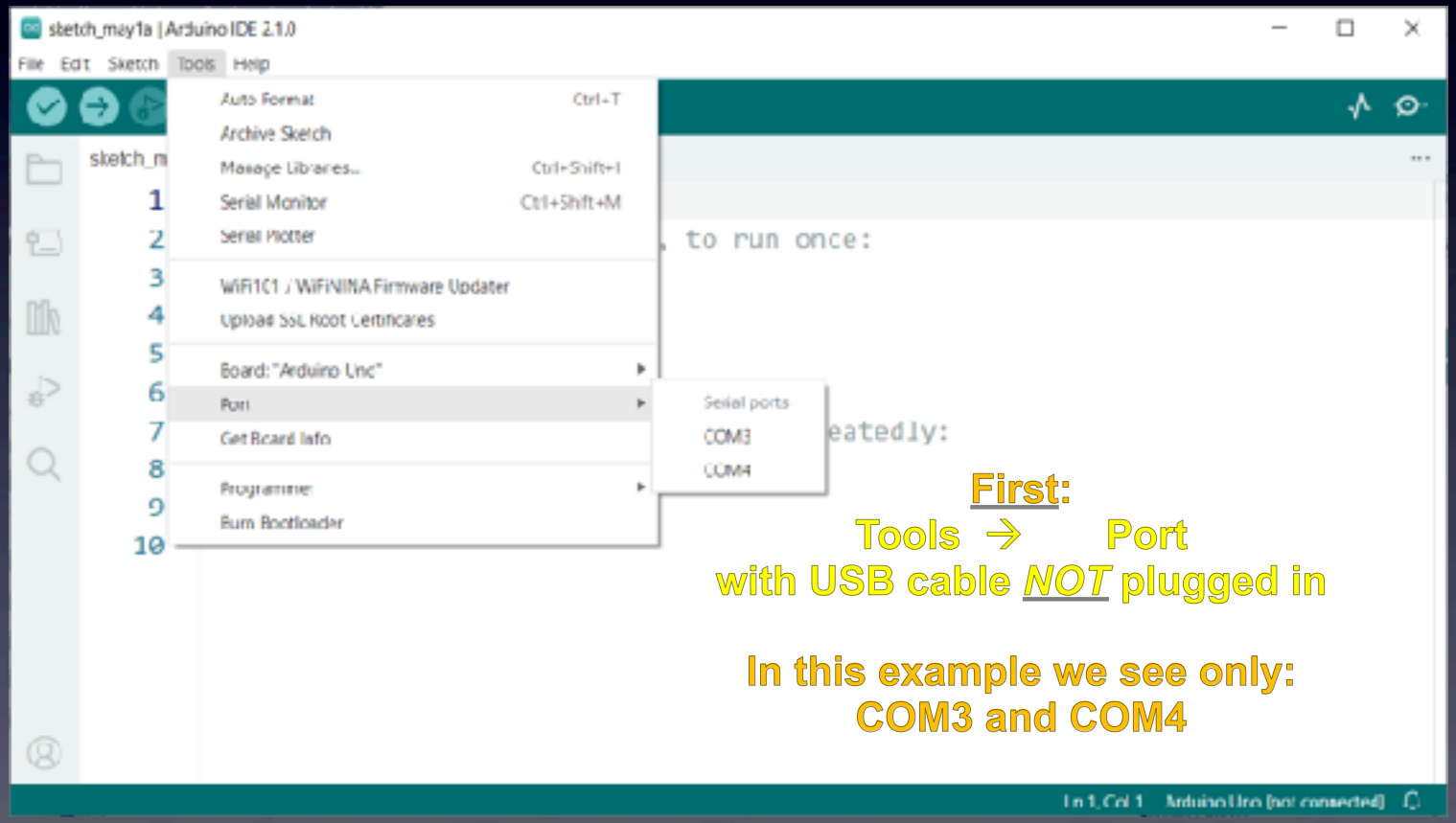


# Arduino

**The first time you start your Arduino software you need to do three things to set things up**

**with USB cable *NOT* plugged in:**

**(2) Choose the Port (this will be different depending on your Operating System)**



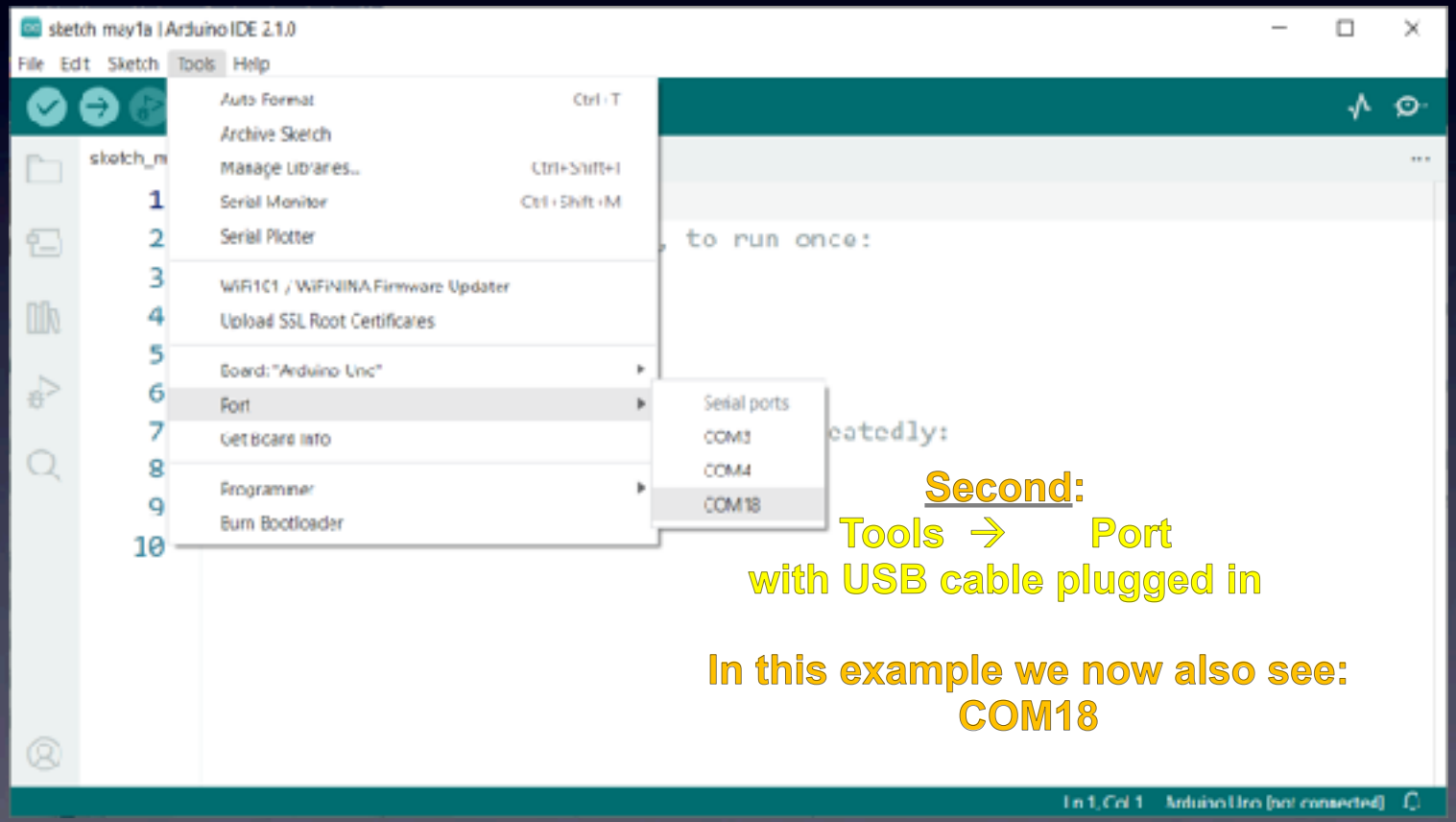
# Arduino

**The first time you start your Arduino software you need to do three things to set things up**

**with USB cable plugged in (with driver installed):**

**(2)**
**Choose the Port (this will be different depending on your Operating System)**

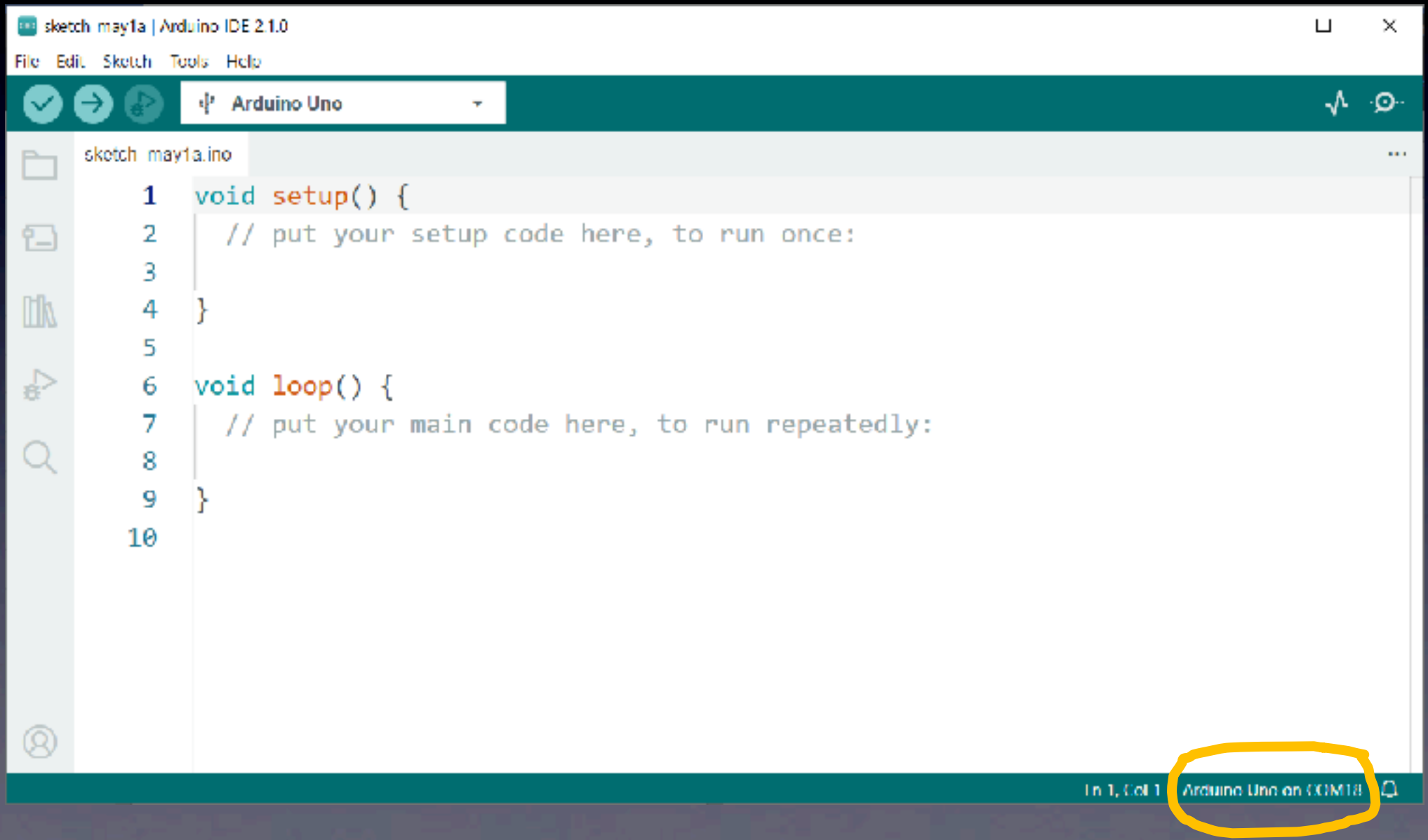
**(After installing the driver for your USB-Serial cable and plugging it in your operating system will see a serial port and it appears here.)**




Choose the new port:
In this example: COM18

# Arduino

## Your Arduino software is almost ready

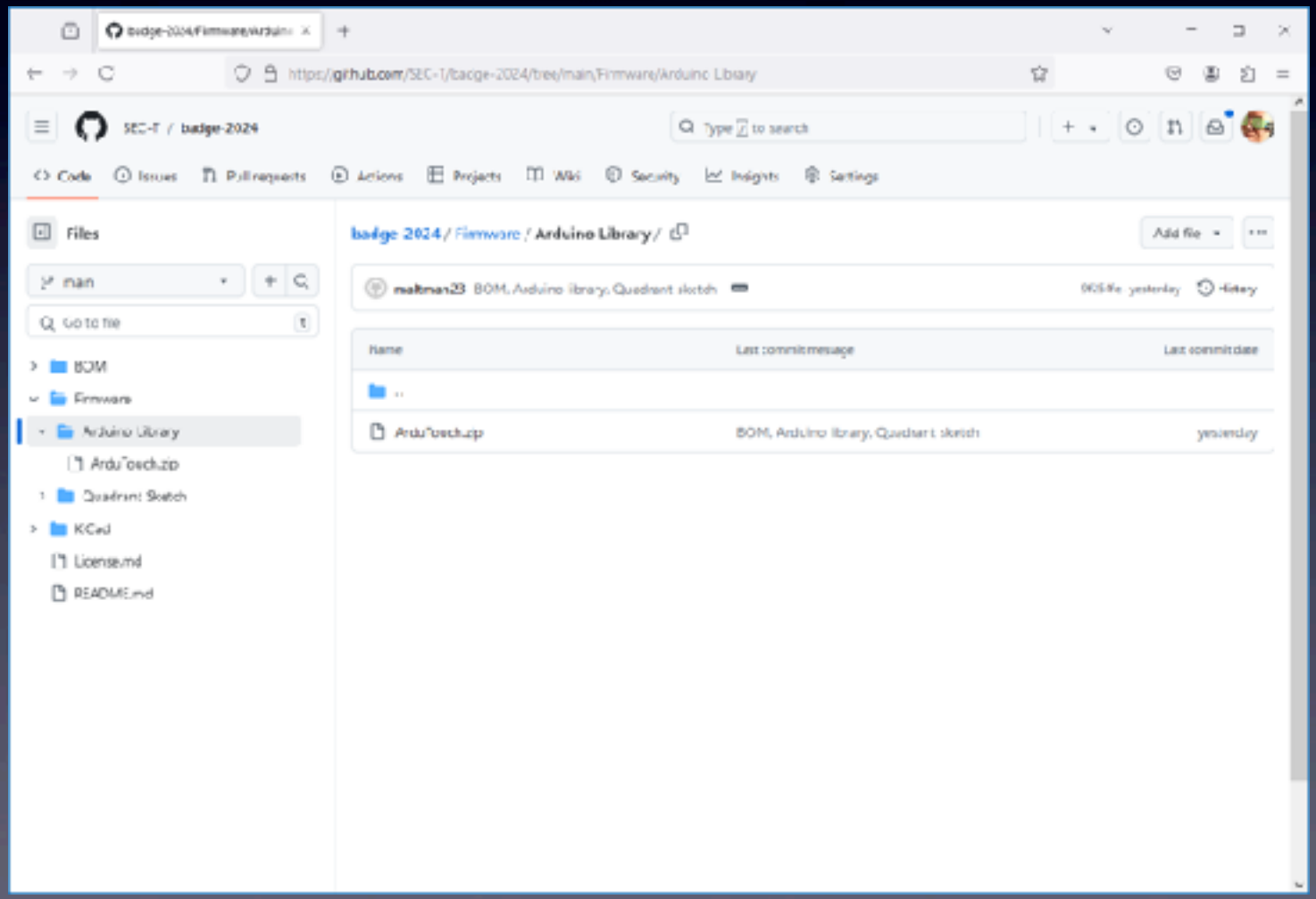


# Arduino

**The first time you start your Arduino software you need to do three things to set things up**

**(3) Install the SEC-T 0x10sion Synth Badge library**

https://github.com/SEC-T/badge-2024/tree/main/Firmware/Arduino%20Library

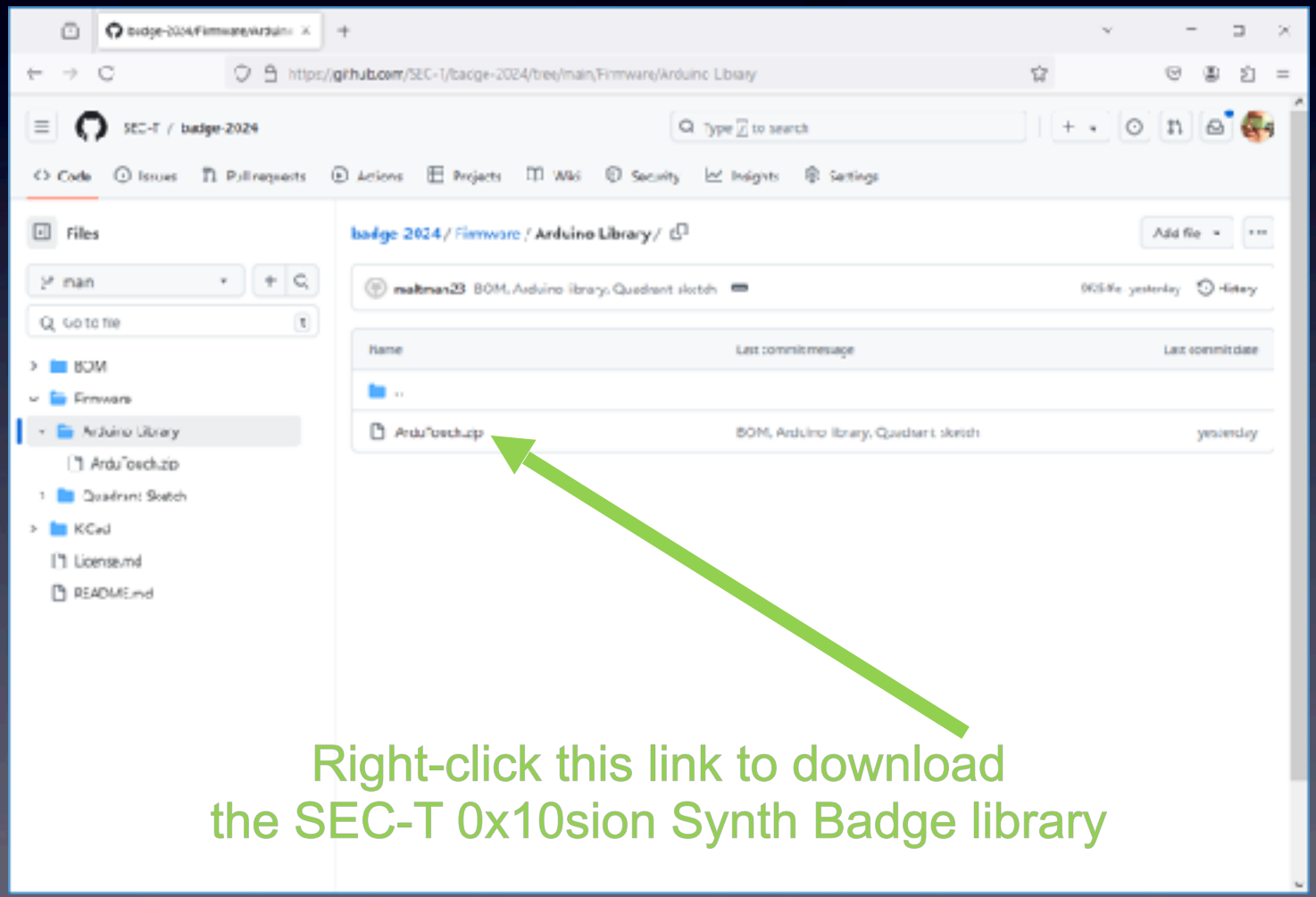


# Arduino

**The first time you start your Arduino software you need to do three things to set things up**

**(3) Install the SEC-T 0x10sion Synth Badge library**

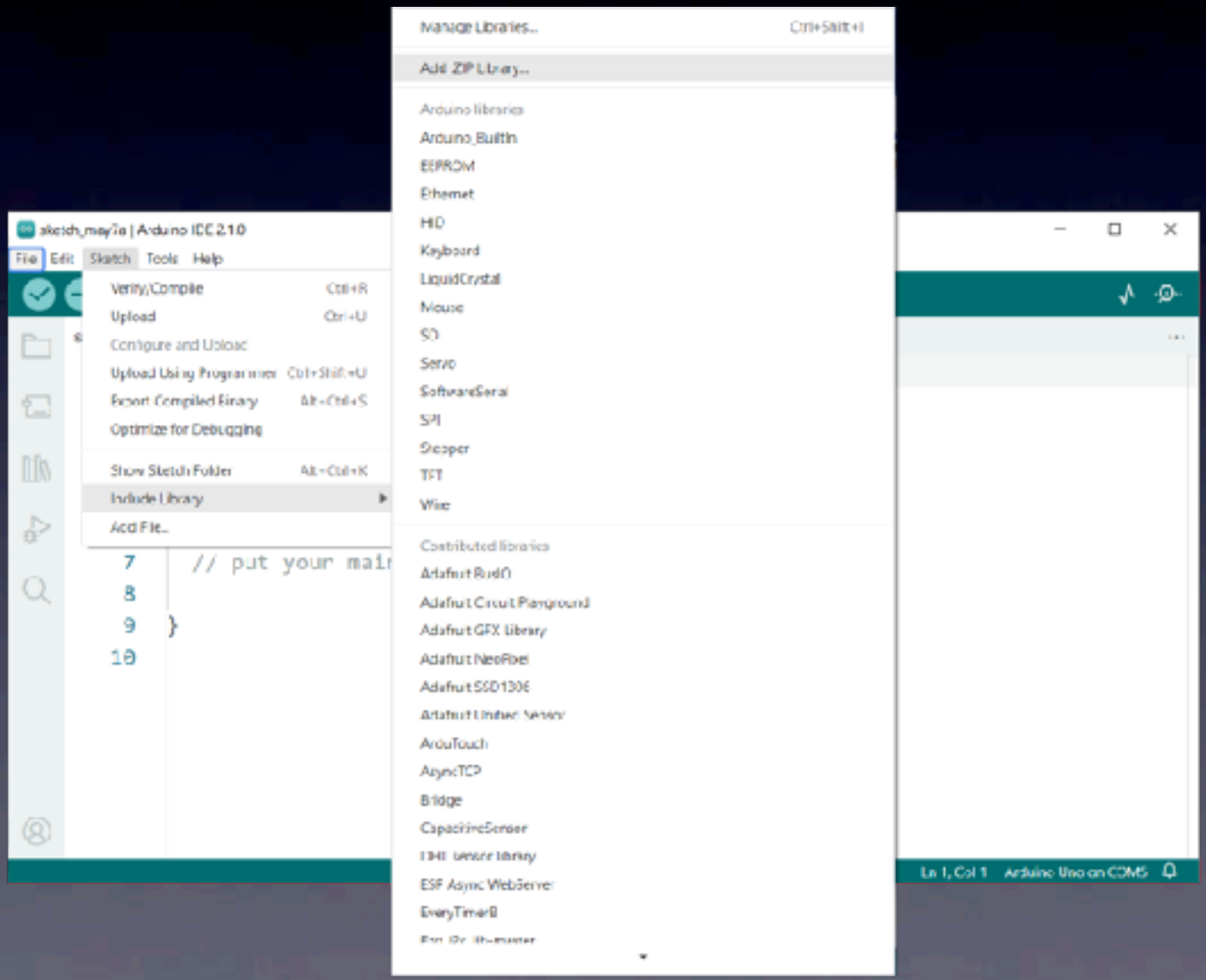
https://github.com/SEC-T/badge-2024/tree/main/Firmware/Arduino%20Library



# Arduino

**The first time you start your Arduino software you need to do three things to set things up**

**(3) Install the SEC-T 0x10sion Synth Badge library**



# Arduino

**The first time you start your Arduino software**
**you need to do three things to set things up**

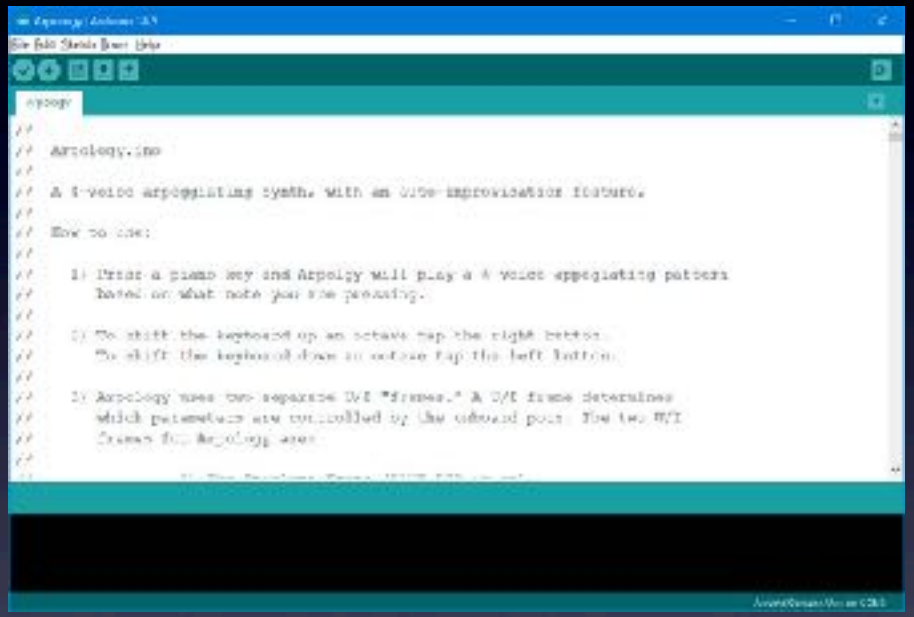
***Your Arduino software is now ready***

to program a synth sketch
into your SEC-T 0x10sion Synth Badge !

# Arduino

**Designed for non-geeky artists**

**Download a new Synth Badge synth “sketch”**



**“Sketch” :**

**an Arduino program**

# Arduino

## Download a new Synth Badge synth “sketch”



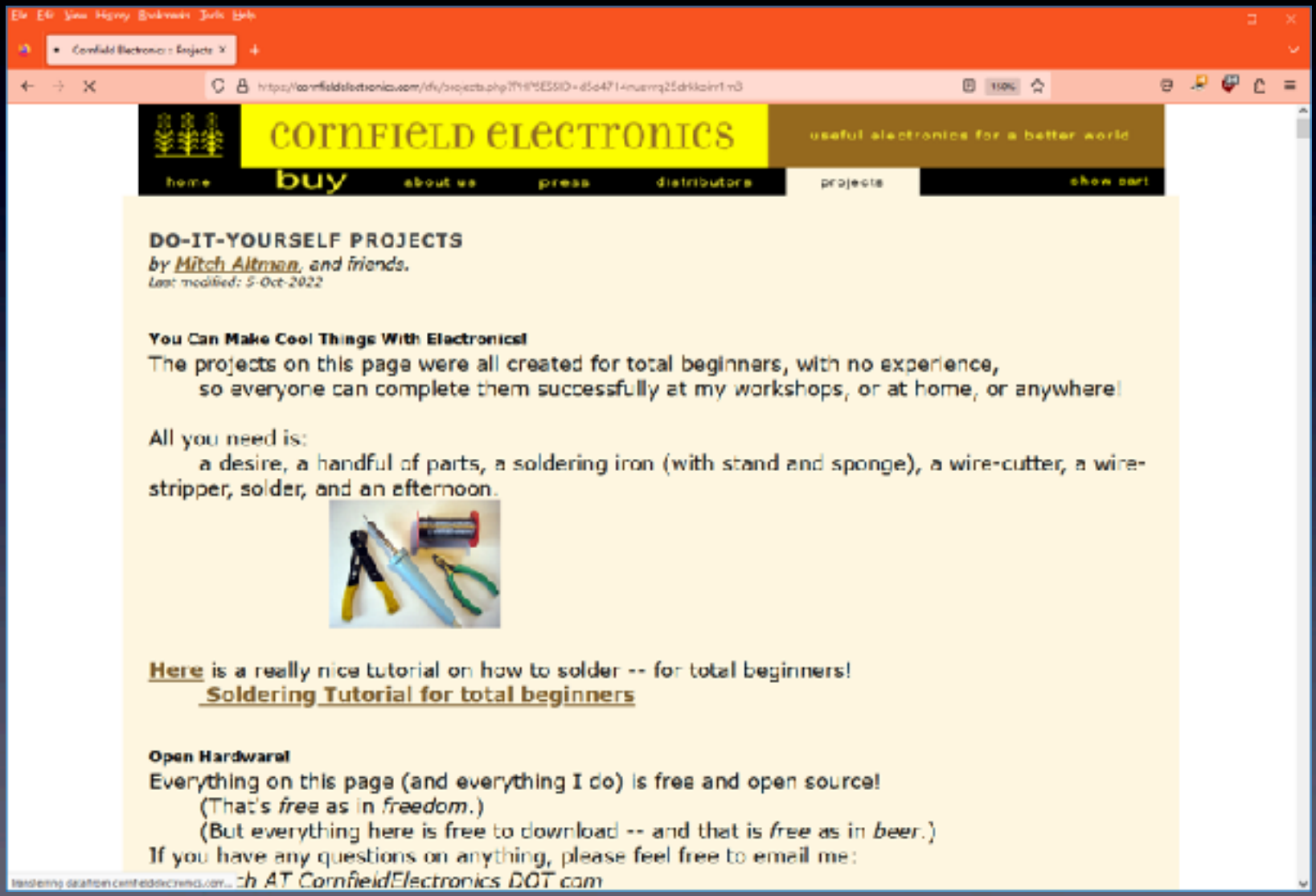
# Arduino

## Download a new Synth Badge synth “sketch”



# Arduino

## Download a new Synth Badge synth “sketch”



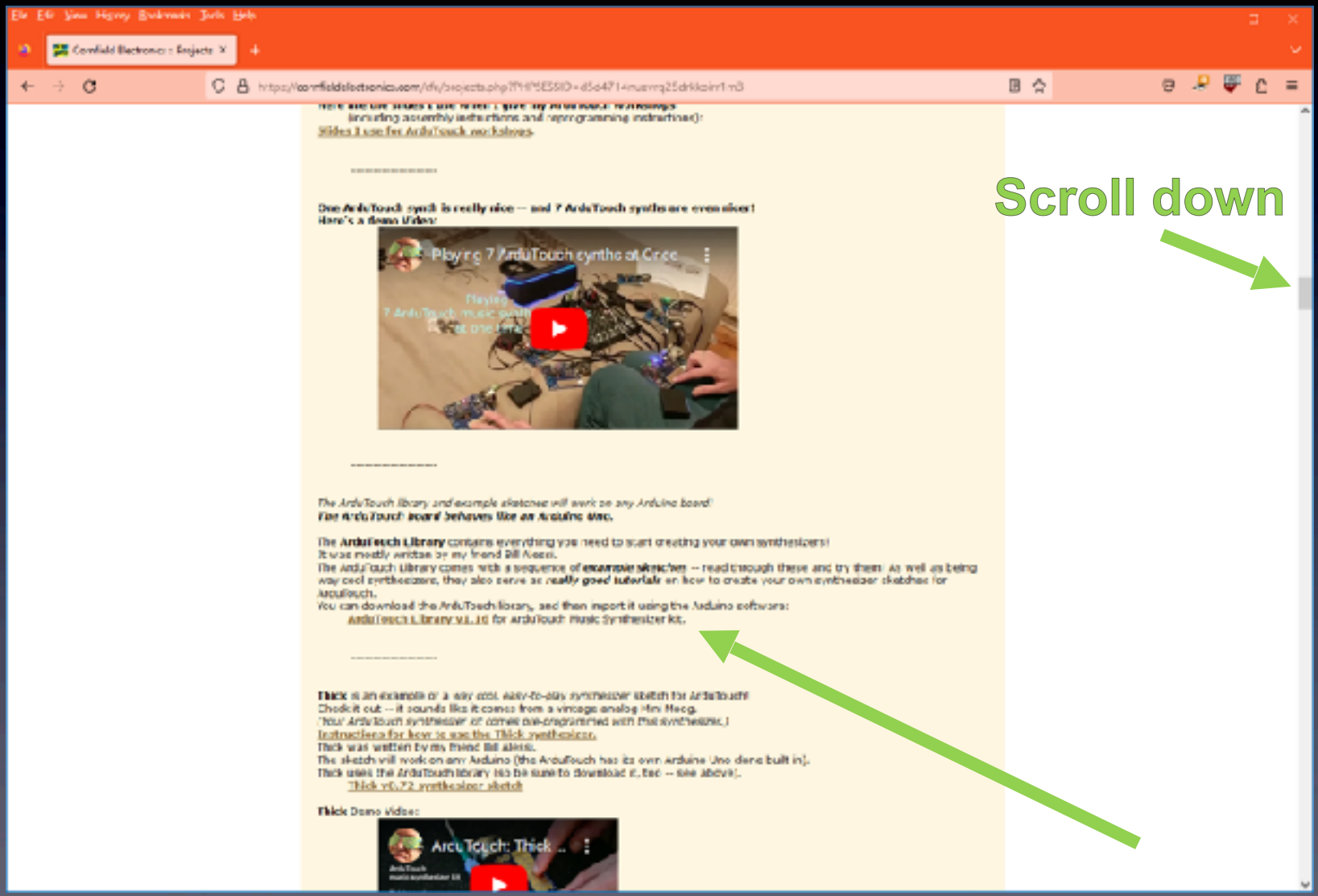
# Arduino

## Download a new Synth Badge synth “sketch”



# Arduino

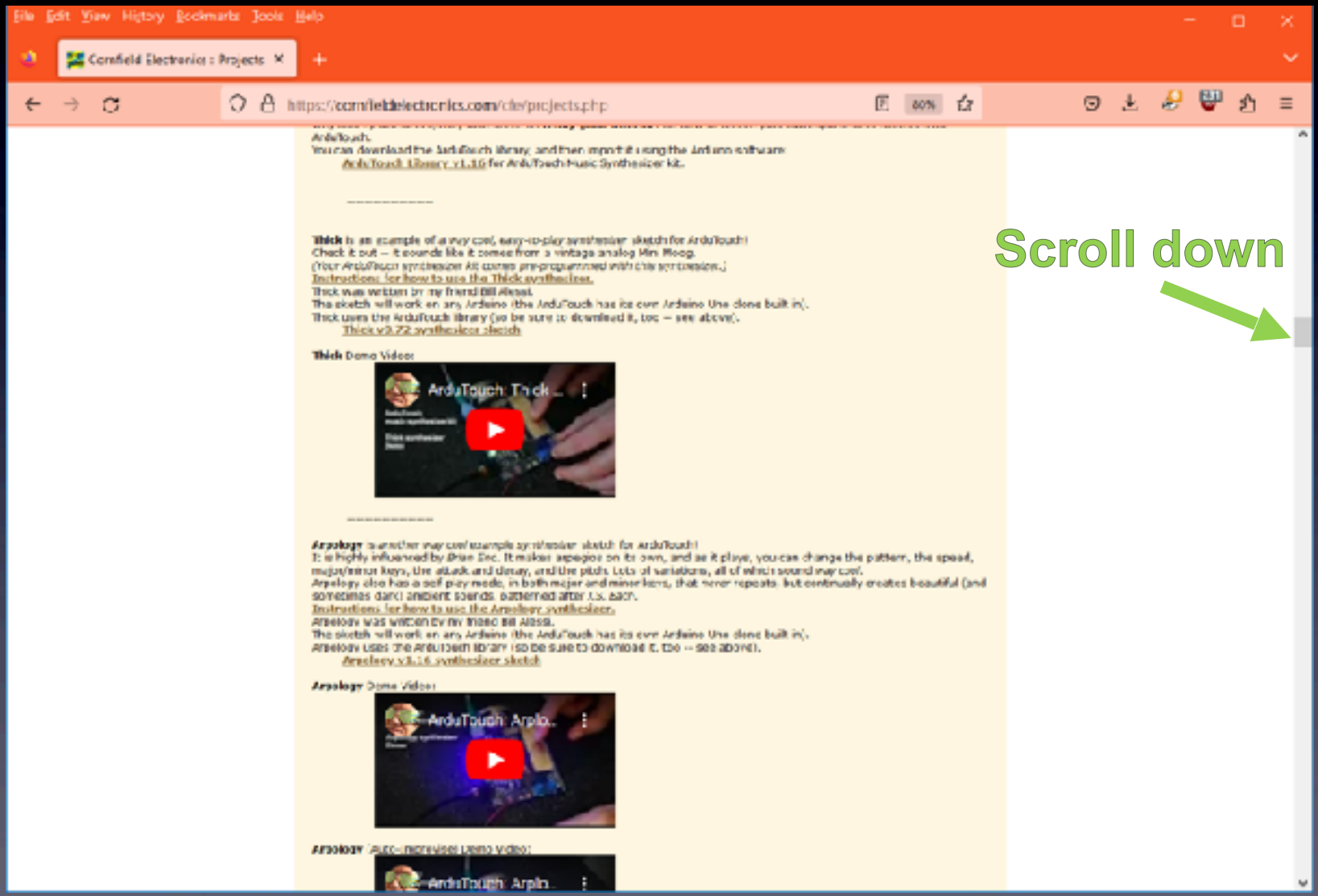
## Download a new Synth Badge synth “sketch”




**NOTE**: *Do NOT use this ArduTouch library for the SEC-T Synth Badge*

# Arduino

## Download a new Synth Badge synth “sketch”



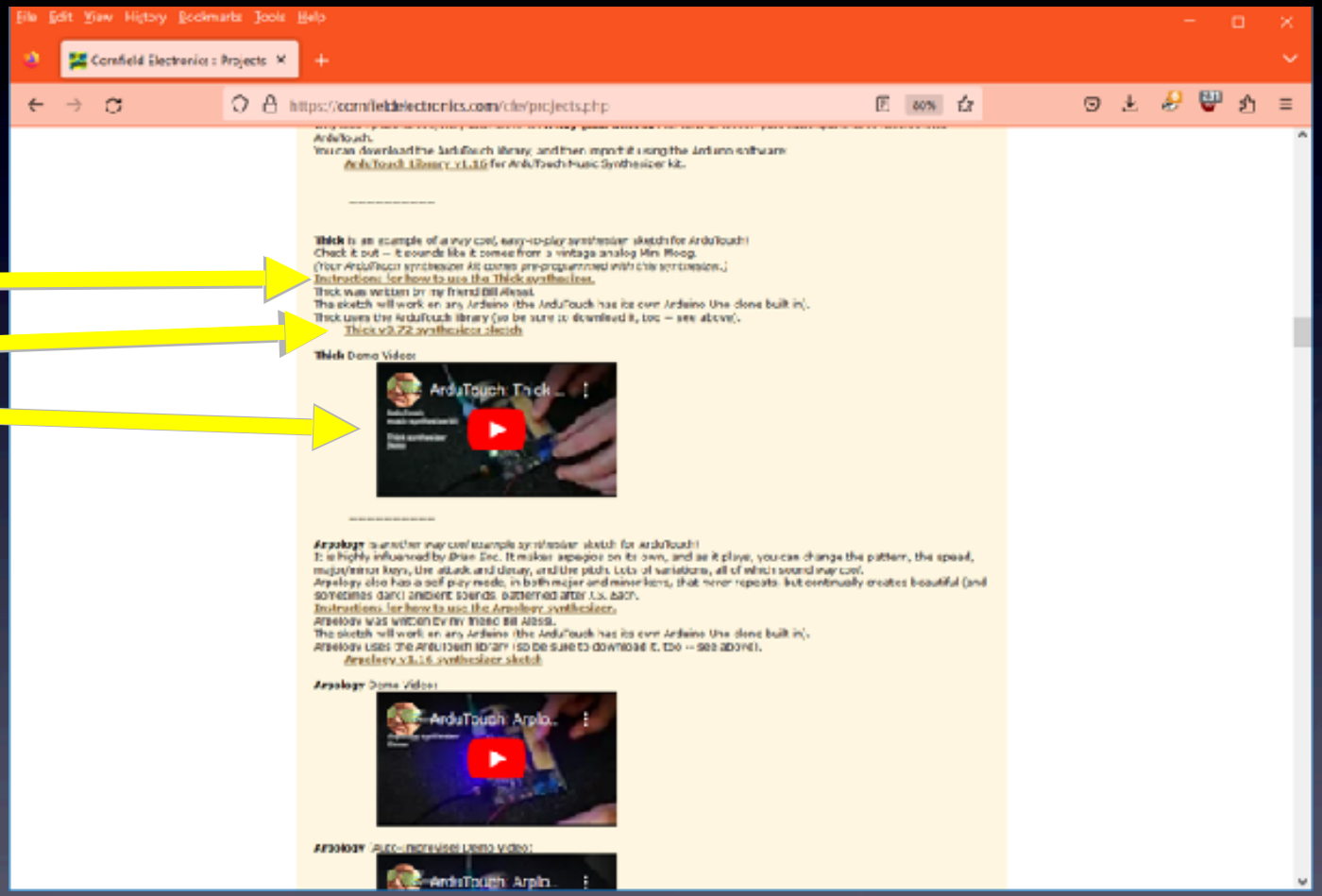

Here you will see several ArduTouch synth sketches

# Arduino

## Download a new Synth Badge synth “sketch”

**Each ArduTouch synth sketch has:**

- Instructions
- Sketch download
- Demo video(s)



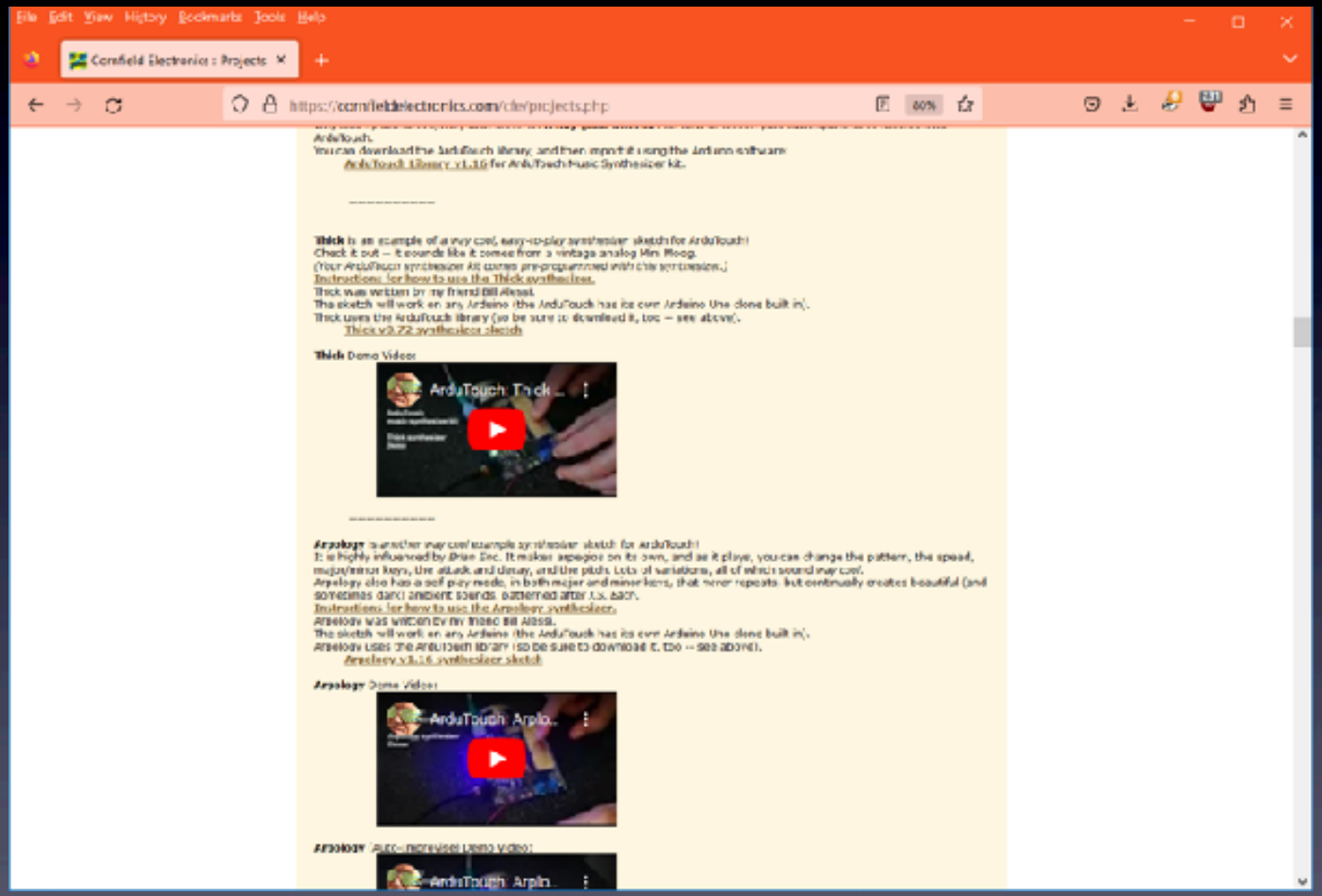

**Here you will see several ArduTouch synth sketches**

# Arduino

## Download a new Synth Badge synth “sketch”

**NOTE:**
**These sketches do <u>NOT</u> have <u>functionality for</u>:**

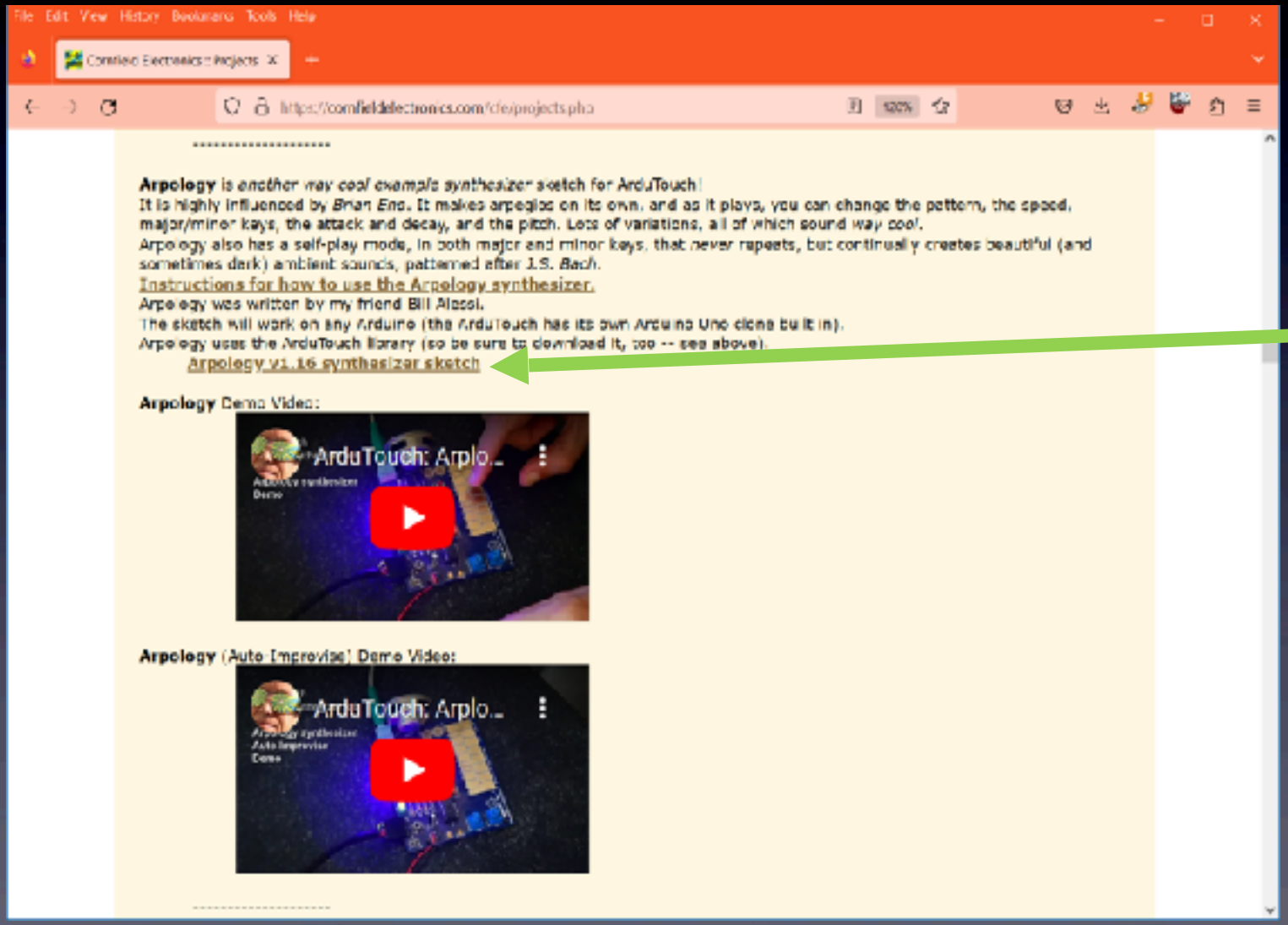
- RGB LEDs
- knobs A and B




**Here you will see several ArduTouch synth sketches**

# Arduino

## Download a new Synth Badge synth “sketch”

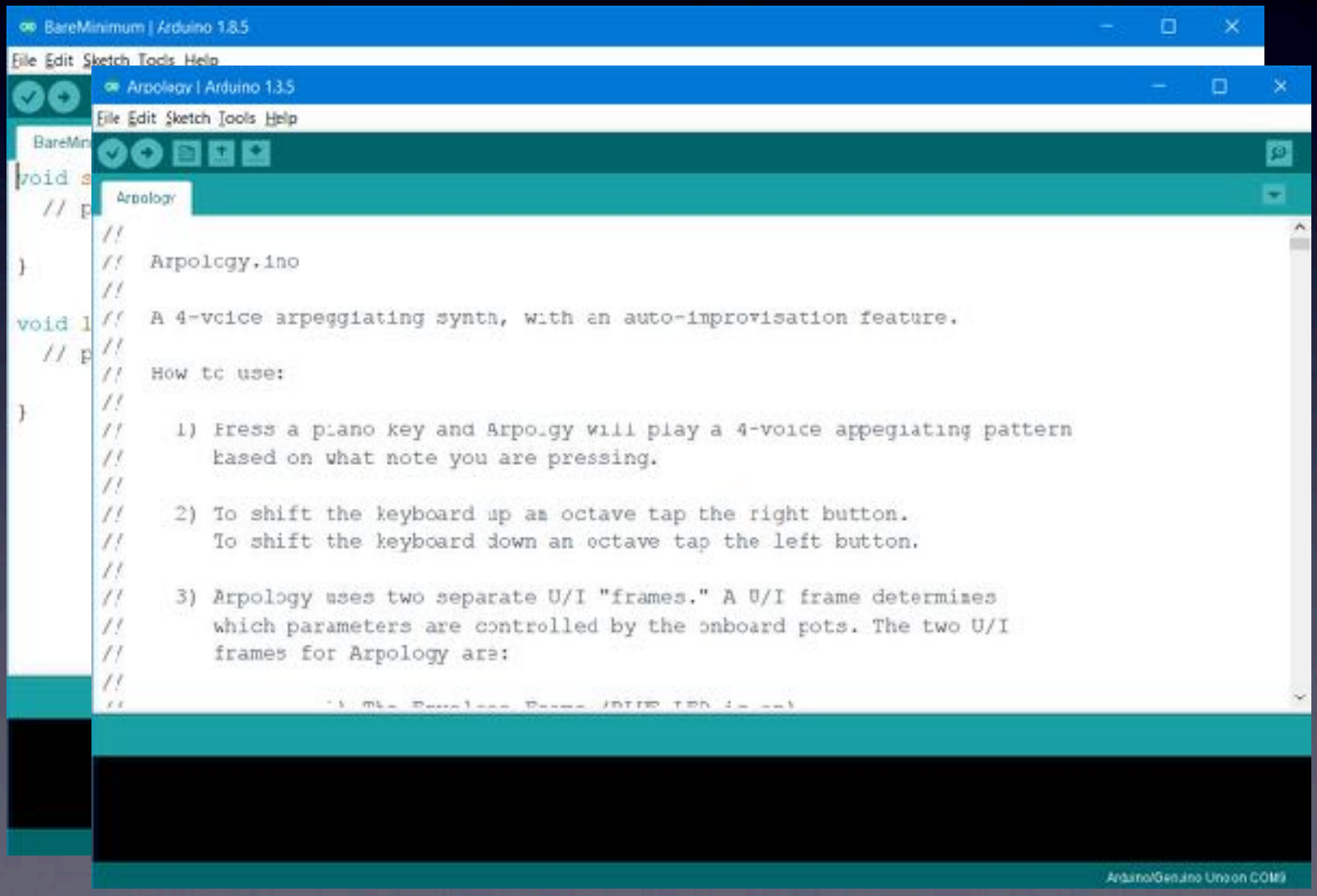



Click the sketch link to download the sketch

# Arduino

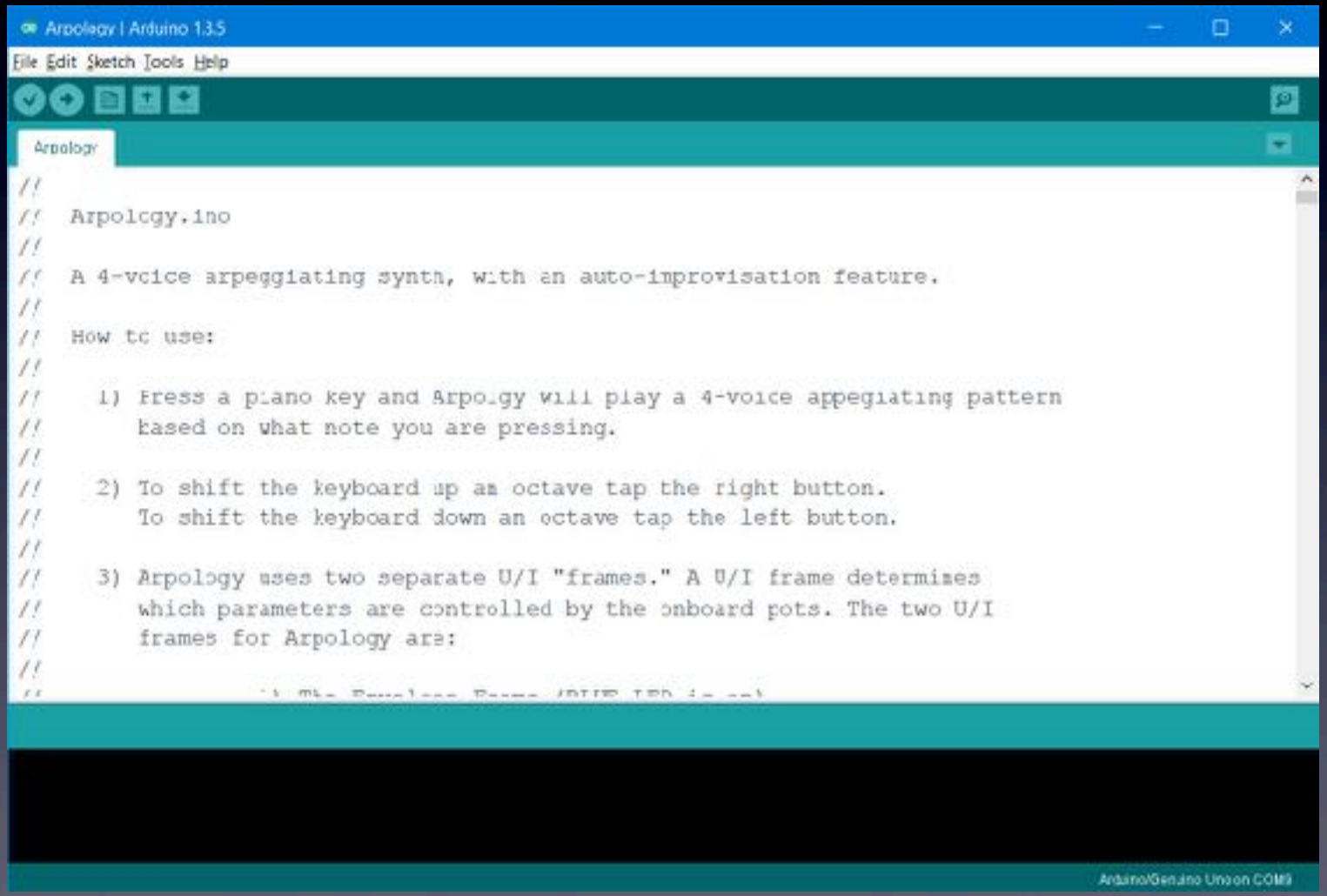
**You can open the ArduTouch / Synth Badge synth sketch:**
**File → Open…**

**(I opened "Arpology here)**



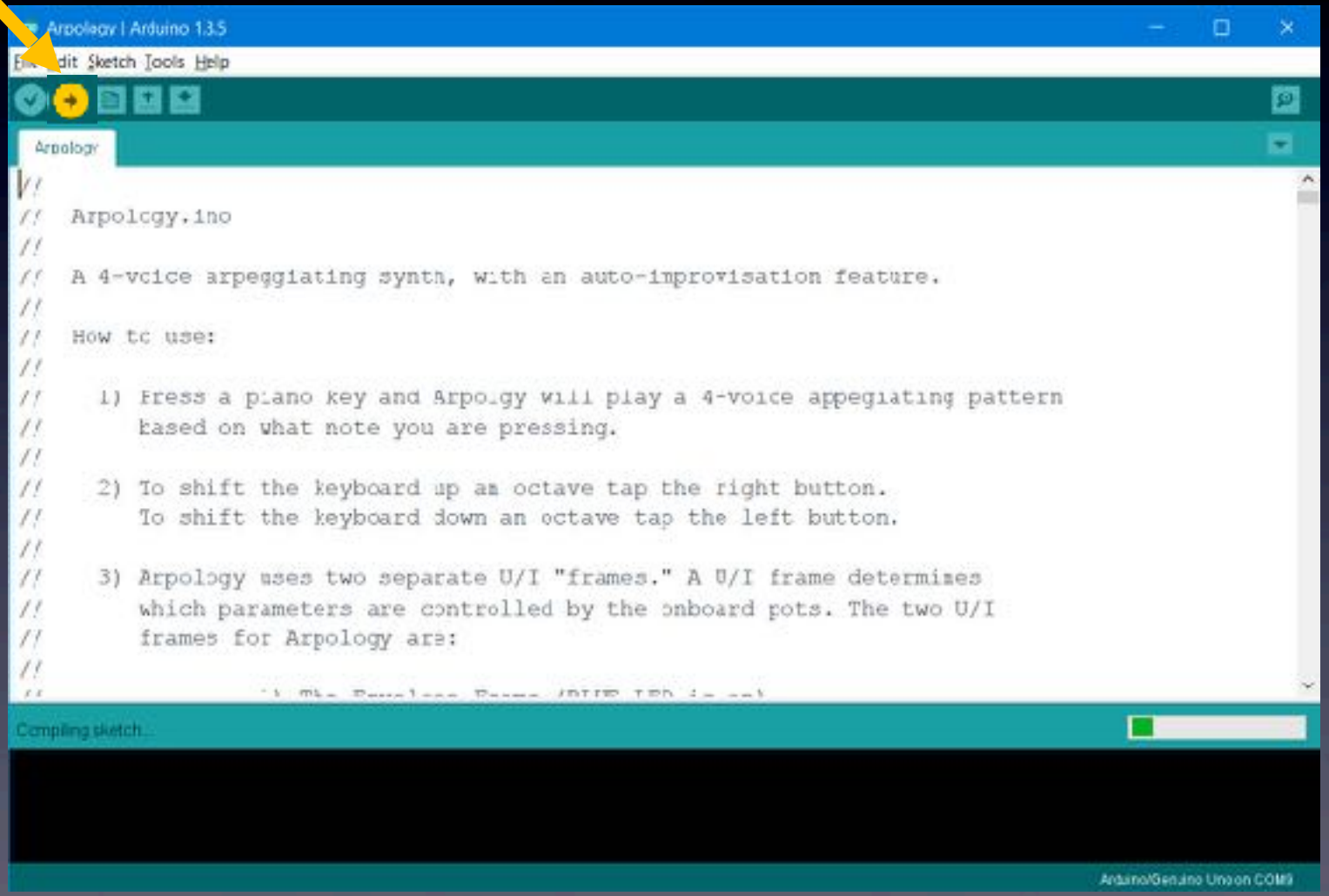
# Arduino

**You can now program your Synth Badge with a new synth sketch !**



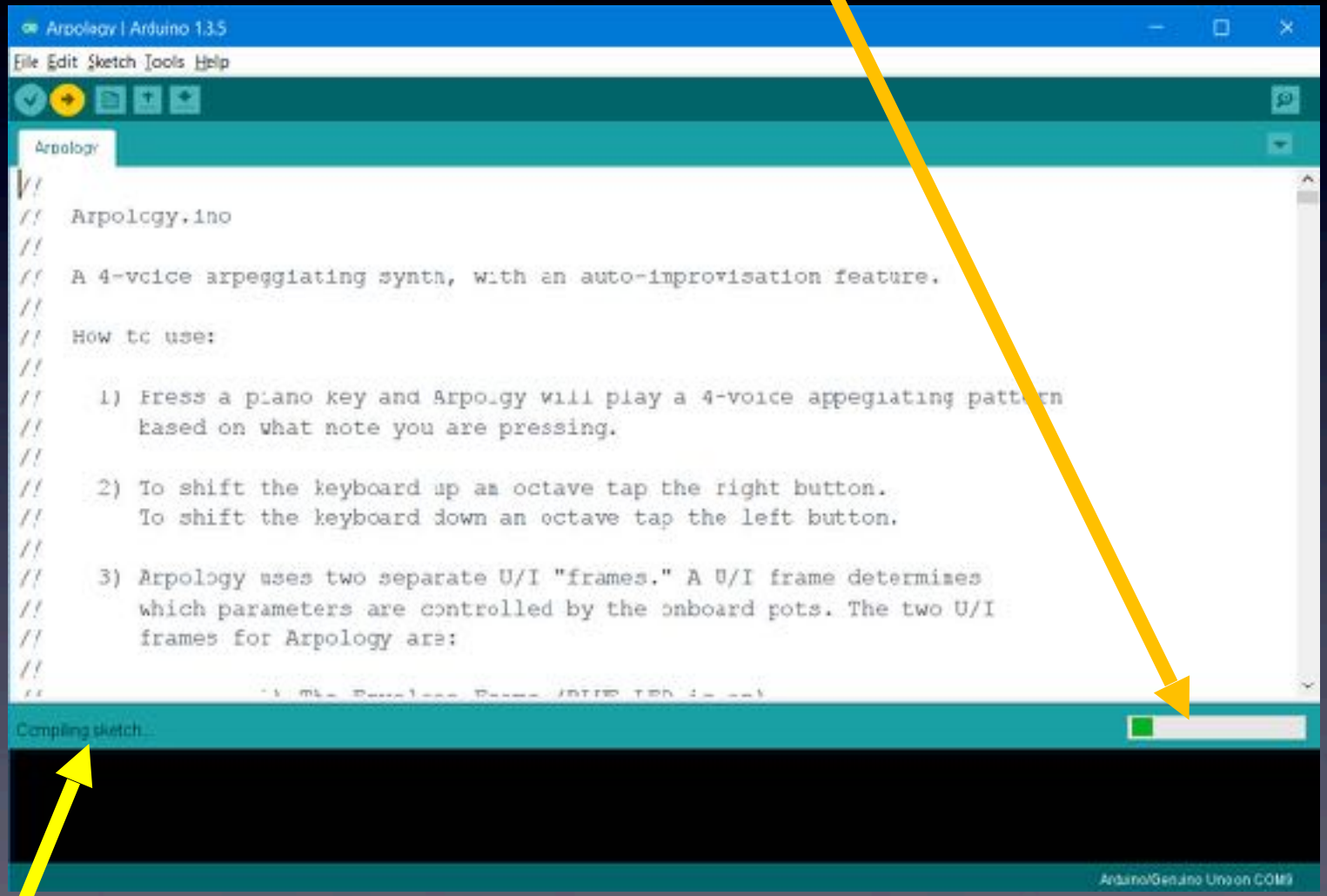
# Arduino

**With the USB-Serial cable connected to your Synth Badge board press the Upload button**



# Arduino

**While uploading, you will see a progress bar…**




**…and when it's completed successfully, it says: "Upload done"**

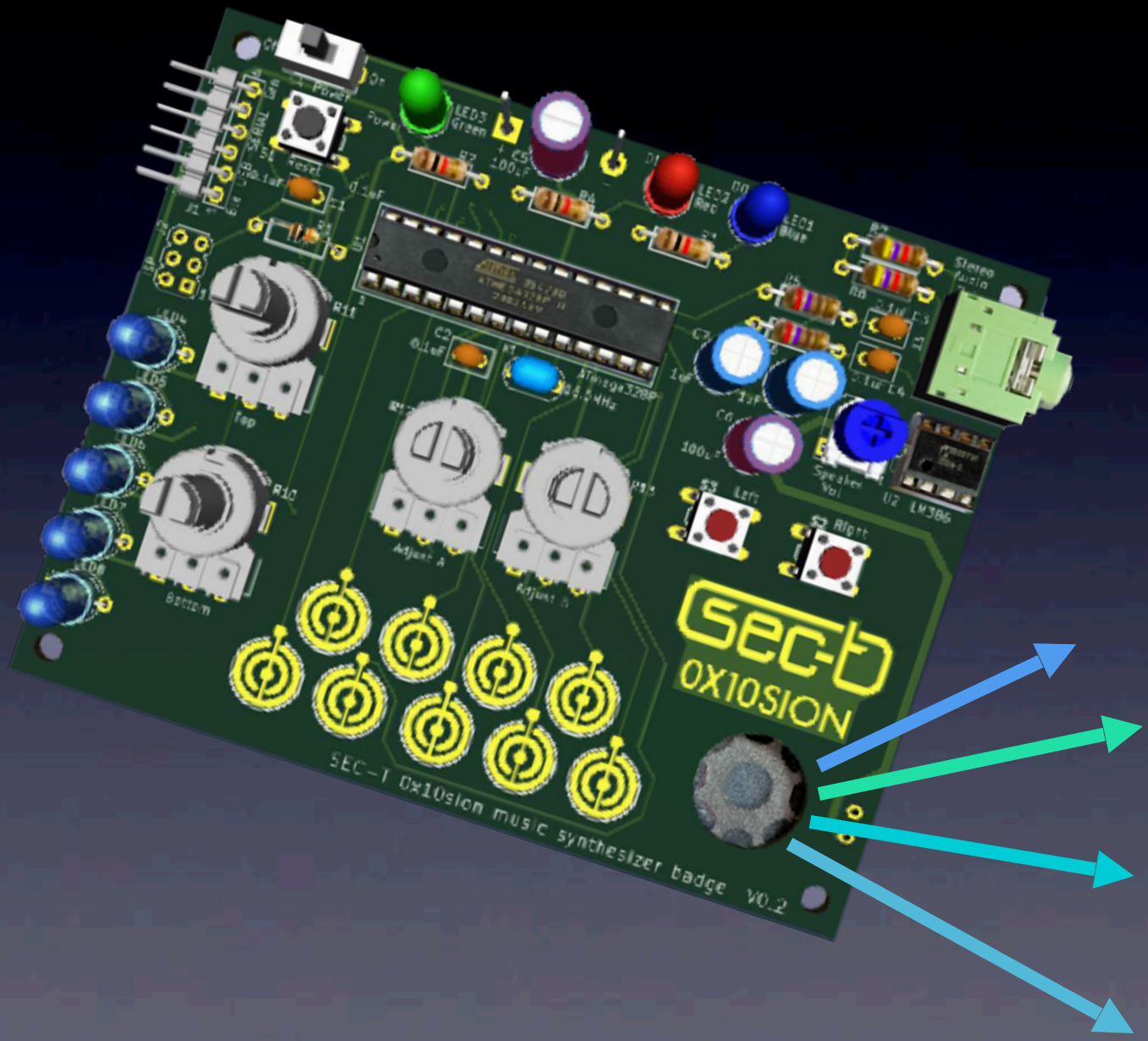
# ArduTouch

**Disconnect your SEC-T 0x10sion Synth Badge board from the USB-Serial cable,**

**turn on the Power switch S4,**

# And…

# Let’s make new noise!



Please Remember:

to

**Wash your hands**

after soldering

12-Sep-2025

# SEC-T 0x10sion
# Music Synthesizer Badge

## Assembly Instructions

(with re-programming how-to)




V 0.2

open source hardware

SEC-T 0X10SION

CC BY-SA 4.0 © 2024 Mitch Altman

CORNFIELD ELECTRONICS